CHAPTER 9

Statistics

9.1 Revision EMCJK

Terminology EMCJM

Measures of central tendency:

Provide information on the data values at the centre of the data set.

- The **mean** is the 'average' value of a data set. It is calculated as

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

  where the x_i are the data and n is the number of data entries. We read $\bar{x}$ as "x bar".

- The **median** is the middle value of an ordered data set. To find the median, we first sort the data in ascending or descending order and then pick out the value in the middle of the sorted list. If the middle is in between two values, the median is the average of those two values.

Measures of dispersion:

Tell us how spread out a data set is. If a measure of dispersion is small, the data are clustered in a small region. If a measure of dispersion is large, the data are spread out over a large region.

- The **range** is the difference between the maximum and minimum values in the data set.

- The **inter-quartile range** is the difference between the first and third quartiles of the data set. The quartiles are computed in a similar way to the median. The median is halfway into the ordered data set and is sometimes also called the second quartile. The first quartile is one quarter of the way into the ordered data set, whereas the third quartile is three quarters of the way into the ordered data set.

  If you begin numbering your ordered data set with the number 1, the formulae for the location of each quartile are as follows:

$$\begin{aligned}
\text{Location of } Q_1 &= \frac{1}{4}(n-1)+1 \\
\text{Location of } Q_2 &= \frac{1}{2}(n-1)+1 \\
\text{Location of } Q_3 &= \frac{3}{4}(n-1)+1
\end{aligned}$$

- The **variance** of the data is the average squared distance between the mean and each data value.
  The variance of the data is

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$$

  in a population of n elements, $\{x_1; x_2; \ldots; x_n\}$, with a mean of $\bar{x}$.

- The **standard deviation** measures how spread out the values in a data set are around the mean. More precisely, it is a measure of the average distance between the values of the data in the set and the mean.
  The standard deviation of the data is

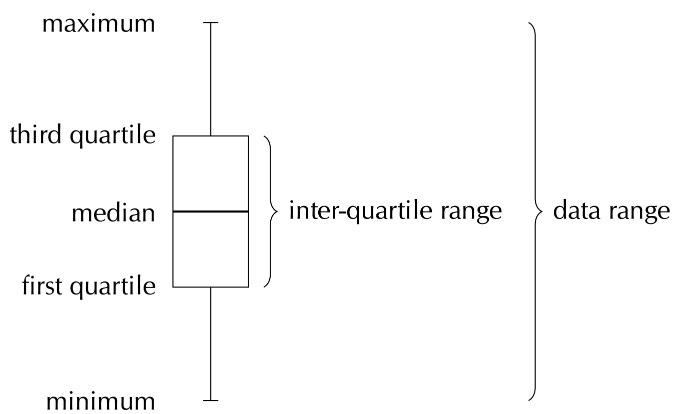
$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}}$$

  in a population of n elements, $\{x_1; x_2; \ldots; x_n\}$, with a mean of $\bar{x}$.

The **five number summary** combines a measure of central tendency, the median, with measures of dispersion, namely the range and the inter-quartile range. More precisely, the five number summary is written in the following order:

- minimum;
- first quartile;
- median;
- third quartile;
- maximum.

The five number summary is often presented visually using a **box and whisker diagram**, illustrated below.



See video: 29BS at www.everythingmaths.co.za

Worked example 1: Five number summary

QUESTION

Draw a box and whisker diagram for the following data set:

$$1{,}25\ ;\ 1{,}5\ ;\ 2{,}5\ ;\ 2{,}5\ ;\ 3{,}1\ ;\ 3{,}2\ ;\ 4{,}1\ ;\ 4{,}25\ ;\ 4{,}75\ ;\ 4{,}8\ ;\ 4{,}95\ ;\ 5{,}1$$

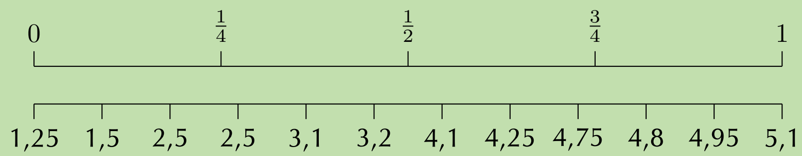
SOLUTION

Step 1: Determine the minimum and maximum

Since the data set is already ordered, we can read off the minimum as the first value ($1{,}25$) and the maximum as the last value ($5{,}1$).

Step 2: Determine the quartiles

There are 12 values in the data set. We can use the figure below or the formulae to determine where the quartiles are located.



Using the figure above we can see that the median is between the sixth and seventh values. We can confirm this using the formula:

$$\begin{aligned}\text{Location of } Q_2 &= \frac{1}{2}(n-1)+1\\ &= \frac{1}{2}(11)+1\\ &= 6{,}5\end{aligned}$$

Therefore, the value of the median is

$$\frac{3{,}2+4{,}1}{2} = 3{,}65$$

The first quartile lies between the third and fourth values. We can confirm this using the formula:

$$\begin{aligned}\text{Location of } Q_1 &= \frac{1}{4}(n-1)+1\\ &= \frac{1}{4}(11)+1\\ &= 3{,}75\end{aligned}$$

Therefore, the value of the first quartile is

$$Q_1 = \frac{2{,}5+2{,}5}{2} = 2{,}5$$

The third quartile lies between the ninth and tenth values. We can confirm this using the formula:

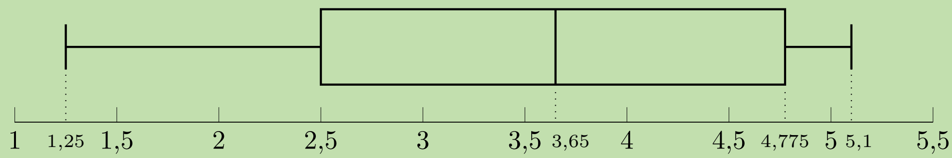
$$\begin{aligned}\text{Location of } Q_3 &= \frac{3}{4}(n-1)+1\\ &= \frac{3}{4}(11)+1\\ &= 9{,}25\end{aligned}$$

Therefore, the value of the third quartile is

$$Q_3 = \frac{4{,}75+4{,}8}{2} = 4{,}775$$

Step 3: Draw the box and whisker diagram

We now have the five number summary as $(1{,}25;\ 2{,}5;\ 3{,}65;\ 4{,}775;\ 5{,}1)$. The box and whisker diagram representing the five number summary is given below.



See video: 29BT at www.everythingmaths.co.za

Worked example 2: Variance and standard deviation

QUESTION

You flip a coin 100 times and it lands on heads 44 times. You then use the same coin and do another 100 flips. This time in lands on heads 49 times. You repeat this experiment a total of 10 times and get the following results for the number of heads.

$$\{44; 49; 52; 62; 53; 48; 54; 49; 46; 51\}$$

For the data set above:

- Calculate the mean.
- Calculate the variance and standard deviation using a table.
- Confirm your answer for the variance and standard deviation using a calculator.

SOLUTION

Step 1: Calculate the mean

The formula for the mean is

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

In this case, we sum the data and divide by 10 to get $\overline{x} = 50{,}8$.

Step 2: Calculate the variance using a table

The formula for the variance is

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n}$$

We first subtract the mean from each data point and then square the result.

x_i	44	49	52	62	53	48	54	49	46	51
$x_i - \overline{x}$	$-6{,}8$	$-1{,}8$	$1{,}2$	$11{,}2$	$2{,}2$	$-2{,}8$	$3{,}2$	$-1{,}8$	$-4{,}8$	$0{,}2$
$(x_i - \overline{x})^2$	$46{,}24$	$3{,}24$	$1{,}44$	$125{,}44$	$4{,}84$	$7{,}84$	$10{,}24$	$3{,}24$	$23{,}04$	$0{,}04$

The variance is the sum of the last row in this table divided by 10, so $\sigma^2 = 22{,}56$.

Step 3: Calculate the variance using a calculator

Using the SHARP EL-531VH calculator:

Using your calculator, change the mode from normal to "Stat x ". Do this by pressing [2ndF] and then 1. This mode enables you to type in univariate data.

Key in the data, row by row:

Enter:	**Press:**	**See:**
44	**DATA**	n = 1
49	**DATA**	n = 2
52	**DATA**	n = 3
62	**DATA**	n = 4
53	**DATA**	n = 5
48	**DATA**	n = 6
54	**DATA**	n = 7
49	**DATA**	n = 8
46	**DATA**	n = 9
51	**DATA**	n = 10

Note: The [**DATA**] button is the same as the [M+] button.

Get the value for σ_x:

Press:	**Press:**	**See:**
RCL	σx	$\sigma x = \pm 4{,}75$

$\therefore \sigma_x = \pm 4{,}75$ and ${\sigma_x}^2 = (4{,}75)^2 = 22{,}56$

Using the CASIO fx-82ZA PLUS calculator:

Switch on the calculator. Press [MODE] and then select STAT by pressing [2]. The following screen will appear:

1:	$1-VAR$	2:	$A+BX$
3:	$_{+}CX^{2}$	4:	lnX
5:	eX	6:	$A.BX$
7:	$A.XB$	8:	$1/X$

Now press [1] for variance and standard deviation. Your screen should look something like this:

	X
1	
2	
3	

Press [44] and then [=] to enter the first x-value under x. Then enter the other values in the same way.

	X
1	44
2	49
3	52

Then press [AC]. The screen clears but the data remains stored.

Now press [SHIFT][1] to get the stats computations screen shown below.

1:	Type	2:	Data
3:	Sum	4:	Var
5:	MinMax		

Choose variance by pressing [4].

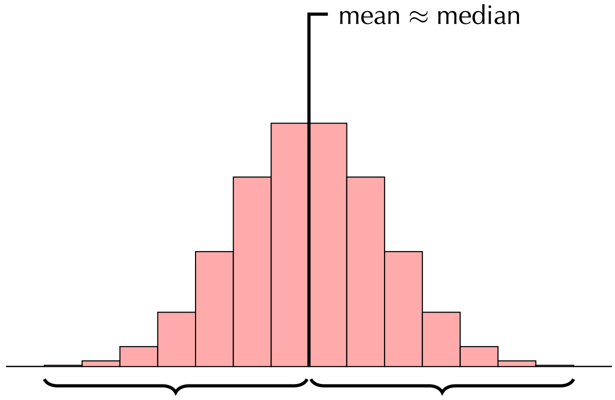
1:	n	2:	$\bar{x}$
3:	σx	4:	sx

Get the value for σ_x:

Press [3] and [=] to get the value of σx $\therefore$ $\sigma_x = \pm 4{,}75$ and ${\sigma_x}^2 = (4{,}75)^2 = 22{,}56$

Last year you learnt about three shapes of data distribution: symmetric, left skewed and right skewed.

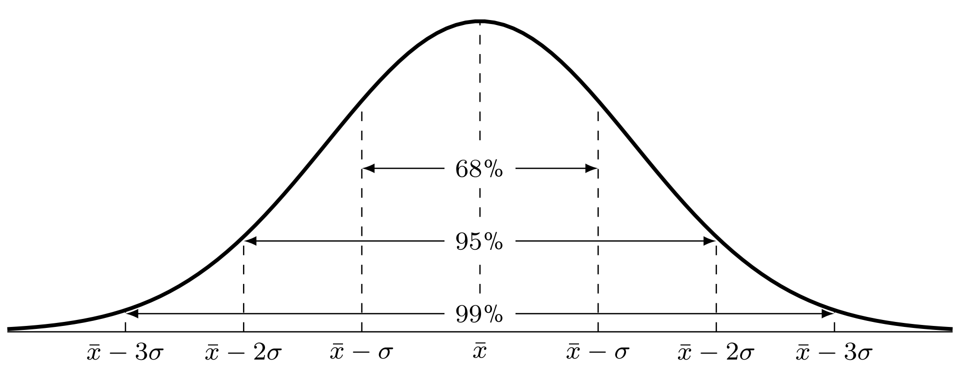
A *symmetric* distribution is one where the left and right hand sides of the distribution are roughly equally balanced around the mean. The histogram below shows a typical symmetric distribution.



For symmetric distributions, the mean is approximately equal to the median and the left and right tails are equally balanced, meaning that they have about the same length.

If large numbers of data are collected from a population, the graph will often have a bell shape. If the data was, say, examination results, a few learners usually get very high marks, a few very low marks and most get a mark in the middle range. This is a common type of symmetric data known as a *normal* distribution. We say a distribution is normal if

- the mean, median and mode are equal.
- it is symmetric around the mean.
- 68% of the sample lies within one standard deviation of the mean, 95% within two standard deviations and 99% within three standard deviations of the mean.



See video: 29BV at www.everythingmaths.co.za

What happens if the test was very easy or very difficult? Then the distribution may not be symmetrical. If extremely high or extremely low scores are added to a distribution, then the mean and median tend to shift towards these scores and the curve becomes skewed.

If the test was very difficult, the mean and median scores are shifted to the left. In this case, we say the distribution is *positively skewed*, or *skewed right*.

A distribution that is skewed right has the following characteristics:

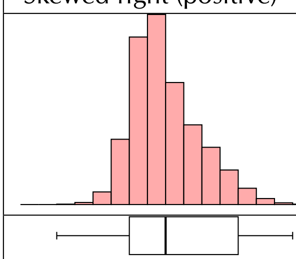
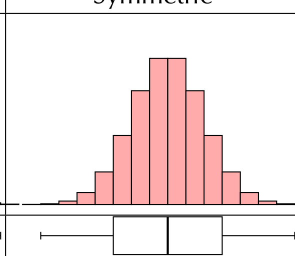
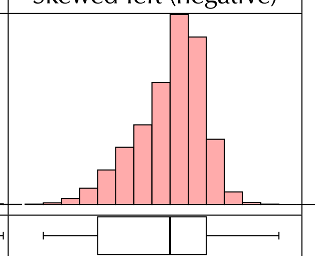
- the mean is typically more than the median;
- the tail of the distribution is longer on the right hand side than on the left hand side; and
- the median is closer to the first quartile than to the third quartile.

If the test was very easy, then many learners would get high scores, and the mean and median of the distribution would be shifted to the right. We say the distribution is *negatively skewed*, or *skewed left*.

A distribution that is skewed right has the following characteristics:

- the mean is typically less than the median;
- the tail of the distribution is longer on the left hand side than on the right hand side; and
- the median is closer to the third quartile than to the first quartile.

The table below summarises the different categories visually.

Skewed right (positive)	Symmetric	Skewed left (negative)
		
mean $>$ median	mean $\approx$ median	mean $<$ median

Worked example 3: Skewed and symmetric data

QUESTION

Three Matric classes wrote a Mathematics test. The test is out of 40 marks and each class has 21 learners. The results of the test are shown in the table on the next page.

Gr. 12A	Gr. 12B	Gr. 12C
4	4	4
8	12	6
8	16	6
12	16	10
12	16	14
12	16	14
12	16	16
16	20	16
16	20	18
16	28	18
16	32	21
32	32	24
32	32	24
32	32	24
36	36	28
36	36	30
36	36	32
40	36	36
40	40	36
40	40	36
40	40	40

1. For each class, determine the five number summary and draw a box and whisker diagram on the same set of axes using an appropriate scale.
2. Determine the mean and standard deviation for each class.
3. Comparing the mean and median values for each class, comment on the distribution of the test marks for each class.

SOLUTION

1. First, we order the data from smallest to largest. This has already been done for us. Then, we divide our data into quartiles:

Quartiles: 0 | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$ | 1

Gr. 12A	4	8	8	12	12	12	12	16	16	16	16	32	32	32	36	36	36	40	40	40	40
Gr. 12B	4	12	16	16	16	16	16	20	20	28	32	32	32	32	36	36	36	36	40	40	40
Gr. 12C	4	6	6	10	14	14	16	16	18	18	21	24	24	24	28	30	32	36	36	36	40

The minimum of each data set is 4. The maximum of each data set is 40.

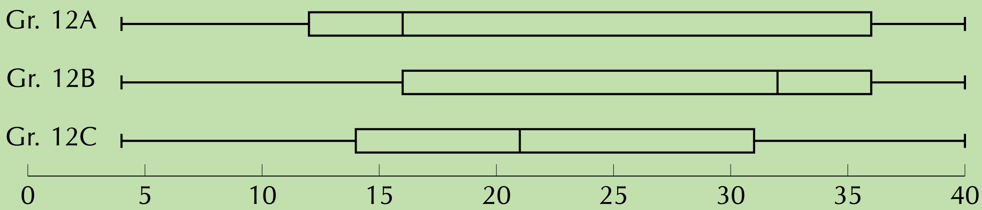
Since there are 21 values in the data set, the median lies on the eleventh mark, making it equal to 16 for Gr. 12A, 32 for Gr. 12B and 21 for Gr. 12C.

The first quartile lies between the fifth and sixth values, making it equal to 12 for Gr. 12A, 16 for Gr. 12B and 14 for Gr. 12C.

The third quartile lies between the 16^{th} and 17^{th} values, making it equal to 36 for Gr. 12A and Gr. 12B, and $\frac{30+32}{2} = 31$ for Gr. 12C.

Therefore, we are able to formulate the following five number summaries and subsequent box and whisker plots:

- Gr. 12A $= [4; 12; 16; 36; 40]$
- Gr. 12B $= [4; 16; 32; 36; 40]$
- Gr. 12C $= [4; 14; 21; 31; 40]$



2. Gr. 12A:

$$\text{mean } (\bar{x}) = \frac{496}{21} = 23{,}6$$

$$\text{standard deviation } (\sigma) = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \pm 12{,}70$$

Gr. 12B:

$$\text{mean } (\bar{x}) = \frac{556}{21} = 26{,}5$$

$$\text{standard deviation } (\sigma) = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \pm 10{,}65$$

Gr. 12C:

$$\text{mean } (\bar{x}) = \frac{453}{21} = 21{,}6$$

$$\text{standard deviation } (\sigma) = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \pm 10{,}54$$

3. If the mean is greater than the median, the data is typically positively skewed and if the mean is less than the median, the data is typically negatively skewed.

Gr. 12A: mean $-$ median $= 23{,}6 - 16 = 7{,}6$. The marks for 12A are therefore positively skewed, meaning that there were many low marks in the class with the high marks being more spread out.

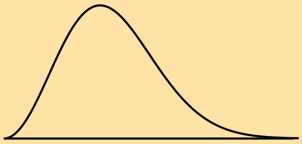
Gr. 12B: mean $-$ median $= 26{,}5 - 32 = -5{,}5$. The marks for 12B are therefore negatively skewed, meaning that there were many high marks in the class with the low marks being more spread out.

Gr. 12C: mean $-$ median $= 21{,}6 - 21 = 0{,}6$. The marks for 12C are therefore normally distributed, meaning that there are as many low marks in the class as there are high marks.

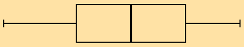
Exercise 9 – 1: Revision

1. State whether each of the following data sets are symmetric, skewed right or skewed left.

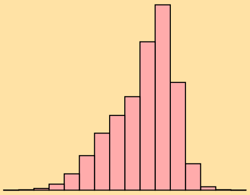
   a) A data set with this distribution:

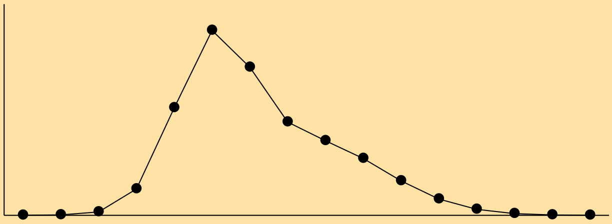
   b) A data set with this box and whisker plot:

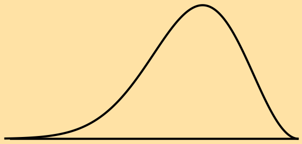
   c) A data set with this histogram:

   d) A data set with this frequency polygon:

   e) A data set with this distribution:

   f) The following data set:
   $105 ; 44 ; 94 ; 149 ; 83 ; 178 ; -4 ; 112 ; 50 ; 188$

2. For the following data sets:

   - Determine the mean and five number summary.
   - Draw the box and whisker plot.
   - Determine the skewness of the data.

   a) $40 ; 45 ; 12 ; 6 ; 9 ; 16 ; 11 ; 7 ; 35 ; 7 ; 31 ; 3$

   b) $65 ; 100 ; 99 ; 21 ; 8 ; 27 ; 21 ; 31 ; 33 ; 31 ; 38 ; 16$

   c) $65 ; 57 ; 77 ; 92 ; 77 ; 58 ; 90 ; 46 ; 11 ; 81$

   d) $1 ; 99 ; 76 ; 76 ; 50 ; 74 ; 83 ; 91 ; 41 ; 17 ; 33$

   e) $0{,}5; -0{,}9 ; -1{,}8 ; 3 ; -0{,}2 ; -5{,}2 ; -1{,}8 ; 0{,}1 ; -1{,}7 ; -2 ; 2{,}2 ; 0{,}5; -0{,}5$

   f) $86 ; 64 ; 25 ; 71 ; 54 ; 44 ; 97 ; 31 ; 78 ; 46 ; 60 ; 86$

3. For the following data sets:
    - Determine the mean.
    - Use a table to determine the variance and the standard deviation.
    - Determine the percentage of data points within one standard deviation of the mean. Round your answer to the nearest percentage point.

    a) $\{9{,}1; 0{,}2; 2{,}8; 2{,}0; 10{,}0; 5{,}8; 9{,}3; 8{,}0\}$
    b) {9; 5; 1; 3; 3; 5; 7; 4; 10; 8}
    c) {81; 22; 63; 12; 100; 28; 54; 26; 50; 44; 4; 32}

4. Use a calculator to determine the
    - mean,
    - variance,
    - and standard deviation

    of the following data sets:

    a) $8 ; 3 ; 10 ; 7 ; 7 ; 1 ; 3 ; 1 ; 3 ; 7$
    b) $4 ; 4 ; 13 ; 9 ; 7 ; 7 ; 2 ; 5 ; 15 ; 4 ; 22 ; 11$
    c) $4{,}38 ; 3{,}83 ; 4{,}99 ; 4{,}05 ; 2{,}88 ; 4{,}83 ; 0{,}88 ; 5{,}33 ; 3{,}49 ; 4{,}10$
    d) $4{,}76 ; -4{,}96 ; -6{,}35 ; -3{,}57 ; 0{,}59 ; -2{,}18 ; -4{,}96 ; -3{,}57 ; -2{,}18 ; 1{,}98$
    e) $7 ; 53 ; 29 ; 42 ; 12 ; 111 ; 122 ; 79 ; 83 ; 5 ; 69 ; 45 ; 23 ; 77$

5. Xolani surveyed the price of a loaf of white bread at two different supermarkets. The data, in rands, are given below.

Supermarket A	3,96	3,76	4,00	3,91	3,69	3,72
Supermarket B	3,97	3,81	3,52	4,08	3,88	3,68

    a) Find the mean price at each supermarket and then state which supermarket has the lower mean.
    b) Find the standard deviation of each supermarket's prices.
    c) Which supermarket has the more consistently priced white bread? Give reasons for your answer.

6. The times for the 8 athletes who swam the 100 m freestyle final at the 2012 London Olympic Games are shown below. All times are in seconds.
    $47{,}52 ; 47{,}53 ; 47{,}80 ; 47{,}84 ; 47{,}88 ; 47{,}92 ; 48{,}04 ; 48{,}44$

    a) Calculate the mean time.
    b) Calculate the standard deviation for the data.
    c) How many of the athletes' times are more than one standard deviation away from the mean?

7. The following data set has a mean of $14{,}7$ and a variance of $10{,}01$.

$$18 \; ; \; 11 \; ; \; 12 \; ; \; a \; ; \; 16 \; ; \; 11 \; ; \; 19 \; ; \; 14 \; ; \; b \; ; \; 13$$

    Calculate the values of a and b.

8. The height of each learner in a class was measured and it was found that the mean height of the class was $1{,}6$ m. At the time, three learners were absent. However, when the heights of the learners who were absent were included in the data for the class, the mean height did not change.

   If the heights of two of the learners who were absent are $1{,}45$ m and $1{,}63$ m, calculate the height of the third learner who was absent. [NSC Paper 3 Feb-March 2013]

9. There are 184 students taking Mathematics in a first-year university class. The marks, out of 100, in the half-yearly examination are normally distributed with a mean of 72 and a standard deviation of 9. [NSC Paper 3 Feb-March 2013]

   a) What percentage of students scored between 72 and 90 marks?
   b) Approximately how many students scored between 45 and 63 marks?

10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 29BW	1b. 29BX	1c. 29BY	1d. 29BZ	1e. 29C2	1f. 29C3
2a. 29C4	2b. 29C5	2c. 29C6	2d. 29C7	2e. 29C8	2f. 29C9
3a. 29CB	3b. 29CC	3c. 29CD	4a. 29CF	4b. 29CG	4c. 29CH
4d. 29CJ	4e. 29CK	5. 29CM	6. 29CN	7. 29CP	8. 29CQ
9. 29CR					

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9.2 Curve fitting EMCJP

Intuitive curve fitting EMCJQ

In Grade 11, we used various means, such as histograms, frequency polygons and ogives, to visualise our data. These are very useful tools to depict **univariate** data, i.e. data with only one variable such as the height of learners in a class.

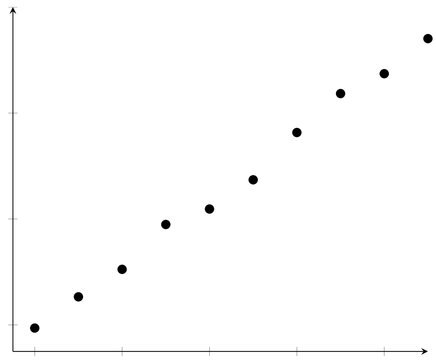
Last year we also learnt about a visual tool called scatter plots. **Scatter plots** are a common way to visualise **bivariate** data, i.e. data with two variables. This allows us to identify the *direction* and *strength* of a relationship between two variables.

We identify the nature of a relationship between two variables by examining if the points on the scatter plot conform to a linear, exponential, quadratic or some other function. The process of fitting functions to data is known as **curve fitting**.

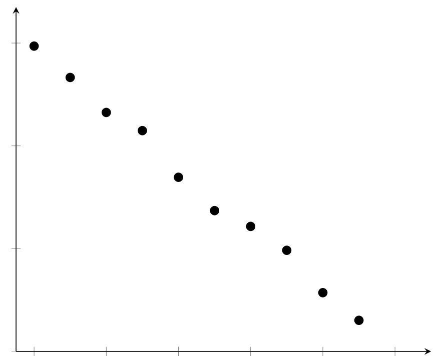
The strength of a relationship can be described as *strong* if the data points conform closely to a function or *weak* if they are further away.

In the case of linear functions, the direction of a relationship is *positive* if high values of one variable occur with high values of the other or *negative* if high values of one variable occur with low values of the other.

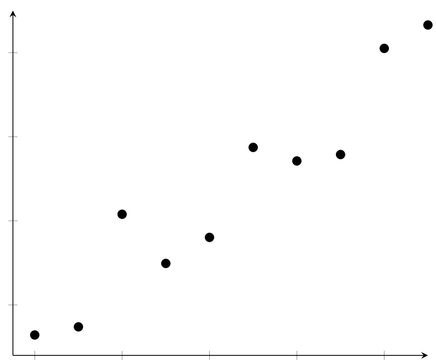
The different relationships are summarised below:



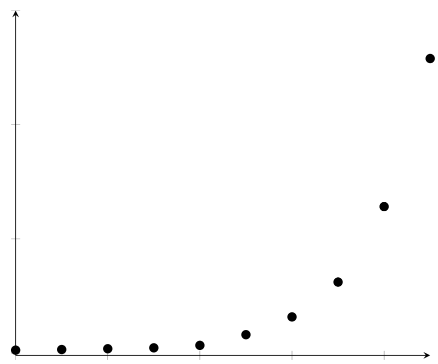
Strong, positive linear relationship



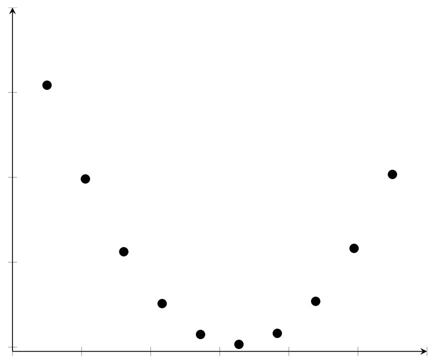
Strong, negative linear relationship



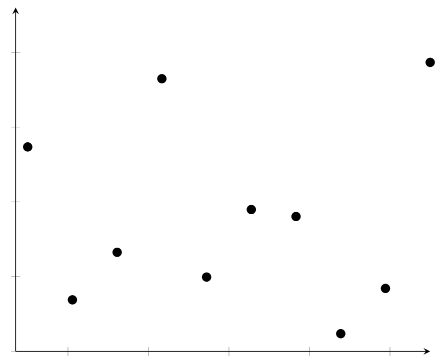
Weak, positive linear relationship



Exponential relationship



Quadratic relationship

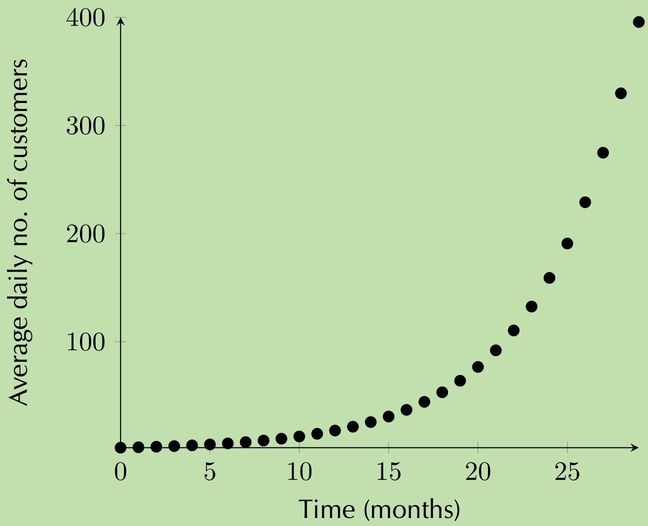


No relationship

Worked example 4: Intuitive curve fitting

QUESTION

Examine the scatter plot below of data collected from a new shop:



- What are the two variables being compared?
- What type of function best fits the data?
- Is the relationship between the two variables strong or weak?
- Is the relationship between the two variables positive or negative?
- Using your answers above, describe the relationship between the two variables in one sentence.

SOLUTION

- The variables being compared are average daily number of customers and time in months.
- The data fit an exponential function.
- The data points appear to fit the curve close to perfectly, so the relationship can be described as very strong.
- As time increases, the number of customers increases, so the relationship can be described as positive.
- There is a very strong, positive, exponential relationship between average daily customers and time in the new shop.

In the worked example above, by plotting the average daily customers and time data of a new shop on a scatter plot, we were able to identify the relationship between the two variables. Once we know the relationship between two variables, we are able to do another very useful thing we can predict values where no data exist.

DEFINITION: *Interpolation and extrapolation*

When we predict values that fall within the range of our data, this is known as **interpolation**. When we predict the values of a variable beyond the range of our data, this is known as **extrapolation**.

Extrapolation must be done with caution unless it is known that the observed relationship continues beyond the range of our data. For example, an exponential function may look linear if we only have the first few data points available but if we extrapolate far enough beyond the initial data points, our predictions will be inaccurate.

In order to interpolate or extrapolate values, we need to find the equation of the function which best fits the data. For linear data, we draw a straight line through the data which best approximates the available data points. This line is known as the **line of best fit** or trend line. Let us try our hand at this in the following example.

Worked example 5: Fitting by hand

QUESTION

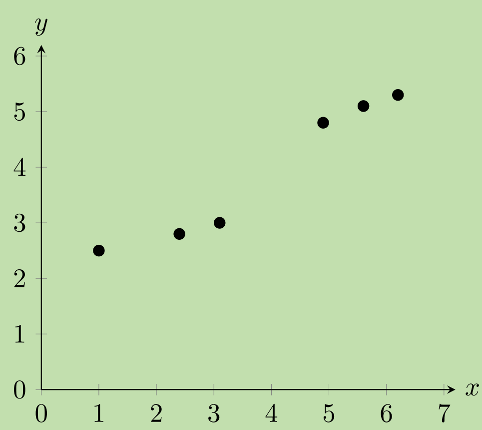
- Use the data below to draw a scatter plot and line of best fit.
- Write down the equation of the line that best seems to fit the data.
- Use your equation to calculate the estimated value for y if $x = 4$.
- Use your equation to calculate the estimated value for x if $y = 6$.

x	1,0	2,4	3,1	4,9	5,6	6,2
y	2,5	2,8	3,0	4,8	5,1	5,3

SOLUTION

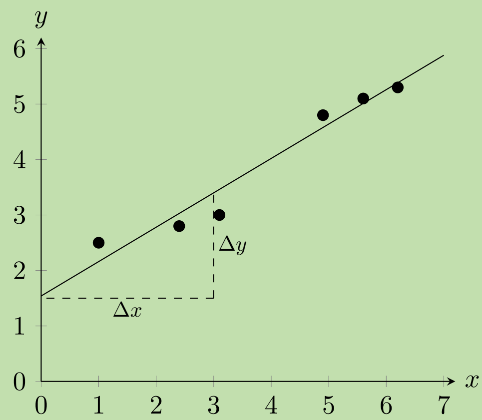
Step 1: Draw the graph

1. Choose a suitable scale for the axes.
2. Draw the axes.
3. Plot the points.



Step 2: Drawing the line of best fit

The next step is to draw a straight line which goes as close to as many points as possible. It is generally best to have as many points above the line as below the line.



Step 3: Calculating the equation of the line

The equation of the line is $y = mx + c$

From the graph we have drawn, we estimate the y-intercept to be $1{,}5$. We estimate that $y = 3{,}5$ when $x = 3$. So we have that points $(3; 3{,}5)$ and $(0; 1{,}5)$ lie on the line. The gradient of the line, m, is given by

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{3{,}5 - 1{,}5}{3 - 0} \\ &= \frac{2}{3} \end{aligned}$$

So we finally have that the equation of the line of best fit is $y = \frac{2}{3}x + 1{,}5$

Step 4: Calculate the unknown values

The equation of the line is $y = \frac{2}{3}x + 1{,}5$ so in order to find the unknown values, we insert the known values into our equation.

For $x = 4$:

$$\begin{aligned} y &= \frac{2}{3} \cdot 4 + 1{,}5 \\ &= 4{,}17 \end{aligned}$$

Since this x-value is within the data range, this is **interpolation**.

For $y = 6$:

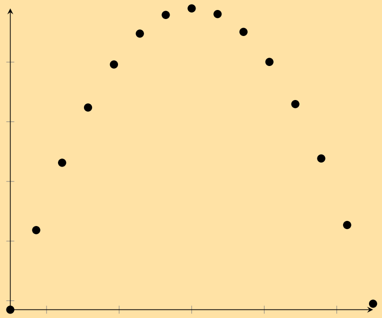
$$\begin{aligned} 6 &= \frac{2}{3} \cdot x + 1{,}5 \\ \therefore x &= (6 - 1{,}5) \times \frac{3}{2} \\ &= 6{,}75 \end{aligned}$$

Since this y-value is outside the data range, this is **extrapolation**.

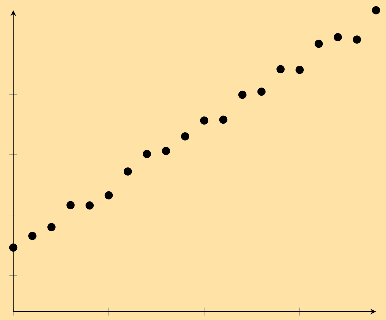
Exercise 9 – 2: Intuitive curve fitting

1. Identify the function (linear, exponential or quadratic) which would best fit the data in each of the scatter plots below:

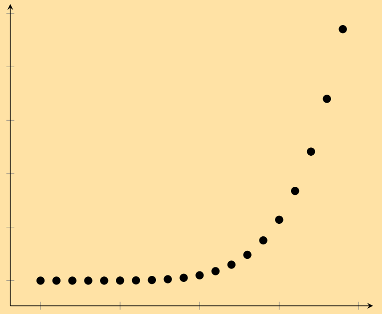
a)



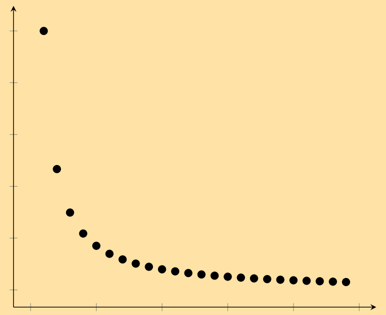
d)



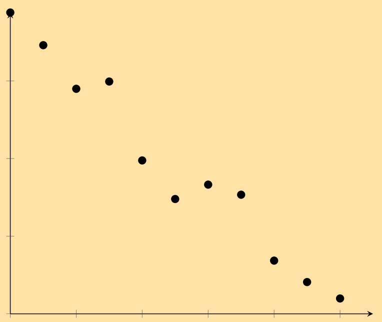
b)



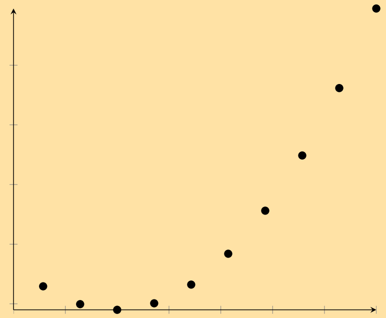
e)



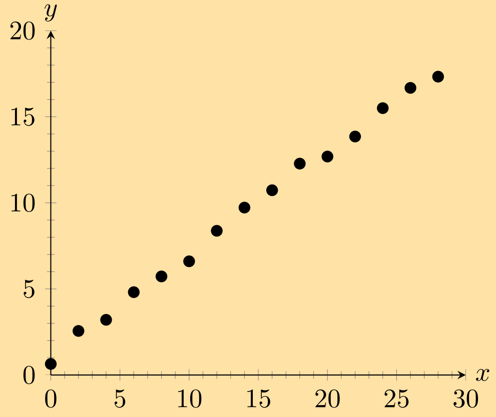
c)



f)



2. Given the scatter plot below, answer the questions that follow on the next page.



a) What type of function fits the data best? Comment on the fit of the function in terms of strength and direction.

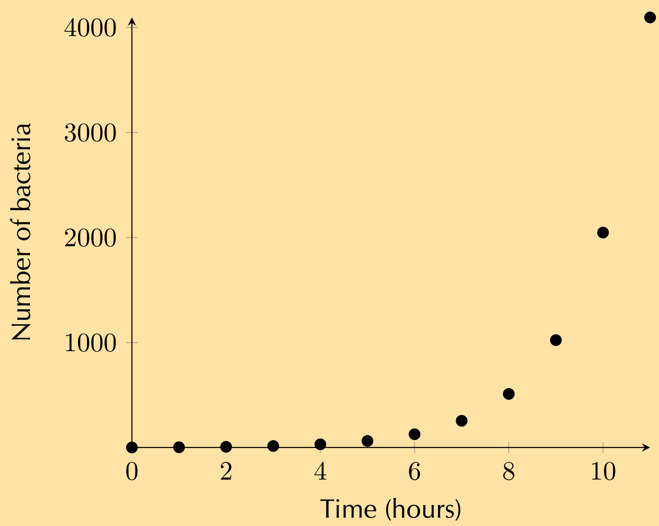
b) Draw a line of best fit through the data and determine the equation for your line.

c) Using your equation, determine the estimated y-value where $x = 25$.

d) Using your equation, determine the estimated x-value where $y = 25$.

3. Tuberculosis (TB) is a disease of the lungs caused by bacteria which are spread through the air when an infected person coughs or sneezes. Drug-resistant TB arises when patients do not take their medication properly. Andile is a scientist studying a new treatment for drug-resistant TB.

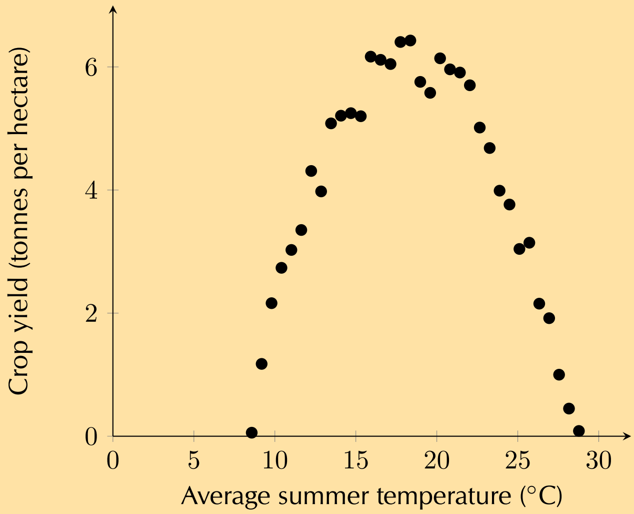
For his research, he needs to grow the TB bacterium. He takes two bacteria and puts them on a plate with nutrients for their growth. He monitors how the number of bacteria increases over time. Look at his data in the scatter plot below and answer the questions that follow.



a) What type of function do you think fits the data best?

b) The equation for bacterial growth is $x_n = x_0(1 + r)^t$ where x_0 is the initial number of bacteria, r is the growth rate per unit time as a proportion of 1, t is time in hours, and x_n is the number of bacteria at time, t. Determine the number of bacteria grown by Andile after 24 hours if the number of bacteria doubles every hour (i.e. the growth rate is 100% per hour).

4. Marelize is a researcher at the Department of Agriculture. She has noticed that farmers across the country have very different crop yields depending on the region. She thinks that this has to do with the different climate in each region. In order to test her idea, she collected data on crop yield and average summer temperatures from a number of farmers. Examine her data on the opposite page and answer the questions that follow.



a) Identify what type of function would fit the data best.

b) Marelize determines that the equation for the function which fits the data best is $y = -0{,}06x^2 + 2{,}2x - 14$. Determine the optimal temperature to grow wheat and the respective crop yield. Round your answer to two decimal places.

5. Dr Dandara is a scientist trying to find a cure for a disease which has an 80% mortality rate, i.e. 80% of people who get the disease will die. He knows of a plant which is used in traditional medicine to treat the disease. He extracts the active ingredient from the plant and tests different dosages (measured in milligrams) on different groups of patients. Examine his data below and complete the questions that follow.

Dosage (mg)	0	25	50	75	100	125	150	175	200
Mortality rate (%)	80	73	63	49	42	32	25	11	5

a) Draw a scatter plot of the data

b) Which function would best fit the data? Describe the fit in terms of strength and direction.

c) Draw a line of best fit through the data and determine the equation of your line.

d) Use your equation to estimate the dosage required for a 0% mortality rate.

e) Dr Dandara decided to administer the estimated dosage required for a 0% mortality rate to a group of infected patients. However, he still found a mortality rate of 5%. Name the statistical technique Dr Dandara used to estimate a mortality rate of 0% and explain why his equation did not accurately predict his experimental results.

6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 29CS 1b. 29CT 1c. 29CV 1d. 29CW 1e. 29CX 1f. 29CY
2. 29CZ 3. 29D2 4. 29D3 5. 29D4

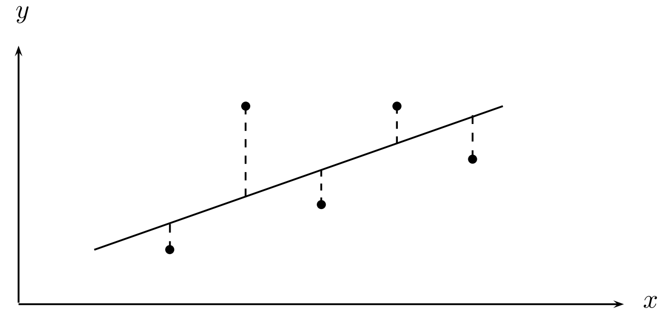
www.everythingmaths.co.za m.everythingmaths.co.za

In the previous worked example and exercises, you drew the line of best fit by hand. This can give us a reasonable approximation of which function best fits the data when the data points are close together. However, you and your classmates may have found that you obtained slightly different answers from one another. In the next section, we will learn about a more precise way of fitting a linear function to data.

Linear regression EMCJR

Linear regression analysis is a statistical technique for finding out exactly which linear function best fits a given set of data. We can find out the equation of the regression line by using an algebraic method called the **least squares method**, available on most scientific calculators. The linear regression equation is written $\hat{y} = a + bx$ (we say y-hat) or $y = A + Bx$. Of course these are both variations of the more familiar equation $y = mx + c$.

The least squares method is very simple. Suppose we guess a line of best fit, then at every data point, we find the distance between the data point and the line. If the line fitted the data perfectly, this distance would be zero for all the data points. The worse the fit, the larger the differences. We then square each of these distances, and add them all together.



The best-fit line is then the line that minimises the sum of the squared distances.

See video: 29D5 at www.everythingmaths.co.za

Suppose we have a data set of n points $\{(x_1; y_1), (x_2; y_2), ..., (x_n; y_n)\}$. We also have a line $f(x) = mx + c$ that we are trying to fit to the data. The distance between the first data point and the line, for example, is distance $= y_1 - f(x_1) = y_1 - (mx_1 + c)$

We now square each of these distances and add them together. Lets call this sum $S(m, c)$. Then we have that

$$
\begin{aligned}
S(m,c) &= (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + \ldots + (y_n - f(x_n))^2 \\
&= \sum_{i=1}^{n} (y_i - f(x_i))^2
\end{aligned}
$$

Thus our problem is to find the value of m and c such that $S(m, c)$ is minimised. Let us call these minimising values b and a respectively. Then the line of best-fit is $f(x) = a + bx$. We can find a and b using calculus, but it is tricky, and we will just give you the result, which is that

$$
\begin{aligned}
b &= \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} (x_i)^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \\
a &= \frac{1}{n}\sum_{i=1}^{n} y_i - \frac{b}{n}\sum_{i=1}^{n} x_i = \bar{y} - b\bar{x}
\end{aligned}
$$

Worked example 6: Method of least squares by hand

QUESTION

In the table below, we have the records of the maintenance costs in rands compared with the age of the appliance in months. We have data for five appliances. Determine the equation for the least squares regression line by hand.

Appliance	1	2	3	4	5
Age (x)	5	10	15	20	30
Cost (y)	90	140	250	300	380

SOLUTION

Appliance	x	y	xy	x^2
1	5	90	450	25
2	10	140	1400	100
3	15	250	3750	225
4	20	300	6000	400
5	30	380	11 400	900
Total	80	1160	23 000	1650

$$
\begin{aligned}
b &= \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2} = \frac{5 \times 23000 - 80 \times 1160}{5 \times 1650 - 80^2} \\
&= 12 \\
a &= \bar{y} - b\bar{x} = \frac{1160}{5} - \frac{12 \times 80}{5} \\
&= 40 \\
\therefore \hat{y} &= 40 + 12x
\end{aligned}
$$

Worked example 7: Using the SHARP EL-531VH calculator

QUESTION

Using a calculator, find the equation of the least squares regression line for the following data:

Days (x)	1	2	3	4	5
Growth in m (y)	1,00	2,50	2,75	3,00	3,50

NB. If you have a CASIO calculator, do the next worked example first. Come back to this worked example once you are done and see if you get the same answer on your calculator.

SOLUTION

Step 1: Getting your calculator ready

Using your calculator, change the mode from normal to "Stat xy". Do this by pressing [2ndF] and then 2. This mode enables you to type in bivariate data.

Step 2: Entering the data

Key in the data row by row:

Enter:	Press:	Enter:	Press:	See:
1	(x, y)	1	**DATA**	n = 1
2	(x, y)	2,5	**DATA**	n = 2
3	(x, y)	2,75	**DATA**	n = 3
4	(x, y)	3,0	**DATA**	n = 4
5	(x, y)	3,5	**DATA**	n = 5

Note: The [(x, y)] button is the same as the [**STO**] button and the [**DATA**] button is the same as the [M+] button.

Step 3: Getting regression results from the calculator

Ask for the values of the regression coefficients a and b.

Press:	Press:	See:
RCL	a	$a = 0{,}9$
RCL	b	$b = 0{,}55$

$\therefore \hat{y} = 0{,}9 + 0{,}55x$

Worked example 8: Using the CASIO fx-82ZA PLUS calculator

QUESTION

Using a calculator determine the least squares line of best fit for the following data set.

Learner	1	2	3	4	5
Chemistry (%)	52	55	86	71	45
Accounting (%)	48	64	95	79	50

For a Chemistry mark of 65%, what mark does the least squares line predict for Accounting?

NB. If you have a SHARP calculator, ensure that you have done the previous worked example first. Once you have completed the previous worked example, attempt this example using your calculator and see if you get the same answer.

SOLUTION

Step 1: Getting your calculator ready

Switch on the calculator. Press [MODE] and then select STAT by pressing [2]. The following screen will appear:

1:	$1 - VAR$	2:	$A + BX$
3:	$_+CX^2$	4:	lnX
5:	eX	6:	$A.BX$
7:	$A.XB$	8:	$1/X$

Now press [2] for linear regression. Your screen should look something like this:

	X	Y
1		
2		
3		

Step 2: Entering the data

Press [52] and then [=] to enter the first mark under x. Then enter the other values, in the same way, for the x-variable (the Chemistry marks) in the order in which they are given in the data set. Then move the cursor across and up and enter 48 under y opposite 52 in the x-column. Continue to enter the other y-values (the Accounting marks) in order so that they pair off correctly with the corresponding x-values.

	X	Y
1	52	48
2	55	64
3	86	95

Then press [AC]. The screen clears but the data remains stored.

Now press [SHIFT][1] to get the stats computations screen shown below.

1:	Type	2:	Data
3:	Sum	4:	Var
5:	Reg	6:	MinMax

Choose Regression by pressing [5].

1:	A	2:	B
3:	r	4:	$\hat{x}$
5:	$\hat{y}$		

Step 3: Getting regression results from the calculator

1. Press [1] and [=] to get the value of the y-intercept, $a = -5{,}065\ldots = -5{,}07$ (to two decimal places)

   Finally, to get the slope, use the following key sequence: [SHIFT][1][5][2][=]. The calculator gives $b = 1{,}169\ldots = 1{,}17$ (to two decimal places)

   The equation of the line of regression is thus:

   $\hat{y} = -5{,}07 + 1{,}17x$

2. Press [AC][65][SHIFT][1][5][5][=]

   This gives a (predicted) Accounting mark of $= 70{,}94 = 71\%$

Exercise 9 – 3: Least squares regression analysis

1. Determine the equation of the least-squares regression line using a table for the data sets below. Round a and b to two decimal places.

a)

x	10	4	9	11	11	6	8	18	9	13
y	1	0	6	3	9	5	9	8	7	15

b)

x	8	12	12	7	6	14	8	14	14	17
y	-5	4	3	-3	-5	-6	-2	0	-4	3

c)

x	-9	3	4	7	13	6	0	8	1	14
y	0	-12	-10	-14	-31	-32	-41	-52	-51	-63

2. Use your calculator to determine the equation of the least squares regression line for the following sets of data:

a)

x	0,16	0,32	3	2,6	6,12	7,68	6,16	8,56	11,24	11,96
y	5,48	10,56	13,4	15,96	15,44	16,6	17,2	22,28	22,04	24,32

b)

x	$-3{,}5$	5,5	4	1	5,5	5	3,5	5,5	7,5	8,5
y	-10	$-20{,}5$	$-30{,}5$	-46	$-46{,}5$	$-64{,}5$	-67	$-76{,}5$	$-83{,}5$	-94

c)

x	2,5	4,5	-2	9	8,5	10	7,5	3	8	15
y	-2	6	11	11,5	17	21	21	30,5	32,5	33,5

d)

x	7,24	8,24	5,34	1,66	0,32	11,46	9,34	14,24	12,9	12,34
y	$-3{,}2$	$-18{,}78$	$-21{,}1$	-32	$-31{,}2$	$-53{,}02$	-53	$-65{,}46$	$-74{,}8$	$-80{,}24$

e)

x	$-0{,}28$	2,32	0,12	4,64	3,08	7,92	5,08	8,96	10,28	7,12
y	$-6{,}88$	$-0{,}32$	3,68	4,8	11,68	19,2	20,96	24,96	29,28	33,28

f)

x	1	1,1	4,8	3,55	2,75	1,95	6,1	8,9	10,35	9,55
y	$-8{,}45$	$-5{,}95$	$-4{,}35$	0,85	$-2{,}95$	$-1{,}8$	0,25	0,05	4,8	$-3{,}05$

g)

x	1,9	1,1	−1,5	1,3	0,95	8,25	10,6	6,2	8,1	8,65
y	7	8,45	0,9	0,1	2,45	4,35	2,2	1,4	0,15	2,05

h)

x	−81,8	73,1	84	92,2	−69,7	−56,1	8,8	80,9	68,4	−40,4
y	10,6	16,1	3,6	4,6	11,9	18,3	16,6	17,6	17,7	24,1

i)

x	2,8	7,4	−2,4	4	11,3	6,9	2,5	1,7	5,4	8,2
y	12,4	13,4	15,3	15,4	16,4	19,2	21,1	19,4	21,3	25

j)

x	5	1,2	8	6	7,4	7,4	6,7	8,7	12,2	14,3
y	−4,2	−13,7	−23,7	−33,5	−43,8	−54,2	−63,9	−73,9	−84,5	−93,5

3. Determine the equation of the least squares regression line given each set of data values below. Round a and b to two decimal places in your final answer.
   a) $n = 10$; $\sum x = 74$; $\sum y = 424$; $\sum xy = 4114{,}51$; $\sum(x^2) = 718{,}86$
   b) $n = 13$; $\bar{x} = 8{,}45$; $\bar{y} = 17{,}83$; $\sum xy = 1879{,}25$; $\sum(x^2) = 855{,}45$
   c) $n = 10$; $\bar{x} = 5{,}77$; $\bar{y} = 17{,}03$; $\overline{xy} = 133{,}817$; $\sigma_x = \pm 3{,}91$
   (Hint: multiply the numerator and denominator of the formula for b by $\frac{1}{n^2}$)
4. The table below shows the average maintenance cost in rands of a certain model of car compared to the age of the car in years.

Age (x)	1	3	5	6	8	9	10
Cost (y)	1000	1500	1600	1800	2000	2400	2600

   a) Draw a scatter plot of the data.
   b) Complete the table below, filling in the totals of each column in the final row:

Age (x)	**Cost** (y)	xy	x^2
1	1000		
3	1500		
5	1600		
6	1800		
8	2000		
9	2400		
10	2600		
$\sum = \ldots$	$\sum = \ldots$	$\sum = \ldots$	$\sum = \ldots$

   c) Use your table to determine the equation of the least squares regression line. Round a and b to two decimal places.
   d) Use your equation to estimate what it would cost to maintain this model of car in its 15$^{\text{th}}$ year.
   e) Use your equation to estimate the age of the car in the year where the maintenance cost totals over R 3000 for the first time.
5. Miss Colly has always maintained that there is a relationship between a learner's ability to understand the language of instruction and their marks in Mathematics. Since she teaches Mathematics through the medium of English, she decides to compare the Mathematics and English marks of her learners in order to investigate the relationship between the two marks. Examine her data below and answer the quesitons on the following page.

English % (x)	28	33	30	45	45	55	55	65	70	76	65	85	90
Mathematics % (y)	35	36	34	45	50	40	60	50	65	85	70	80	90

a) Complete the table below, filling in the totals of each column in the final row:

English % (x)	Mathematics % (y)	xy	x^2
28	35		
33	36		
30	34		
45	45		
45	50		
55	40		
65	50		
70	65		
76	85		
65	70		
85	80		
90	90		
$\sum = \ldots$	$\sum = \ldots$	$\sum = \ldots$	$\sum = \ldots$

b) Use your table to determine the equation of the least squares regression line. Round a and b to two decimal places.

c) Use your equation to estimate the Mathematics mark of a learner who obtained 50% for English, correct to two decimal places.

d) Use your equation to estimate the English mark of a learner who obtained 75% for Mathematics, correct to two decimal places.

6. Foot lengths and heights of ten students are given in the table below.

Height (cm)	170	163	131	181	146	134	166	172	185	153
Foot length (cm)	27	23	20	28	22	20	24	26	29	22

a) Using foot length as your x-variable, draw a scatter plot of the data.

b) Identify and describe any trends shown in the scatter plot.

c) Find the equation of the least squares line using the formulae and draw the line on your graph. Round a and b to two decimal places in your final answer.

d) Confirm your calculations above by finding the least squares regression line using a calculator.

e) Use your equation to predict the height of a student with a foot length of $21{,}6$ cm.

f) Use your equation to predict the foot length of a student 190 cm tall, correct to two decimal places.

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 29D6 1b. 29D7 1c. 29D8 2a. 29D9 2b. 29DB 2c. 29DC
2d. 29DD 2e. 29DF 2f. 29DG 2g. 29DH 2h. 29DJ 2i. 29DK
2j. 29DM 3a. 29DN 3b. 29DP 3c. 29DQ 4. 29DR 5. 29DS
6. 29DT

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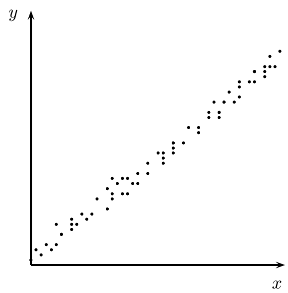
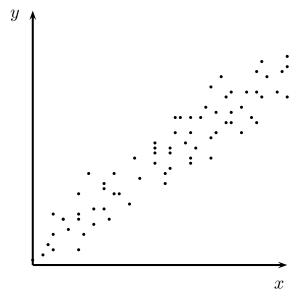
Now that we have a precise technique for finding the line of best fit, we still do not know how well our line of best fit really fits our data. We can fit a least squares regression line to any bivariate data, even if the two variables do not show a linear relationship. If the fit is not "good", our assumption of the a and b values in $\hat{y} = a + bx$ might be incorrect. Next, we will learn of a quantitative measure to determine how well our line really fits our data.

9.3 Correlation EMCJS

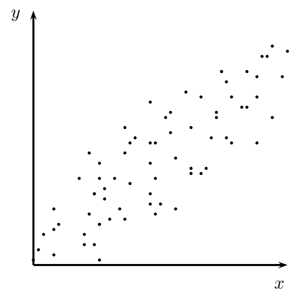
The linear correlation coefficient, r, is a measure which tells us the strength and direction of a relationship between two variables. The correlation coefficient $r \in [-1; 1]$. When $r = -1$, there is perfect negative correlation, when $r = 0$, there is no correlation and when $r = 1$ there is perfect positive correlation.



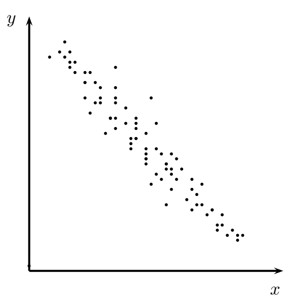
Positive, strong
$r \approx 0{,}9$



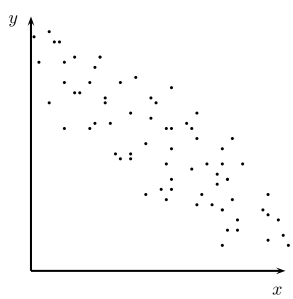
Positive, fairly strong
$r \approx 0{,}7$



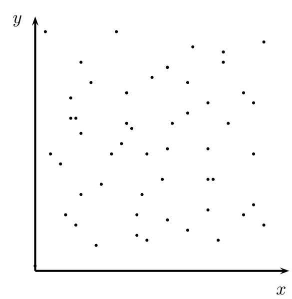
Positive, weak
$r \approx 0{,}4$



Negative, fairly strong
$r \approx -0{,}7$



Negative, weak
$r \approx -0{,}4$



No correlation
$r = 0$

The linear correlation coefficient r can be calculated using the formula $r = b\dfrac{\sigma_x}{\sigma_y}$

- where b is the gradient of the least squares regression line,
- σ_x is the standard deviation of the x-values and
- σ_y is the standard deviation of the y-values.

This is known as the Pearson's product moment correlation coefficient. It is much easier to do on a calculator where you simply follow the procedure to calculate the regression equation, and go on to find r.

In general:

Positive	Strength	Negative
$r = 0$	no correlation	$r = 0$
$0 < r < 0{,}25$	very weak	$-0{,}25 < r < 0$
$0{,}25 < r < 0{,}5$	weak	$-0{,}5 < r < -0{,}25$
$0{,}5 < r < 0{,}75$	moderate	$-0{,}75 < r < -0{,}5$
$0{,}75 < r < 0{,}9$	strong	$-0{,}9 < r < -0{,}75$
$0{,}9 < r < 1$	very strong	$-1 < r < -0.9$
$r = 1$	perfect correlation	$r = -1$

NOTE:
Correlation does not imply causation! Just because two variables are correlated does not mean that they are causally linked, i.e. if A and B are correlated, that does not mean A causes B, or vice versa. This is a common mistake made by many people, especially journalists looking for their next juicy story.

For example, ice cream sales and shark attacks are correlated. This does not mean that the sale of ice cream is somehow causing more shark attacks. Instead, a simpler explanation is that the warmer it is, the more likely people are to buy ice cream and the more likely people are to go to the beach as well, thus increasing the likelihood of a shark attack.

See video: 29DV at www.everythingmaths.co.za

Worked example 9: The correlation coefficient

QUESTION

A cardiologist wanted to test the relationship between resting heart rate and the peak heart rate during exercise. Heart rate is measured in beats per minute (bpm). The following set of data was generated from 12 study participants after they had run on a treadmill at 10 km/h for 10 minutes.

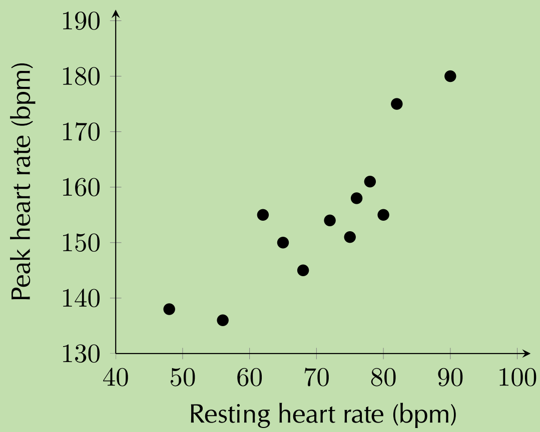
Resting heart rate	48	56	90	65	75	78	80	72	82	76	68	62
Peak heart rate	138	136	180	150	151	161	155	154	175	158	145	155

- Draw a scatter plot of the data. Use resting heart rate as your x-variable.
- Use your calculator to determine the equation of the line of best fit.
- Estimate what the heart rate of a person with a resting heart rate of 70 bpm will be after exercise.
- Without using your calculator, find the correlation coefficient, r. Confirm your answer using your calculator.
- What can you conclude regarding the relationship between resting heart rate and the heart rate after exercise?

SOLUTION

Step 1: Draw the scatter plot

1. Choose a suitable scale for the axes.
2. Draw the axes.
3. Plot the points.



Step 2: Calculate the equation of the line of best fit

As you learnt previously, use your calculator to determine the values for a and b.

$a = 86{,}75$

$b = 0{,}96$

Therefore, the equation for the line of best fit is $y = 86{,}75 + 0{,}96x$

Step 3: Calculate the estimated value for y

If $x = 70$, using our equation, the estimated value for y is:

$$y = 86{,}75 + 0{,}96 \times 70 = 153{,}95$$

Step 4: Calculate the correlation co-efficient

The formula for r is:

$$r = b\frac{\sigma_x}{\sigma_y}$$

We already know the value of b and you know how to calculate b by hand from worked example 5, so we are just left to determine the value for σ_x and σ_y. The formula for standard deviation is:

$$\sigma_x = \frac{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}{n}$$

First, you need to determine $\bar{x}$ and $\bar{y}$ and then complete a table like the one below.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 71$$

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = 154{,}83 \text{ (rounded to two decimal places)}$$

Resting heart rate (x)	Peak heart rate (y)	$(x - \bar{x})^2$	$(y - \bar{y})^2$
48	138	529	283,25
56	136	225	354,57
90	180	361	633,53
65	150	36	23,33
75	151	16	14,67
78	161	49	38,07
80	155	81	0,03
72	154	1	0,69
82	175	121	406,83
76	158	25	10,05
68	145	9s	96,63
62	155	81	0,03
$\sum = 852$	$\sum = 1858$	$\sum = 1534$	$\sum = 1861{,}68$

$$\sigma_x = \frac{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}{n} = \frac{\sqrt{1534}}{12} = \pm 3{,}26$$

$$\sigma_y = \frac{\sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}{n} = \frac{\sqrt{1861{,}68}}{12} = \pm 3{,}60$$

$$b = 0{,}96$$

$$\therefore r = 0{,}96 \times \frac{3{,}26}{3{,}60}$$

$$= 0{,}87$$

Step 5: Confirm your answer using your calculator

Once you know the method for finding the equation of the best line of fit on your calculator, finding the value for r is trivial. After you have entered all your x and y values into your calculator, in STAT mode:

- on a SHARP calculator: press [RCL] then [r] (the same key as [÷])
- on a CASIO calculator: press [SHIFT] then [STAT], [5], [3] then [=]

Step 6: Comment on the correlation coefficient

$$r = 0{,}87$$

Therefore, there is a strong, positive, linear relationship between resting heart rate and peak heart rate during exercise. This means that the higher your resting heart rate, the higher your peak heart rate during exercise is likely to be.

Exercise 9 – 4: Correlation coefficient

1. Determine the correlation coefficient by hand for the following data sets and comment on the strength and direction of the correlation. Round your answers to two decimal places.

a)

x	5	8	13	10	14	15	17	12	18	13
y	5	8	3	8	7	5	3	-1	4	-1

b)

x	7	3	11	7	7	6	9	12	10	15
y	13	23	32	45	50	55	67	69	85	90

c)

x	3	10	7	6	11	16	17	15	17	20
y	6	24	30	38	53	56	65	75	91	103

2. Using your calculator, determine the value of the correlation coefficient to two decimal places for the following data sets and describe the strength and direction of the correlation.

a)

x	0,1	0,8	1,2	3,4	6,5	3,9	6,4	7,4	9,9	8,5
y	$-5{,}1$	-10	$-17{,}3$	$-24{,}9$	$-31{,}9$	$-38{,}6$	-42	-55	-62	$-64{,}8$

b)

x	-26	-34	-51	-14	50	-57	-11	-10	36	-35
y	-66	-10	-26	-51	-58	-56	45	-142	-149	-30

c)

x	101	-398	103	204	105	606	807	-992	609	-790
y	-300	98	-704	-906	-8	690	-12	686	984	-18

d)

x	101	82	-7	-6	45	-94	-23	78	-11	0
y	111	-74	21	106	51	26	21	86	-29	66

e)

x	-3	5	-4	0	-2	9	10	11	17	9
y	24	18	21	30	31	39	48	59	56	54

3. Calculate and describe the direction and strength of r for each of the sets of data values below. Round all r-values to two decimal places.

a) $b = -1{,}88;\ \sigma_x^2 = 48{,}62;\ \sigma_y^2 = 736{,}54.$

b) $a = 32{,}19;\ x = 4{,}3;\ \bar{y} = 36{,}6;\ \sum_{i=1}^{n}(x_i - \bar{x})^2 = 620{,}1;\ \sum_{i=1}^{n}(y_i - \bar{y})^2 = 2636{,}4.$

4. The geography teacher, Mr Chadwick, gave the data set below to his class to illustrate the concept that average temperature depends on how far a place is from the equator (known as the latitude). There are 90 degrees between the equator and the North Pole. The equator is defined as 0 degrees. Examine the data set below and answer the questions that follow.

City	**Degrees N (x)**	**Ave. temp. (y)**	xy	x^2	$(x-\bar{x})^2$	$(y-\bar{y})^2$
Cairo	43	22				
Berlin	53	19				
London	40	18				
Lagos	6	32				
Jerusalem	31	23				
Madrid	40	28				
Brussels	51	18				
Istanbul	39	23				
Boston	43	23				
Montreal	45	22				
Total:						

a) Copy and complete the table.
b) Using your table, determine the equation of the least squares regression line. Round a and b to two decimal places in your final answer.
c) Use your calculator to confirm your equation for the least squares regression line.
d) Using your table, determine the value of the correlation coefficient to two decimal places.
e) What can you deduce about the relationship between how far north a city is and its average temperature?
f) Estimate the latitude of Paris if it has an average temperature of $25°$C

5. A taxi driver recorded the number of kilometres his taxi travelled per trip and his fuel cost per kilometre in Rands. Examine the table of his data below and answer the questions that follow.

Distance (x)	3	5	7	9	11	13	15	17	20	25	30
Cost (y)	$2{,}8$	$2{,}5$	$2{,}46$	$2{,}42$	$2{,}4$	$2{,}36$	$2{,}32$	$2{,}3$	$2{,}25$	$2{,}22$	2

a) Draw a scatter plot of the data.
b) Use your calculator to determine the equation of the least squares regression line and draw this line on your scatter plot. Round a and b to two decimal places in your final answer.
c) Using your calculator, determine the correlation coefficient to two decimal places.
d) Describe the relationship between the distance travelled per trip and the fuel cost per kilometre.
e) Predict the distance travelled if the cost per kilometre is R $1{,}75$.

6. The time taken, in seconds, to complete a task and the number of errors made on the task were recorded for a sample of 10 primary school learners. The data is shown in the table below. [Adapted from NSC Paper 3 Feb-March 2013]

Time taken to complete task (in seconds)	23	21	19	9	15	22	17	14	21	18
Number of errors made	2	4	5	9	7	3	7	8	3	5

a) Draw a scatter plot of the data.
b) What is the influence of more time taken to complete the task on the number of errors made?
c) Determine the equation of the least squares regression line and draw this line on your scatter plot. Round a and b to two decimal places in your final answer.
d) Determine the correlation coefficient to two decimal places.
e) Predict the number of errors that will be made by a learner who takes 13 seconds to complete this task.
f) Comment on the strength of the relationship between the variables.

7. A recording company investigates the relationship between the number of times a CD is played by a national radio station and the national sales of the same CD in the following week. The data below was collected for a random sample of 10 CDs. The sales figures are rounded to the nearest 50. [NSC Paper 3 November 2012]

Number of times CD is played	47	34	40	34	33	50	28	53	25	46
Weekly sales of the CD	3950	2500	3700	2800	2900	3750	2300	4400	2200	3400

a) Draw a scatter plot of the data.
b) Determine the equation of the least squares regression line.
c) Calculate the correlation coefficient.
d) Predict, correct to the nearest 50, the weekly sales for a CD that was played 45 times by the radio station in the previous week.
e) Comment on the strength of the relationship between the variables.

8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 29DW 1b. 29DX 1c. 29DY 2a. 29DZ 2b. 29F2 2c. 29F3
2d. 29F4 2e. 29F5 3a. 29F6 3b. 29F7 4. 29F8 5. 29F9
6. 29FB 7. 29FC

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9.4 Summary EMCJT

- **Curve fitting** is the process of fitting functions to data.
- **Intuitive** curve fitting is performed by visually interpreting if the points on the scatter plot conform to a linear, exponential, quadratic or some other function.
- The **line of best fit** or trend line is a straight line through the data which best approximates the available data points. This allows for the estimation of missing data values.
- **Interpolation** is the technique used to predict values that fall within the range of the available data.
- **Extrapolation** is the technique used to predict the value of variables beyond the range of the available data.
- **Linear regression analysis** is a statistical technique of finding out exactly which linear function best fits a given set of data.
- The **least squares method** is an algebraic method of finding the linear regression equation. The linear regression equation is written $\hat{y} = a + bx$, where

$$b = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} (x_i)^2 - (\sum_{i=1}^{n} x_i)^2}$$

$$a = \frac{1}{n}\sum_{i=1}^{n} y_i - \frac{b}{n}\sum_{i=1}^{n} x_i = \bar{y} - b\bar{x}$$

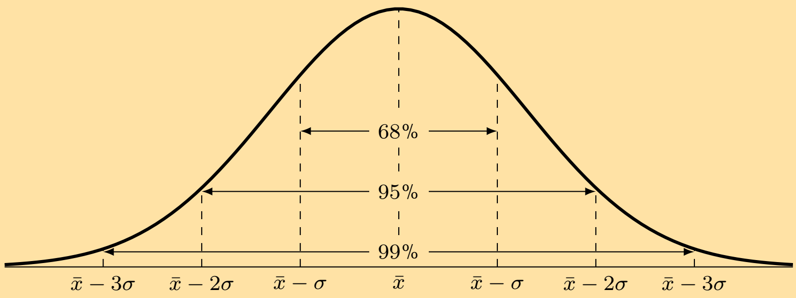
- The **linear correlation coefficient**, r, is a measure which tells us the strength and direction of a relationship between two variables, determined using the equation:

$$r = b(\frac{\sigma_x}{\sigma_y})$$

- The correlation coefficient $r \in [-1; 1]$. When $r = -1$, there is perfect negative correlation, when $r = 0$, there is no correlation and when $r = 1$, there is perfect positive correlation.

Exercise 9 – 5: End of chapter exercises

1. The number of SMS messages sent by a group of teenagers was recorded over a period of a week. The data was found to be normally distributed with a mean of 140 messages and a standard deviation of 12 messages. [NSC Paper 3 Feb-March 2012]



   Answer the following questions with reference to the information provided in the graph:

   a) What percentage of teenagers sent less than 128 messages?

   b) What percentage of teenagers sent between 116 and 152 messages?

2. A company produces sweets using a machine which runs for a few hours per day. The number of hours running the machine and the number of sweets produced are recorded.

Machine hours	Sweets produced
3,80	275
4,23	287
4,37	291
4,10	281
4,17	286

   Find the linear regression equation for the data, and estimate the machine hours needed to make 300 sweets.

3. The profits of a new shop are recorded over the first 6 months. The owner wants to predict his future sales. The profits by month so far have been R 90 000; R 93 000; R 99 500; R 102 000; R 101 300; R 109 000.

   a) Calculate the linear regression function for the data, using profit as your y-variable. Round a and b to two decimal places.

   b) Give an estimate of the profits for the next two months.

   c) The owner wants a profit of R 130 000. Estimate how many months this will take.

4. A fast food company produces hamburgers. The number of hamburgers made and the costs are recorded over a week. Examine the data below and answer the questions on the following page.

Hamburgers made	Costs
495	R 2382
550	R 2442
515	R 2484
500	R 2400
480	R 2370
530	R 2448
585	R 2805

a) Find the linear regression function that best fits the data. Use hamburgers made as your x-variable and round a and b to two decimal places.

b) Calculate the value of the correlation coefficient, correct to two decimal places, and comment on the strength and direction of the correlation.

c) If the total cost in a day is R 2500, estimate the number of hamburgers produced. Round your answer down to the nearest whole number.

d) What is the cost of 490 hamburgers?

5. A collection of data related to an investigation into biceps length and height of students was recorded in the table below. Answer the questions to follow.

Length of right biceps (cm)	**Height (cm)**
25,5	163,3
26,1	164,9
23,7	165,5
26,4	173,7
27,5	174,4
24	156
22,6	155,3
27,1	169,3

a) Draw a scatter plot of the data set.

b) Calculate equation of the line of regression.

c) Draw the regression line onto the graph.

d) Calculate the correlation coefficient r

e) What conclusion can you reach, regarding the relationship between the length of the right biceps and height of the students in the data set?

6. A class wrote two tests, and the marks for each were recorded in the table below. Full marks in the first test was 50, and the second test was out of 30.

Learner	**Test 1**	**Test 2**
	(Full marks: 50)	(Full marks: 30)
1	42	25
2	32	19
3	31	20
4	42	26
5	35	23
6	23	14
7	43	24
8	23	12
9	24	14
10	15	10
11	19	11
12	13	10
13	36	22
14	29	17
15	29	17
16	25	16
17	29	18
18	17	
19	30	19
20	28	17

a) Is there a strong correlation between the marks for the first and second test? Show why or why not.

b) One of the learners (in Row 18) did not write the second test. Given her mark for the first test, calculate an expected mark for the second test. Round the mark up to the nearest whole number.

7. Lindiwe works for Eskom, the South African power distributor. She knows that on hot days more electricity than average is used to cool houses. In order to accurately predict how much more electricity needs to be produced, she wants to determine the precise nature of the relationship between temperature and electricity usage.

The data below shows the peak temperature in degrees Celsius on ten consecutive days during summer and the average number of units of electricity used by a number of households. Examine her data and answer the questions that follow.

Peak temp. (y)	32	40	30	28	25	38	36	20	24	26
Ave. no. of units (x)	37	45	35	30	20	40	38	15	20	22

a) Draw a scatter plot of the data.

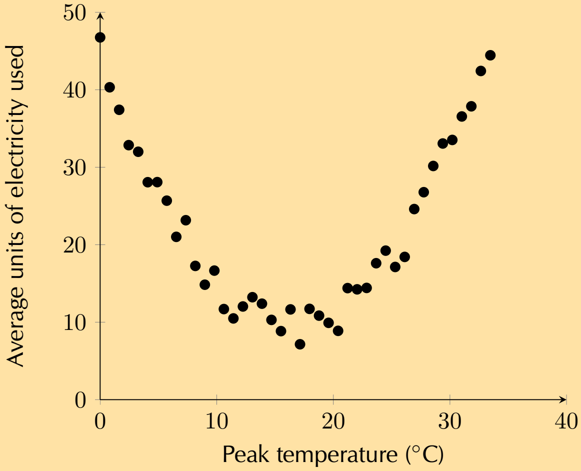
b) Using the formulae for a and b, determine the equation of the least squares line.

c) Determine the value of the correlation coefficient, r, by hand.

d) What can Lindiwe conclude about the relationship between peak temperature and the number of electricity units used?

e) Predict the average number of units of electricity used by a household on a day with a peak temperature of 45°C. Give your answer correct to the nearest unit and identify what this type of prediction is called.

f) Lindiwe suspected that the relationship between temperature and electricity consumption was not linear for all temperatures. She then decided to collect data for peak temperatures down to 0°C. Examine the graph of her data below and identify which type of function would best fit the data and describe the nature of the relationship between temperature and electricity for the newly available data.



g) Lindiwe is asked by her superiors to determine which day is best to perform maintenance on one of their power plants. She determined that the equation $y = 0{,}13x^2 - 4{,}3x + 45$ best fit her data. Use her equation to estimate the peak temperature and average no. of units used on the day when the least amount of electricity generation is required.

8. Below is a list of data concerning 12 countries and their respective carbon dioxide (CO_2) emission levels per person per annum (measured in tonnes) and the gross domestic product (GDP is a measure of products produced and services delivered within a country in a year) per person (in US dollars). Data sourced from the World Bank and the US Department of Energy's Carbon Dioxide Information Analysis Center.

	CO_2 emmissions per capita (x)	GDP per capita (y)
South Africa	8,8	11 440
Thailand	4,1	9815
Italy	7,5	32 512
Australia	18,3	44 462
China	5,3	9233
India	1,4	3876
Canada	15,3	42 693
United Kingdom	8,5	35 819
United States	17,2	49 965
Saudi Arabia	16,1	24 571
Iran	7,3	11 395
Indonesia	1,8	4956

a) Draw a scatter plot of the data set.

b) Draw your estimate of the line of best fit on your scatter plot and determine the equation of your line of best fit.

c) Use your calculator to determine the equation for the least squares regression line. Round a and b to two decimal places in your final answer.

d) Use your calculator to determine the correlation coefficient, r. Round your answer to two decimal places.

e) What conclusion can you reach regarding the relationship between CO_2 emissions per annum and GDP per capita for the countries in the data set?

f) Kenya has a GDP per capita of $1712. Use your equation of the least squares regression line to estimate the annual CO_2 emissions of Kenya correct to two decimal places.

9. A group of students attended a course in Statistics on Saturdays over a period of 10 months. The number of Saturdays on which a student was absent was recorded against the final mark the student obtained. The information is shown in the table below. [Adapted from NSC Paper 3 Feb-March 2012]

Number of Saturdays absent	0	1	2	2	3	3	5	6	7
Final mark (as %)	96	91	78	83	75	62	70	68	56

a) Draw a scatter plot of the data.

b) Determine the equation of the least squares line and draw it on your scatter plot.

c) Calculate the correlation coefficient.

d) Comment on the trend of the data.

e) Predict the final mark of a student who was absent for four Saturdays.

10. Grant and Christie are training for a half-marathon together in 8 weeks time. Christie is much fitter than Grant but she has challenged him to beat her time at the race. Grant has begun a rigid training programme to try and improve his time.

    Time taken to complete a half marathon was recorded each Sunday. The first recorded Sunday is denoted as week 1. The half-marathon takes place on the eighth Sunday, i.e. week 8. Examine the data set in the table below and answer the questions the follow.

Week	1	2	3	4	5	6
Grant's time (HH:MM)	02:01	01:59	01:55	01:53	01:47	01:42
Christie's time (HH:MM)	01:40	01:42	01:38	01:39	01:37	01:35

    a) Draw a scatter plot of the data sets. Include Grant and Christie's data on the same set of axes. Use a $\bullet$ to denote Grant's data points and $\times$ to denote Christie's data points. Convert all times to minutes.
    b) Comment on and compare any trends that you observe in the data.
    c) Determine the equations of the least squares regression lines for Grant's data and Christie's data. Draw these lines on your scatter plot. Use a different colour for each.
    d) Calculate the correlation coefficient and comment on the fit for each data set.
    e) Assuming the observed trends continue, will Grant beat Christie at the race?
    f) Assuming the observed trends continue, extrapolate the week in which Grant will be able to run a half-marathon in less time than Christie.

11. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29FD 2. 29FF 3. 29FG 4. 29FH 5. 29FJ 6. 29FK
7. 29FM 8. 29FN 9. 29FP 10. 29FQ

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CHAPTER 10

Probability

10.1 Revision EMCJV

Terminology EMCJW

Outcome: a single observation of an uncertain or random process (called an *experiment*). For example, when you accidentally drop a book, it might fall on its cover, on its back or on its side. Each of these options is a possible outcome.

Sample space of an experiment: the set of all possible outcomes of the experiment. For example, the sample space when you roll a single 6-sided die is the set $\{1;2;3;4;5;6\}$. For a given experiment, there is exactly one sample space. The sample space is denoted by the letter S.

Event: a set of outcomes of an experiment. For example, if you have a standard deck of 52 cards, an event may be picking a spade card or a king card.

Probability of an event: a real number between and inclusive of 0 and 1 that describes how likely it is that the event will occur. A probability of 0 means the outcome of the experiment will never be in the event set. A probability of 1 means the outcome of the experiment will always be in the event set. When all possible outcomes of an experiment have equal chance of occurring, the probability of an event is the number of outcomes in the event set as a fraction of the number of outcomes in the sample space. To calculate a probability, you divide the number of favourable outcomes by the total number of possible outcomes.

Relative frequency of an event: the number of times that the event occurs during experimental trials, divided by the total number of trials conducted. For example, if we flip a coin 10 times and it landed on heads 3 times, then the relative frequency of the heads event is $\frac{3}{10} = 0{,}3$.

Union of events: the set of all outcomes that occur in at least one of the events. For 2 events called A and B, we write the union as "A or B". Another way of writing the union is using set notation: $A \cup B$. For example, if A is all the countries in Africa and B is all the countries in Europe, A or B is all the countries in Africa and Europe.

Intersection of events: the set of all outcomes that occur in all of the events. For 2 events called A and B, we write the intersection as "A and B". Another way of writing the intersection is using set notation: $A \cap B$. For example, if A is soccer players and B is cricket players, A and B refers to those who play both soccer and cricket.

Mutually exclusive events: events with no outcomes in common, that is $(A$ and $B)$ is an empty set. Mutually exclusive events can never occur simultaneously. For example the event that a number is even and the event that the same number is odd are mutually exclusive, since a number can never be both even and odd.

Complementary events: two mutually exclusive events that together contain all the outcomes in the sample space. For an event called A, we write the complement as "not A". Another way of writing the complement is as A'.

Dependent and independent events: two events, A and B, are **independent** if the outcome of the first event does not influence the outcome of the second event. For example, if you flip a coin and it lands on tails and flip it again and it lands on heads, neither outcome influences the other. Two events, C and D, are **dependent** if the outcome of one event influences the outcome of the other. For example, if your lunchbox contains 3 sandwiches and 2 apples, when you eat one of the items, this reduces the number of choices you have when deciding to eat a second item.

See video: 29FR at www.everythingmaths.co.za

10.2 Identities EMCJX

The **addition rule** (also called the sum rule) for any 2 events, A and B is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

This rule relates the probabilities of 2 events with the probabilities of their union and intersection.

The **addition rule for 2 mutually exclusive events** is

$$P(A \text{ or } B) = P(A) + P(B)$$

This rule is a special case of the previous rule. Because the events are mutually exclusive, $P(A \text{ and } B) = 0$.

The **complementary rule** is

$$P(\text{not } A) = 1 - P(A)$$

This rule is a special case of the previous rule. Since A and (not A) are complementary, $P(A \text{ or } (\text{not } A)) = 1$.

The **product rule** for independent events A and B is:

$$P(A \text{ and } B) = P(A) \times P(B)$$

If two events A and B are **dependent** then:

$$P(A \text{ and } B) \neq P(A) \times P(B)$$

WARNING!

Just because two events are mutually exclusive does not necessarily mean that they are independent. To test whether events are mutually exclusive, always check that $P(A$ and $B) = 0$. To test whether events are independent, always check that $P(A$ and $B) = P(A) \times P(B)$. See the exercises below for examples of events that are mutually exclusive and independent in different combinations.

Worked example 1: Dependent and independent events

QUESTION

Write down which of the following events are dependent and which are independent:

1. The student council chooses a head student and then a deputy head student.
2. A bag contains blue marbles and red marbles. You take a red marble out of the bag and then throw it back in again before you take another marble out of the bag.

SOLUTION

Step 1: Ask the question: Did the available choices change for the second event because of the first event?

1. Yes, because after selecting the head student there are fewer council members available to choose for the deputy head student position. Therefore the two events are dependent.
2. No, because when you throw the first marble back into the bag, there are the same number and colour composition of choices for the second marble. Therefore the two events are independent.

Worked example 2: Independent and dependent events

QUESTION

A bag contains 3 yellow and 4 black beads. We remove a random bead from the bag, record its colour and put it back into the bag. We then remove another random bead from the bag and record its colour.

1. What is the probability that the first bead is yellow?
2. What is the probability that the second bead is black?
3. What is the probability that the first bead is yellow and the second bead is black?
4. Are the first bead being yellow and the second bead being black independent events?

SOLUTION

Step 1: Probability of a yellow bead first

Since there is a total of 7 beads, of which 3 are yellow, the probability of getting a yellow bead is

$$P(\text{first bead yellow}) = \frac{3}{7}$$

Step 2: Probability of a black bead second

The problem states that the first bead is placed back into the bag before we take the second bead. This means that when we draw the second bead, there are again a total of 7 beads in the bag, of which 4 are black. Therefore the probability of drawing a black bead is

$$P(\text{second bead black}) = \frac{4}{7}$$

Step 3: Probability of yellow first and black second

When drawing two beads from the bag, there are 4 possibilities. We can get

- a yellow bead and then another yellow bead;
- a yellow bead and then a black bead;
- a black bead and then a yellow bead;
- a black bead and then another black bead.

We want to know the probability of the second outcome, where we have to get a yellow bead first. Since there are 3 yellow beads and 7 beads in total, there are $\frac{3}{7}$ ways to get a yellow bead first. Now we put the first bead back, so there are again 3 yellow beads and 4 black beads in the bag. Therefore there are $\frac{4}{7}$ ways to get a black bead second if the first bead was yellow. This means that there are

$$\frac{3}{7} \times \frac{4}{7} = \frac{12}{49}$$

ways to get a yellow bead first and a black bead second. So, the probability of getting a yellow bead first and a black bead second is $\frac{12}{49}$.

Step 4: Dependent or independent?

According to the definition, events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

In this problem:

- $P(\text{first bead yellow}) = \frac{3}{7}$
- $P(\text{second bead black}) = \frac{4}{7}$
- $P(\text{first bead yellow and second bead black}) = \frac{12}{49}$

Since $\frac{12}{49} = \frac{3}{7} \times \frac{4}{7}$, the events are independent.

See video: 29FS at www.everythingmaths.co.za

Worked example 3: Independent and dependent events

QUESTION

In the previous example, we picked a random bead and put it back into the bag before continuing. This is called **sampling with replacement**. In this worked example, we will follow the same process, except that we will not put the first bead back into the bag. This is called **sampling without replacement**.

So, from a bag with 3 red and 5 green beads, we remove a random bead and record its colour. Then, without putting back the first bead, we remove another random bead from the bag and record its colour.

1. What is the probability that the first bead is red?
2. What is the probability that the second bead is green?
3. What is the probability that the first bead is red and the second bead is green?
4. Are the first bead being red and the second bead being green independent events?

SOLUTION

Step 1: Count the number of outcomes

We will examine the number ways in which we can get the different possible outcomes when removing 2 beads. The possible outcomes are

- a red bead and then another red bead (RR);
- a red bead and then a green bead (RG);
- a green bead and then a red bead (GR);
- a green bead and then another green bead (GG).

For the first outcome, we have to get a red bead first. Since there are 3 red beads and 8 beads in total, there are $\frac{3}{8}$ ways to get a red bead first. After we have taken out a red bead, there are now 2 red beads and 5 green beads left. Therefore there are $\frac{2}{7}$ ways to get a red bead second if the first bead was also red. This means that there are

$$\frac{3}{8} \times \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$$

ways to get a red bead first and a red bead second. The probability of the first outcome is $\frac{3}{28}$.

For the second outcome, we have to get a red bead first. As in the first outcome, there are $\frac{3}{8}$ ways to get a red bead first; and there are now 2 red beads and 5 green beads left. Therefore there are $\frac{5}{7}$ ways to get a green bead second if the first bead was red. This means that there are

$$\frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

ways to get a red bead first and a green bead second. The probability of the second outcome is $\frac{15}{56}$.

In the third outcome, the first bead is green and the second bead is red. There are $\frac{5}{8}$ ways to get a green bead first; and there are now 4 green beads and 3 red beads left. Therefore there are $\frac{3}{7}$ ways to get a red bead second if the first bead was green. This means that there are

$$\frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$$

ways to get a red bead first and a green bead second. The probability of the third outcome is $\frac{15}{56}$.

In the fourth outcome, the first and second beads are both green. Since there are 5 green beads and 8 beads in total, there are $\frac{5}{8}$ ways to get a green bead first. After we have removed a green bead, 3 red beads and 4 green beads remain in the bag. Therefore there are $\frac{4}{7}$ ways to get a green bead second if the first bead was also green. This means that there are

$$\frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

ways to get a green bead first and a green bead second. Therefore the probability of the fourth outcome is $\frac{5}{14}$.

To summarise, these are the possible outcomes and their probabilities:

- first bead red and second bead red (RR): $\frac{3}{28}$;
- first bead red and second bead green (RG): $\frac{15}{56}$;
- first bead green and second bead red (GR): $\frac{15}{56}$;
- first bead green and second bead green (GG): $\frac{5}{14}$.

Step 2: Probability of a red bead first

To determine the probability of getting a red bead on the first draw, we look at all of the outcomes that contain a red bead first. These are

- a red bead and then another red bead (RR);
- a red bead and then a green bead (RG).

The probability of the first outcome is $\frac{3}{28}$ and the probability of the second outcome is $\frac{15}{56}$. By adding these two probabilities, we see that the probability of getting a red bead first is

$$P(\text{first bead red}) = \frac{3}{28} + \frac{15}{56} = \frac{6}{56} + \frac{15}{56} = \frac{21}{56} = \frac{3}{8}$$

This is the same as in the previous worked example, which should not be too surprising since the probability of the first bead being red is not affected by whether or not we put it back into the bag before drawing the second bead.

Step 3: Probability of a green bead second

To determine the probability of getting a green bead on the second draw, we look at all of the outcomes that contain a green bead second. These are

- a red bead and then a green bead (RG);
- a green bead and then another green bead (GG).

The probability of the first outcome is $\frac{15}{56}$ and the probability of the second outcome is $\frac{5}{14}$. By adding these two probabilities, we see that the probability of getting a green bead second is

$$P(\text{second bead green}) = \frac{15}{56} + \frac{5}{14} = \frac{15}{56} + \frac{20}{56} = \frac{35}{56} = \frac{5}{8}$$

Step 4: Probability of red first and green second

We have already calculated the probability that the first bead is red and the second bead is green (RG). It is $\frac{15}{56}$.

Step 5: Dependent or independent?

According to the definition, events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

In this problem:

- $P(\text{first bead red}) = \frac{3}{8}$
- $P(\text{second bead green}) = \frac{5}{8}$
- $P(\text{first bead red and second bead green}) = \frac{15}{56}$

Since $\frac{3}{8} \times \frac{5}{8} = \frac{15}{64} \neq \frac{15}{56}$, the events are dependent.

Worked example 4: The addition rule for 2 mutually exclusive events

QUESTION

A sample space, S, consists of all natural numbers less than 16. A is the event of drawing an even number at random. B is the event of randomly drawing a prime number. Are A and B mutually exclusive events? Prove this using the addition rule.

SOLUTION

Step 1: Write down the sample space

The sample space contains all the natural numbers less than 16.

$$S = \{1; 2; 3; 4; 5; 6; 7; 8; 9; 10; 11; 12; 13; 14; 15\}$$

Step 2: Write down the events

The even natural numbers less than 16 are

$$A = \{2; 4; 6; 8; 10; 12; 14\}$$

The prime numbers less than 16 are

$$B = \{2; 3; 5; 7; 11; 13\}$$

We can already see from writing down our event sets that A and B share the event 2 and are thus not mutually exclusive. However, the question asked us to prove this using the addition rule so let's go ahead and do that.

Step 3: Compute the probabilities

The probability of an event is the number of outcomes in the event set divided by the number of outcomes in the sample space. There are 15 outcomes in the sample space.

1. Since there are 7 outcomes in the A event set, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{7}{15}$.
2. Since there are 6 outcomes in the B event set, $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{6}{15} = \dfrac{2}{5}$.
3. The event that is a prime number or an even number is the union of the above two event sets. There are 12 elements in the union of the event sets, so $P(A \text{ or } B) = \dfrac{n(A \text{ or } B)}{n(S)} = \dfrac{12}{15}$.

Step 4: Are the two events mutually exclusive?

To test whether two events are mutually exclusive, we can use the **addition rule**. For two mutually exclusive events,

$$P(A \text{ and } B) \text{ is an empty set, therefore } P(A \text{ or } B) = P(A) + P(B)$$

$$\text{Since } P(A \text{ or } B) = \frac{12}{15} \text{ and } P(A) + P(B) = \frac{6}{15} + \frac{7}{15} = \frac{13}{15}$$
$$P(A \text{ or } B) \neq P(A) + P(B)$$

Therefore the intersection of A and B is nonzero. This means that the events A and B are not mutually exclusive.

Worked example 5: The addition rule

QUESTION

The probability that a person drinks tea is $0{,}5$. The probability that a person drinks coffee is $0{,}4$. The probability that a person drinks tea, coffee or both is $0{,}8$. Determine the probability that a person drinks tea and coffee.

SOLUTION

Step 1: Determine if the events are mutually exclusive
Let the probability that a person drinks tea $= P(T)$ and the probability that a person drinks coffee $= P(C)$.

From the information provided in the question, we know that:

- $P(T) = 0{,}5$
- $P(C) = 0{,}4$
- $P(T \text{ or } C) = 0{,}8$
- $P(T) + P(C) = 0{,}5 + 0{,}4 = 0{,}9$
- Therefore $P(T \text{ or } C) \neq P(T) + P(C)$

Therefore the events are not mutually exclusive.

Step 2: Compute the probability that a person drinks tea and coffee
Using the addition rule, we know that:

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ \therefore P(T \text{ or } C) &= P(T) + P(C) - P(T \text{ and } C) \\ 0{,}8 &= 0{,}4 + 0{,}5 - P(T \text{ and } C) \\ \therefore P(T \text{ and } C) &= 0{,}4 + 0{,}5 - 0{,}8 = 0{,}1 \end{aligned}$$

Therefore the probability that a person drinks tea and coffee is $0{,}1$.

See video: 29FT at www.everythingmaths.co.za

Worked example 6: The complementary rule

QUESTION

Joe wants to open a tuckshop at his school but is not sure which cool drinks to stock. Before opening, he interviewed a sample of learners to determine what types of cool drinks they like. From his research, he determined that the probability that a learner drinks cola is $0{,}3$, the probability that a learner drinks lemonade is $0{,}6$ and the probability that a learner drinks neither is $0{,}2$. Determine:

- the probability that a learner drinks cola and lemonade.
- the probability that a learner drinks only cola or only lemonade.

SOLUTION

Step 1: Determine the probability that a learner drinks cola or lemonade

Let the probability that a learner drinks cola $= P(C)$ and the probability that a learner drinks lemonade $= P(L)$.

From the information provided in the question, we know that:

- $P(C) = 0{,}3$
- $P(L) = 0{,}6$
- $P(\text{not } (C \text{ or } L)) = 0{,}2$

Using the complementary rule:

$$\begin{aligned} P(\text{not } (C \text{ or } L)) &= 1 - P(C \text{ or } L) \\ \therefore P(C \text{ or } L) &= 1 - P(\text{not } (C \text{ or } L)) \\ &= 1 - 0{,}2 \\ &= 0{,}8 \end{aligned}$$

Step 2: Calculate the probability that a learner drinks cola and lemonade

Using the addition rule:

$$\begin{aligned} P(C \text{ or } L) &= P(C) + P(L) - P(C \text{ and } L) \\ \therefore P(C \text{ and } L) &= P(C) + P(L) - P(C \text{ or } L) \\ &= 0{,}3 + 0{,}6 - 0{,}8 \\ &= 0{,}1 \end{aligned}$$

The probability that a learner drinks both cola and lemonade is $0{,}1$.

Step 3: Determine the probability that a learner drinks only cola or only lemonade

This question requires us to calculate the probability that a learner likes lemonade or cola but not both of them. We can write this as:

$$P(\text{only } C \text{ or only } L) = P(C \text{ or } L) - P(C \text{ and } L)$$

since a learner can like either cola or lemonade but not both.

We already know $P(C \text{ or } L) = 0{,}8$ and $P(C \text{ and } L) = 0{,}1$, therefore the probability of a learner drinking only cola or only lemonade is $0{,}7$.

Exercise 10 – 1: The product and addition rules

1. Determine whether the following events are dependent or independent and give a reason for your answer:
   a) Joan has a box of yellow, green and orange sweets. She takes out a yellow sweet and eats it. Then, she chooses another sweet and eats it.
   b) Vuzi throws a die twice.
   c) Celia chooses a card at random from a deck of 52 cards. She is unhappy with her choice, so she places the card back in the deck, shuffles it and chooses a second card.

d) Thandi has a bag of beads. She randomly chooses a yellow bead, looks at it and then puts it back in the bag. Then she randomly chooses another bead and sees that it is red and puts it back in the bag.

e) Mark has a container with calculators. Some of them work and some are broken. He randomly chooses a calculator and sees that it does not work and throws it away. He then chooses another calculator, sees that it works and keeps it.

2. Given that $P(A) = 0{,}7$; $P(B) = 0{,}4$ and $P(A \text{ and } B) = 0{,}28$,

   a) are events A and B mutually exclusive? Give a reason for your answer.

   b) are the events A and B independent? Give a reason for your answer.

3. In the following examples, are A and B dependent or independent?

   a) $P(A) = 0{,}2; P(B) = 0{,}7$ and $P(A \text{ and } B) = 0{,}21$

   b) $P(A) = 0{,}2; P(B) = 0{,}7$ and $P(B \text{ and } A) = 0{,}14$.

4. $n(A) = 5; n(B) = 4; n(S) = 20$ and $n(A \text{ or } B) = 8$.

   a) Are A and B mutually exclusive?

   b) Are A and B independent?

5. Simon rolls a die twice. What is the probability of getting:

   a) two threes.

   b) a prime number then an even number.

   c) no threes.

   d) only one three.

   e) at least one three.

6. The Mandalay Secondary soccer team has to win both of their next two matches in order to qualify for the finals. The probability that Mandalay Secondary will win their first soccer match against Ihlumelo High is $\frac{2}{5}$ and the probability of winning their second soccer match against Masiphumelele Secondary is $\frac{3}{7}$. Assume each match is an independent event.

   a) What is the probability they will progress to the finals?

   b) What is the probability they will not win either match?

   c) What is the probability they will win only one of their matches?

   d) You were asked to assume that the matches are independent events but this is unlikely in reality. What are some factors you think may result in the outcome of the matches being dependent?

7. A pencil bag contains 2 red pens and 4 green pens. A pen is drawn from the bag and then replaced before a second pen is drawn. Calculate:

   a) The probability of drawing a red pen first if a green pen is drawn second.

   b) The probability of drawing a green pen second if the first pen drawn was red.

   c) The probability of drawing a red pen first and a green pen second.

8. A lunch box contains 4 sandwiches and 2 apples. Vuyele chooses a food item randomly and eats it. He then chooses another food item randomly and eats that. Determine the following:

   a) The probability that the first item is a sandwich.

   b) The probability that the first item is a sandwich and the second item is an apple.

c) The probability that the second item is an apple.
d) Are the events in a) and c) dependent? Confirm your answer with a calculation.

9. Given that $P(A) = 0{,}5$; $P(B) = 0{,}4$ and $P(A \text{ or } B) = 0{,}7$, determine by calculation whether events A and B are:
   a) mutually exclusive
   b) independent
10. A and B are two events in a sample space where $P(A) = 0{,}3$; $P(A \text{ or } B) = 0{,}8$ and $P(B) = k$. Determine the value of k if:
    a) A and B are mutually exclusive
    b) A and B are independent
11. A and B are two events in sample space S where $n(S) = 36$; $n(A) = 9$; $n(B) = 4$ and $n(\text{not } (A \text{ or } B)) = 24$. Determine:
    a) $P(A \text{ or } B)$
    b) $P(A \text{ and } B)$
    c) whether events A and B independent. Justify your answer with a calculation.
12. The probability that a Mathematics teacher is absent from school on a certain day is $0{,}2$. The probability that the Science teacher will be absent that same day is $0{,}3$.
    a) Do you think these two events are independent? Give a reason for your answer.
    b) Assuming the events are independent, what is the probability that the Mathematics teacher or the Science teacher is absent?
    c) What is the probability that neither the Mathematics teacher nor the Science teacher is absent?
13. Langa Cricket Club plays two cricket matches against different clubs. The probability of winning the first match is $\frac{3}{5}$ and the probability of winning the second match is $\frac{4}{9}$. Assuming the results of the matches are independent, calculate the probability that Langa Cricket Club will:
    a) win both matches.
    b) not win the first match.
    c) win one or both of the two matches.
    d) win neither match.
    e) not win the first match and win the second match.
14. Two teams are working on the final problem at a Mathematics Olympiad. They have 10 minutes remaining to finish the problem. The probability that team A will finish the problem in time is 40% and the probability that team B will finish the problem in time is 25%. Calculate the probability that both teams will finish before they run out of time.
15. Thabo and Julia were arguing about whether people prefer tea or coffee. Thabo suggested that they do a survey to settle the dispute. In total, they surveyed 24 people and found that 8 of them preferred to drink coffee and 12 of them preferred to drink tea. The number of people who drink tea, coffee or both is 16. Determine:
    a) the probability that a person drinks tea, coffee or both.

b) the probability that a person drinks neither tea nor coffee.
c) the probability that a person drinks coffee and tea.
d) the probability that a person does not drink coffee.
e) whether the event that a person drinks coffee and the event that a person drinks tea are independent.

16. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29FV	2. 29FW	3. 29FX	4. 29FY	5. 29FZ	6. 29G2
7. 29G3	8. 29G4	9. 29G5	10. 29G6	11. 29G7	12. 29G8
13. 29G9	14. 29GB	15. 29GC			

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10.3 Tools and Techniques EMCJY

Venn diagrams are used to show how events are related to one another. A Venn diagram can be very helpful when doing calculations with probabilities. In a Venn diagram each event is represented by a shape, often a circle or a rectangle. The region inside the shape represents the outcomes included in the event and the region outside the shape represents the outcomes that are not in the event.

Tree diagrams are useful for organising and visualising the different possible outcomes of a sequence of events. Each branch in the tree shows an outcome of an event, along with the probability of that outcome. For each possible outcome of the first event, we draw a line where we write down the probability of that outcome and the state of the world if that outcome happened. Then, for each possible outcome of the second event we do the same thing. The probability of a sequence of outcomes is calculated as the product of the probabilities along the branches of the sequence.

Two-way contingency tables are a tool for keeping a record of the counts or percentages in a probability problem. Two-way contingency tables are especially helpful for figuring out whether events are dependent or independent.

Worked example 7: Venn diagrams

QUESTION

There are 200 boys in Grade 12 at Marist Brothers High School. Their participation in sport can be broken down as follows:

- 107 play rugby
- 90 play soccer
- 63 play cricket
- 35 play rugby and soccer
- 23 play rugby and cricket
- 15 play rugby, soccer and cricket
- 190 boys play rugby, soccer or cricket

1. How many boys do not play any of these sports?
2. Draw a Venn diagram to illustrate the given information and use it to answer the following questions:
   a) How many boys play soccer and cricket, but not rugby?
   b) What is the probability that a randomly chosen Grade 12 boy at Marist Brothers High School will take part in at least two of the sports: rugby, soccer or cricket? Give your answer correct to 3 decimal places.

SOLUTION

Step 1: Calculate the number of boys playing none of the given sports

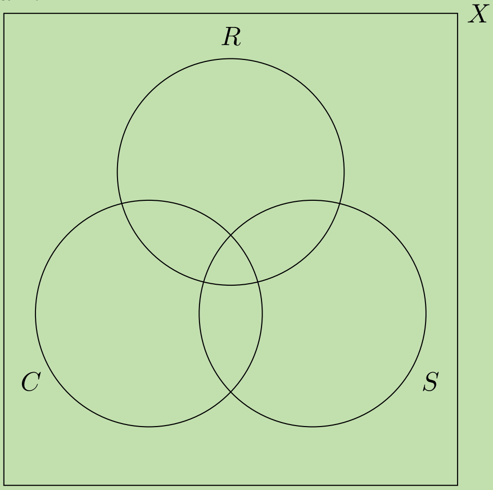
In order to calculate the number of boys playing none of the sports, we subtract the number of boys playing any of the three sports from the total number of boys in the sample space.

$$\text{Not rugby, cricket or soccer } = 200 - 190 = 10$$

Therefore 10 boys do not play rugby, cricket or soccer.

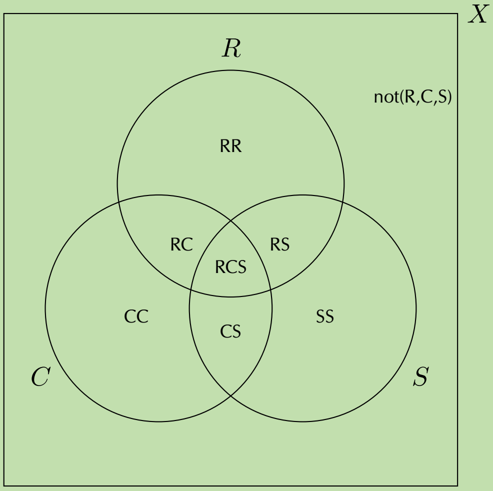
Step 2: Draw the outline of the Venn diagram

Let X = the sample space; R = rugby; S = soccer and C = cricket. Put this information on a Venn diagram:



Step 3: Calculate the counts for the different groupings

The following groupings exist:



Rugby, cricket and soccer: RCS

$$RCS = 15$$

Rugby and soccer but not cricket: RS

$$\begin{aligned} RS &= (R \text{ and } S) - RCS \\ &= 35 - 15 = 20 \end{aligned}$$

Rugby and cricket but not soccer: RC

$$\begin{aligned} RC &= (R \text{ and } C) - RCS \\ &= 23 - 15 = 8 \end{aligned}$$

Cricket and soccer but not rugby: CS

$$\begin{aligned} CS &= (S \text{ and } C) - RCS \\ \text{Let } (S \text{ and } C) &= x \\ \text{Therefore } CS &= x - 15 \end{aligned}$$

Only rugby: RR

$$\begin{aligned} RR &= R - RS - RC - RCS \\ &= 107 - 20 - 8 - 15 \\ &= 64 \end{aligned}$$

Only soccer: SS

$$\begin{aligned} SS &= S - RS - CS - RCS \\ &= 90 - 20 - (x - 15) - 15 \\ &= 70 - x \end{aligned}$$

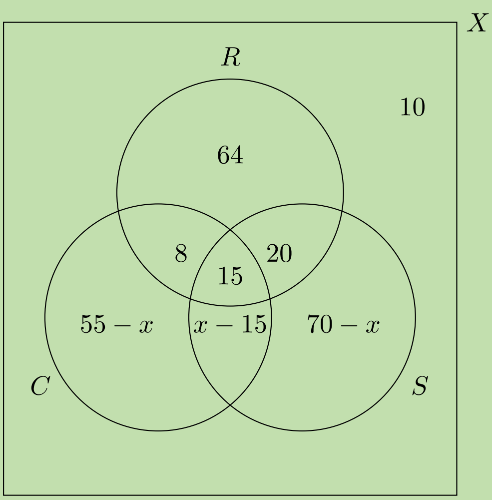
Only cricket: CC

$$\begin{aligned} CC &= C - RC - CS - RCS \\ &= 63 - 8 - (x - 15) - 15 \\ &= 55 - x \end{aligned}$$

Not rugby, cricket or soccer: not(R, C, S)

$$\text{not}(R, C, S) = 10$$

Step 4: Fill in the counts on the Venn diagram



Step 5: Calculate the unknown values

Since 190 of the boys play at least one of the sports, using the values on our Venn diagram, we can set up the following equation to solve for x.

$$\begin{aligned} 64 + 8 + 15 + 20 + (x - 15) + (70 - x) + (55 - x) &= 190 \\ 217 - x &= 190 \\ \text{Therefore } x &= 27 \end{aligned}$$

We know that:

$$\begin{aligned}
\text{Cricket and soccer but not rugby } (CS) &= x - 15 \\
\text{Therefore } CS &= 27 - 15 \\
&= 12
\end{aligned}$$

Therefore there are 12 boys who play cricket and soccer but not rugby.

Step 6: Calculate the probability that a randomly chosen Grade 12 boy plays at least two of the given sports

We know the number of boys who play two or more of rugby, cricket or soccer and we know the total number of boys. Therefore, we can calculate the probability using the following equation:

$$\begin{aligned}
P(\text{at least two sports}) &= \frac{n(RC) + n(RS) + n(CS) + n(RCS)}{n(X)} \\
&= \frac{8 + 15 + 20 + 12}{200} \\
&= \frac{55}{200} = 0{,}275
\end{aligned}$$

Therefore the probability that a randomly chosen Grade 12 boy plays at least 2 of either rugby, cricket or soccer $= 0{,}275$ or $27{,}5\%$

See video: 29GD at www.everythingmaths.co.za

Worked example 8: Tree diagrams

QUESTION

The probability that the floor of a supermarket will be wet when it opens in the morning is 30% and there is a 10% probability of the floor being very wet. The probability that a person will slip and fall if the floor is dry is 12% and a person is three times as likely to fall if the floor is wet. If the floor is very wet, the probability that a person will fall is $0{,}6$. Draw a tree diagram to represent the given information, showing the probabilities of each outcome, and use it to answer the following questions:

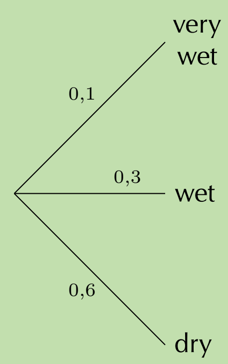
1. What is the probability that a person will fall on any given day?
2. What is the probability that a person will not fall on any given day?
3. Are the events of the floor being dry and a person falling independent? Justify your answer with a calculation.

SOLUTION

Step 1: Identify the events

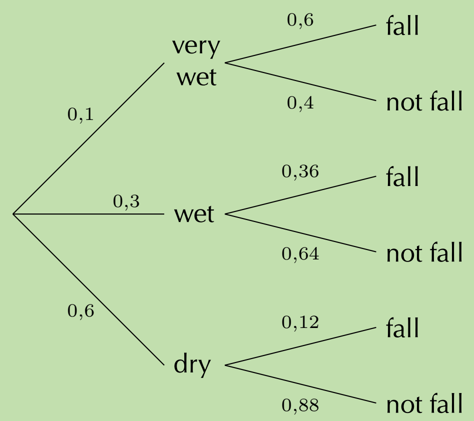
There are three outcomes for the floor, namely, dry, wet and very wet, and two outcomes for a person, namely fall or not fall.

Step 2: Draw the first level of the tree diagram



This tree diagram shows the possible outcomes and probabilities of the status of the floor.

Step 3: Draw the second level of the tree diagram



This tree diagram shows the possible outcomes and probabilities based on whether the floor is very wet, wet or dry. Remember that the sum of the probabilities for any set of branches is 1. Use this as a logical check whenever you are constructing a tree diagram.

Step 4: Compute the probabilities of the various outcomes

We can calculate the probability of each outcome by multiplying the probabilities along the path from the start of the tree to the end of the branch containing the desired outcome.

- $P(\text{very wet and fall}) = 0{,}1 \times 0{,}6 = 0{,}06$
- $P(\text{very wet and not fall}) = 0{,}1 \times 0{,}4 = 0{,}04$
- $P(\text{wet and fall}) = 0{,}3 \times 0{,}36 = 0{,}108$
- $P(\text{wet and not fall}) = 0{,}3 \times 0{,}64 = 0{,}192$
- $P(\text{dry and fall}) = 0{,}6 \times 0{,}12 = 0{,}072$
- $P(\text{dry and not fall}) = 0{,}6 \times 0{,}88 = 0{,}528$

Step 5: Compute the probability of falling or not falling

We can calculate the probability of falling or not falling by adding the probabilities of all the desired outcomes.

- $P(\text{fall}) = 0{,}06 + 0{,}108 + 0{,}072 = 0{,}24$
- $P(\text{not fall}) = 0{,}04 + 0{,}192 + 0{,}528 = 0{,}76$

Therefore the probability of falling on a given day is 24% and the probability of not falling is 76%.

Step 6: Determine whether the floor being dry and a person falling are independent events

Logically, it appears that these events are dependent but the question asked us to prove this using a calculation. We can do this using the rule for independent events:

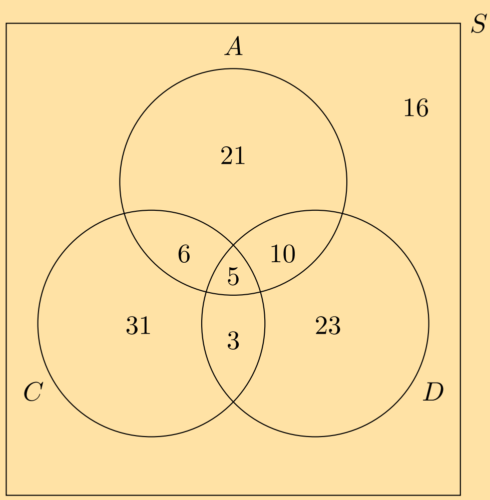
$$P(A \text{ and } B) = P(A) \times P(B)$$

$$\begin{aligned} P(\text{dry and fall}) &= 0{,}072 \\ P(\text{dry}) \times P(\text{fall}) &= 0{,}6 \times 0{,}24 \\ &= 0{,}144 \\ \text{Therefore } P(\text{dry and fall}) &\neq P(\text{dry}) \times P(\text{fall}) \end{aligned}$$

Therefore we can conclude that the floor being dry and a person falling are dependent events.

Exercise 10 – 2: Venn and tree diagrams

1. A survey was done on a group of learners to determine which type of TV shows they enjoy: action, comedy or drama. Let A = action, C = comedy and D = drama. The results of the survey are shown in the Venn diagram below.



Study the Venn diagram and determine the following:

a) the total number of learners surveyed
b) the number of learners who do not enjoy any of the mentioned types of TV shows
c) $P(\text{not } A)$
d) $P(A \text{ or } D)$
e) $P(A \text{ and } C \text{ and } D)$
f) $P(\text{not } (A \text{ and } D))$
g) $P(A \text{ or not } C)$
h) $P(\text{not } (A \text{ or } C))$
i) the probability of a learner enjoying at least two types of TV shows
j) Describe, in words, the meaning of each of the questions c) to h) in the context of this problem.

2. At Thandokulu Secondary School, there are are 320 learners in Grade 12, 270 of whom take one or more of Mathematics, History and Economics. The subject choice is such that everybody who takes Physical Sciences must also take Mathematics and nobody who takes Physical Sciences can take History or Economics. The following is known about the number of learners who take these subjects:

- 70 take History
- 50 take Economics
- 120 take Physical Sciences
- 200 take Mathematics
- 20 take Mathematics and History
- 10 take History and Economics
- 25 take Mathematics and Economics
- x learners take Mathematics and History and Economics

a) Represent the information above in a Venn diagram. Let Mathematics be M, History be H, Physical Sciences be P and Economics be E.
b) Determine the number of learners, x, who take Mathematics, History and Economics.
c) Determine $P(\text{not } (M \text{ or } H \text{ or } E))$ and state in words what your answer means.
d) Determine the probability that a learner takes at least two of these subjects.

3. A group of 200 people were asked about the kind of sports they watch on television. The information collected is given below:

- 180 watch rugby, cricket or soccer
- 5 watch rugby, cricket and soccer
- 25 watch rugby and cricket
- 30 watch rugby and soccer
- 100 watch rugby
- 65 watch cricket
- 80 watch soccer
- x watch cricket and soccer but not rugby

a) Represent all the information in a Venn diagram. Let rugby watchers = R, cricket watchers = C and soccer watchers = F.
b) Find the value of x.
c) Determine $P(\text{not } (R \text{ or } F \text{ or } C))$
d) Determine $P(R \text{ or } F \text{ or not } C)$
e) Are watching cricket and watching rugby independent events? Confirm your answer using a calculation.

4. There are 25 boys and 15 girls in the English class. Each lesson, two learners are randomly chosen to do an oral.

a) Represent the composition of the English class in a tree diagram. Include all possible outcomes and probabilities.
b) Calculate the probability that a boy and a girl are chosen to do an oral in any particular lesson.
c) Calculate the probability that at least one of the learners chosen to do an oral in any particular lesson is male.
d) Are the events picking a boy first and picking a girl second independent or dependent? Justify your answer with a calculation.

5. During July in Cape Town, the probability that it will rain on a randomly chosen day is $\frac{4}{5}$. Gladys either walks to school or gets a ride with her parents in their car. If it rains, the probability that Gladys' parents will take her to school by car is $\frac{5}{6}$. If it does not rain, the probability that Gladys' parents will take her to school by car is $\frac{1}{12}$.

a) Represent the above information in a tree diagram. On your diagram show all the possible outcomes and respective probabilities.
b) What is the probability that it is a rainy day and Gladys walks to school?
c) What is the probability that Gladys' parents take her to school by car?

6. There are two types of property burglaries: burglary of private residences and burglary of business premises. In Metropolis, burglary of a private residence is four times as likely as that of a business premises. The following statistics for each type of burglary were obtained from the Metropolis Police Department:

Burglary of private residences

Following a burglary:

- 25% of criminals are arrested within 48 hours.
- 15% of criminals are arrested after 48 hours.
- 60% of criminals are never arrested for that particular burglary.

Burglary of business premises

Following a burglary:

- 36% of criminals are arrested within 48 hours.
- 54% of criminals are arrested after 48 hours.
- 10% of criminals are never arrested for that particular burglary.

a) Represent the information above in a tree diagram, showing all outcomes and respective probabilities.
b) Calculate the probability that a private home is burgled and nobody is arrested.
c) Calculate the probability that burglars of private homes and business premises are arrested.

d) Use your answer in the previous question to construct a tree diagram to calculate the probability that a burglar is arrested after at most three burglaries.

e) Determine after how many burglaries a burglar has at least a

i. 90% chance of being arrested.

ii. 99% chance of being arrested.

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29GF 2. 29GG 3. 29GH 4. 29GJ 5. 29GK 6. 29GM

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Worked example 9: Two-way contingency tables

QUESTION

The table below shows the results of testing two different treatments on 240 fruit trees which have a disease causing the trees to die. Treatment A involves the careful removal of infected branches and treatment B involves removing infected branches as well as spraying the tree with antibiotic.

	Tree dies within 4 years	Tree lives >4 years	Total
Treatment A	70	50	
Treatment B			
Total	90	150	

1. Fill in the missing values on the table.
2. What is the probability the a tree received treatment B?
3. What is the probability that a tree will live beyond 4 years?
4. What is the probability that a tree is given treatment B and lives beyond 4 years?
5. Of the trees who were given treatment B, what is the probability that a tree lives beyond 4 years?
6. Are a tree given treatment B and living beyond 4 years independent events? Justify your answer with a calculation.

SOLUTION

Step 1: Complete the contingency table

Since each column has to sum to its total, we can work out the number of trees which fall into each category for treatments A and B. Then, we can add each row to get the totals on the right hand side of the table.

	Tree dies within 4 years	Tree lives >4 years	Total
Treatment A	70	50	120
Treatment B	20	100	120
Total	90	150	240

Step 2: Compute the required probabilities

For the second question, we need to determine the probability that a tree receives treatment B. This means that we do not include treatment A in this calculation. So, the probability that treatment B is given to a tree is the ratio between the number of trees that received treatment B and the total number of trees.

$$\begin{aligned} P(\text{treatment B}) &= \frac{n(\text{treatment B})}{n(\text{total trees})} \\ &= \frac{120}{240} = \frac{1}{2} \end{aligned}$$

Similarly for the third question, the probability that a tree will live beyond 4 years:

$$\begin{aligned} P(\text{lives beyond 4 years}) &= \frac{n(\text{lives } > 4 \text{ years})}{n(\text{total trees})} \\ &= \frac{150}{240} = \frac{5}{8} \end{aligned}$$

In the fourth question, we need to determine the probability that a tree receives treatment B and lives beyond 4 years.

$$\begin{aligned} P(\text{treatment B and lives } > 4 \text{ years}) &= \frac{n(\text{treatment B and lives } > 4 \text{ years})}{n(\text{total trees})} \\ &= \frac{100}{240} = \frac{5}{12} \end{aligned}$$

In the fifth question, there is a subtle change from the fourth question. Here, we need to determine the probability that of the trees which received treatment B, a tree lives beyond 4 years. This means we are only concerned with those trees which received treatment B. We no longer need to care about the trees given treatment A, so our denominator needs to be adjusted accordingly.

$$\begin{aligned} P(\text{lives } > 4 \text{ years having received treatment B}) &= \frac{n(\text{treatment B and lives } > 4 \text{ years})}{n(\text{total treatment B})} \\ &= \frac{100}{120} = \frac{5}{6} \end{aligned}$$

Step 3: Independence

We need to determine whether a tree given treatment B and living beyond 4 years are dependent or independent events. According to the definition, two events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

$$P(\text{treatment B}) \times P(\text{lives } > 4 \text{ years}) = \frac{1}{2} \times \frac{5}{8} = \frac{5}{16}$$

$$P(\text{treatment B and lives } > 4 \text{ years}) = \frac{5}{12}$$

From these probabilities we can see that

$$P(\text{treatment B and lives } > 4 \text{ years}) \neq P(\text{treatment B}) \times P(\text{lives } > 4 \text{ years})$$

and therefore the treatment of a tree with treatment B and living beyond 4 years are dependent events.

Exercise 10 – 3: Contingency tables

1. A number of drivers were asked about the number of motor vehicle accidents they were involved in over the last 10 years. Part of the data collected is shown in the table below.

	$\leq$ 2 accidents	$>$ 2 accidents	Total
Female	210	90	
Male			
Total	350	150	500

   a) What are the variables investigated here and what is the purpose of the research?
   b) Complete the table.
   c) Determine whether gender and number of accidents are independent using a calculation.

2. Researchers conducted a study to test how effective a certain inoculation is at preventing malaria. Part of their data is shown below:

	Malaria	No malaria	Total
Male	a	b	216
Female	c	d	648
Total	108	756	864

   a) Calculate the probability that a randomly selected study participant will be female.
   b) Calculate the probability that a randomly selected study participant will have malaria.
   c) If being female and having malaria are independent events, calculate the value c.
   d) Using the value of c, fill in the missing values on the table.

3. The reaction time of 400 drivers during an emergency stop was tested. Within the study cohort (the group of people being studied), the probability that a driver chosen at random was 40 years old or younger is 0,3 and the probability of a reaction time less than 1,5 seconds is 0,7.

   a) Calculate the number of drivers who are 40 years old or younger.
   b) Calculate the number of drivers who have a reaction time of less than 1,5 seconds.
   c) If age and reaction time are independent events, calculate the number of drivers 40 years old and younger with a reaction time of less than 1,5 seconds.
   d) Complete the table below.

	Reaction time $<$ 1,5 s	Reaction time $>$ 1,5 s	Total
$\leq$40 years			
$>$ 40 years			
Total			400

4. A new treatment for influenza (the flu) was tested on a number of patients to determine if it was better than a placebo (a pill with no therapeutic value). The table below shows the results three days after treatment:

	Flu	No flu	Total
Placebo	228	60	
Treatment			
Total	240	312	

   a) Complete the table.
   b) Calculate the probability of a patient receiving the treatment.
   c) Calculate the probability of a patient having no flu after three days.
   d) Calculate the probability of a patient receiving the treatment and having no flu after three days.
   e) Using a calculation, determine whether a patient receiving the treatment and having no flu after three days are dependent or independent events.
   f) Calculate the probability that a patient receiving treatment will have no flu after three days.
   g) Calculate the probability that a patient receiving a placebo will have no flu after three days.
   h) Comparing you answers in f) and g), would you recommend the use of the new treatment for patients suffering from influenza?
   i) A hospital is trying to decide whether to purchase the new treatment. The new treatment is much more expensive than the old treatment. According to the hospital records, of the $72\ 024$ flu patients that have been treated with the old treatment, only 3200 still had the flu three days after treatment.
      - Construct a two-way contingency table comparing the old treatment data with the new treatment data.
      - Using the data from your table, advise the hospital whether to purchase the new treatment or not.

5. Human immunodeficiency virus (HIV) affects 10% of the South African population.
   a) If a test for HIV has a $99{,}9\%$ accuracy rate (i.e. $99{,}9\%$ of the time the test is correct, $0{,}1\%$ of the time, the test returns a false result), draw a two-way contingency table showing the expected results if $10\ 000$ of the general population are tested.
   b) Calculate the probability that a person who tests positive for HIV does not have the disease, correct to two decimal places.
   c) In practice, a person who tests positive for HIV is always tested a second time. Calculate the probability that an HIV-negative person will test positive after two tests, correct to four decimal places.

6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29GN 2. 29GP 3. 29GQ 4. 29GR 5. 29GS

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10.4 The fundamental counting principle EMCJZ

Mathematics began with counting. Initially, fingers, beans and buttons were used to help with counting, but these are only practical for small numbers. What happens when a large number of items must be counted?

This section focuses on how to use mathematical techniques to count different assortments of items.

Introduction EMCK2

An important aspect of probability theory is the ability to determine the total number of possible outcomes when multiple events are considered.

For example, what is the total number of possible outcomes when a die is rolled and then a coin is tossed? The roll of a die has six possible outcomes ($1; 2; 3; 4; 5$ or 6) and the toss of a coin, 2 outcomes (heads or tails). The sample space (total possible outcomes) can be represented as follows:

$$S = \left\{ \begin{array}{cccccc} (1;H); & (2;H); & (3;H); & (4;H); & (5;H); & (6;H); \\ (1;T); & (2;T); & (3;T); & (4;T); & (5;T); & (6;T) \end{array} \right\}$$

Therefore there are 12 possible outcomes.

The use of lists, tables and tree diagrams is only feasible for events with a few outcomes. When the number of outcomes grows, it is not practical to list the different possibilities and the fundamental counting principle is used instead.

DEFINITION: *The fundamental counting principle*

The fundamental counting principle states that if there are $n(A)$ outcomes in event A and $n(B)$ outcomes in event B, then there are $n(A) \times n(B)$ outcomes in event A and event B combined.

If we apply this principle to our previous example, we can easily calculate the number of possible outcomes by multiplying the number of possible die rolls with the number of outcomes of tossing a coin: $6 \times 2 = 12$ outcomes. This allows us to formulate the following:

If there n_1 possible outcomes for event A and n_2 outcomes for event B, then the total possible number of outcomes for both events is $n_1 \times n_2$

This can be generalised to k events, where k is the number of events. The total number of outcomes for k events is:

$$n_1 \times n_2 \times n_3 \times \cdots \times n_k$$

NOTE:
The order in which the experiments are done does not affect the total number of possible outcomes.

Worked example 10: Choices without repetition

QUESTION

A take-away has a 4-piece lunch special which consists of a sandwich, soup, dessert and drink for R $25{,}00$. They offer the following choices for:

Sandwich: chicken mayonnaise, cheese and tomato, tuna mayonnaise, ham and lettuce

Soup: tomato, chicken noodle, vegetable

Dessert: ice-cream, piece of cake

Drink: tea, coffee, Coke, Fanta, Sprite

How many possible meals are there?

SOLUTION

Step 1: Determine how many parts to the meal there are

There are 4 parts: sandwich, soup, dessert and drink.

Step 2: Identify how many choices there are for each part

Meal component	Sandwich	Soup	Dessert	Drink
Number of choices	4	3	2	5

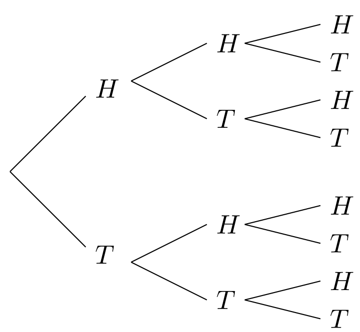
Step 3: Use the fundamental counting principle to determine how many different meals are possible

$4 \times 3 \times 2 \times 5 = 120$

So there are 120 possible meals.

In the previous example, there were a different number of options for each choice. But what happens when the number of choices is unchanged each time you choose?

For example, if a coin is flipped three times, what is the total number of different results? Each time a coin is flipped, there are two possible outcomes, namely heads or tails. The coin is flipped 3 times. We can use a tree diagram to determine the total number of possible outcomes:



From the tree diagram, we can see that there is a total of 8 different possible outcomes.

Drawing a tree diagram is possible for three different coin flips, but as soon as the number of events increases, the total number of possible outcomes increases to the point where drawing a tree diagram is impractical.

For example, think about what a tree diagram would look like if we were to flip a coin six times. In this case, using the fundamental counting principle is a far easier option. We know that each time a coin is flipped that there are two possible outcomes. So if we flip a coin six times, the total number of possible outcomes is equivalent to multiplying 2 by itself six times:

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$$

Another example is if you have the letters A, B, C, and D and you wish to discover the number of ways of arranging them in three-letter patterns if repetition is allowed, such as ABA, DCA, BBB etc. You will find that there are 64 ways. This is because for the first letter of the pattern, you can choose any of the four available letters, for the second letter of the pattern, you can choose any of the four letters, and for the final letter of the pattern you can choose any of the four letters. Multiplying the number of available choices for each letter in the pattern gives the total available arrangements of letters:

$$4 \times 4 \times 4 = 4^3 = 64$$

This allows us to formulate the following:

When you have n objects to choose from and you choose from them r times, then the total number of possibilities is

$$n \times n \times n \ldots \times n \ \ (r \text{ times}) = n^r$$

Worked example 11: Choices with repetition

QUESTION

A school plays a series of 6 soccer matches. For each match there are 3 possibilities: a win, a draw or a loss. How many possible results are there for the series?

SOLUTION

Step 1: Determine how many outcomes you have to choose from for each event

There are 3 outcomes for each match: win, draw or lose.

Step 2: Determine the number of events

There are 6 matches, therefore the number of events is 6.

Step 3: Determine the total number of possible outcomes

There are 3 possible outcomes for each of the 6 events. Therefore, the total number of possible outcomes for the series of matches is

$$3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^6 = 729$$

Exercise 10 – 4: Number of possible outcomes if repetition is allowed

1. Tarryn has five different skirts, four different tops and three pairs of shoes. Assuming that all the colours complement each other, how many different outfits can she put together?

2. In a multiple-choice question paper of 20 questions the answers can be A, B, C or D. How many different ways are there of answering the question paper?

3. A debit card requires a five digit personal identification number (PIN) consisting of digits from 0 to 9. The digits may be repeated. How many possible PINs are there?

4. The province of Gauteng ran out of unique number plates in 2010. Prior to 2010, the number plates were formulated using the style LLLDDDGP, where L is any letter of the alphabet excluding vowels and Q, and D is a digit between 0 and 9. The new style the Gauteng government introduced is LLDDLLGP. How many more possible number plates are there using the new style when compared to the old style?

5. A gift basket consists of one CD, one book, one box of sweets, one packet of nuts and one bottle of fruit juice. The person who makes the gift basket can choose from five different CDs, eight different books, three different boxes of sweets, four kinds of nuts and six flavours of fruit juice. How many different gift baskets can be produced?

6. The code for a safe is of the form XXXXYYY where X is any number from 0 to 9 and Y represents the letters of the alphabet. How many codes are possible for each of the following cases:

   a) the digits and letters of the alphabet can be repeated.
   b) the digits and letters of the alphabet can be repeated, but the code may not contain a zero or any of the vowels in the alphabet.
   c) the digits and letters of the alphabet can be repeated, but the digits may only be prime numbers and the letters X, Y and Z are excluded from the code.

7. A restaurant offers four choices of starter, eight choices for the main meal and six choices for dessert. A customer can choose to eat just one course, two different courses or all three courses. Assuming that all courses are available, how many different meal options does the restaurant offer?

8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29GV 2. 29GW 3. 29GX 4. 29GY 5. 29GZ 6. 29H2
7. 29H3

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10.5 Factorial notation EMCK3

Worked example 12: The arrangement of outcomes without repetition

QUESTION

Eight athletes take part in a 400 m race. In how many different ways can all 8 places in the race be arranged?

SOLUTION

Any of the 8 athletes can come first in the race. Now there are only 7 athletes left to be second, because an athlete cannot be both second and first in the race. After second place, there are only 6 athletes left for the third place, 5 athletes for the fourth place, 4 athletes for the fifth place, 3 athletes for the sixth place, 2 athletes for the seventh place and 1 athlete for the eighth place. Therefore the number of ways that the athletes can be ordered is as follows:

$$8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40\ 320$$

As in the example above, it is a common occurrence in counting problems that the outcome of the first event reduces the number of possible outcomes for the second event by exactly 1, and the outcome of the second event reduces the possible outcomes for the third event by 1 more, etc.

As this sort of problem occurs so frequently, we have a special notation to represent the answer. For an integer, n, the notation $n!$ (read n factorial) represents: $n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$

This allows us to formulate the following:

The total number of possible arrangements of n different objects is

$$n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1 = n!$$

with the following definition: 0! = 1.

Worked example 13: Factorial notation

QUESTION

1. Determine $12!$
2. Show that $\frac{8!}{4!} = 8 \times 7 \times 6 \times 5$
3. Show that $\frac{n!}{(n-1)!} = n$

SOLUTION

1. We know from the definition of a factorial that $12! = 12 \times 11 \times 10 \times \ldots \times 3 \times 2 \times 1$. However, it can be quite tedious to work this out by calculating each multiplication step on paper or typing each step into your calculator. Fortunately, there is a button on your calculator which makes this much easier. To use your calculator to work out the factorial of a number:
   - Input the number.
   - Press SHIFT on your CASIO or 2ndF on your SHARP calculator.
   - Then press $x!$ on your CASIO or $n!$ on your SHARP calculator.
   - Finally, press equals to calculate the answer.

   If we follow these steps for $12!$, we get the answer $479\ 001\ 600$.
2. Expand the factorial notation:
$$\frac{8!}{4!} = \frac{8 \times 7 \times 6 \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}}{\cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}} = 8 \times 7 \times 6 \times 5 = \text{RHS}$$
3. Expand the factorial notation:
$$\frac{n!}{(n-1)!} = \frac{n \times \cancel{(n-1)} \times \cancel{(n-2)} \times \cancel{(n-3)} \times \ldots \times \cancel{3} \times \cancel{2} \times \cancel{1}}{\cancel{(n-1)} \times \cancel{(n-2)} \times \cancel{(n-3)} \times \ldots \times \cancel{3} \times \cancel{2} \times \cancel{1}} = n$$
   If $n = 1$, we get $\frac{1!}{0!}$. This is a special case. Both $1!$ and $0! = 1$, therefore $\frac{1!}{0!} = 1$ so our identity still holds.

Exercise 10 – 5: Factorial notation

1. Work out the following without using a calculator:

   a) $3!$

   b) $6!$

   c) $2!3!$

   d) $8!$

   e) $\dfrac{6!}{3!}$

   f) $6! + 4! - 3!$

   g) $\dfrac{6! - 2!}{2!}$

   h) $\dfrac{2! + 3!}{5!}$

   i) $\dfrac{2! + 3! - 5!}{3! - 2!}$

   j) $(3!)^3$

   k) $\dfrac{3! \times 4!}{2!}$

2. Calculate the following using a calculator:

   a) $\dfrac{12!}{2!}$

   b) $\dfrac{10!}{20!}$

   c) $\dfrac{10! + 12!}{5! + 6!}$

   d) $5!(2! + 3!)$

   e) $(4!)^2(3!)^2$

3. Show that the following is true:

a) $\dfrac{n!}{(n-2)!} = n^2 - n$

b) $\dfrac{(n-1)!}{n!} = \dfrac{1}{n}$

c) $\dfrac{(n-2)!}{(n-1)!} = \dfrac{1}{n-1}$ for $n > 1$

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 29H4	1b. 29H5	1c. 29H6	1d. 29H7	1e. 29H8	1f. 29H9
1g. 29HB	1h. 29HC	1i. 29HD	1j. 29HF	1k. 29HG	2a. 29HH
2b. 29HJ	2c. 29HK	2d. 29HM	2e. 29HN	3a. 29HP	3b. 29HQ
3c. 29HR					

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10.6 Application to counting problems EMCK4

Worked example 14: Further arrangement of outcomes without repetition

QUESTION

Eight athletes take part in a 400 m race. In how many different ways can the first three places be arranged?

SOLUTION

Eight different athletes can occupy the first 3 places. For the first place, there are 8 different choices. For the second place there are 7 different choices and for the third place there are 6 different choices. Therefore 8 different athletes can occupy the first three places in:

$$8 \times 7 \times 6 = 336 \text{ ways}$$

Worked example 15: Arrangement of objects with constraints

QUESTION

In how many ways can seven boys of different ages be seated on a bench if:

1. the youngest boy sits next to the oldest boy?
2. the youngest and the oldest boys must not sit next to each other?

SOLUTION

1. This question is a little different to the previous problems of arrangements without repetition. In this question, we have the constraint that the youngest boy and the oldest boy must sit together. The easiest way to think about this, is to see each set of objects which have to be together as a single object to arrange.

   If we let boy $= B$ and let the number subscript indicate order of age, we can view the objects to arrange as follows:

$$\begin{array}{cccccc} (B_1;B_7); & (B_2); & (B_3); & (B_4); & (B_5); & (B_6) \\ 1 & 2 & 3 & 4 & 5 & 6 \end{array}$$

   If the the youngest and oldest boys are treated as a single object, there are six different objects to arrange so there are $6!$ different arrangements. However, the youngest and oldest boys can be arranged in $2!$ different ways and still be together:

$$(B_1;B_7) \quad \text{or} \quad (B_7;B_1)$$

   Therefore there are:

$$6! \times 2! = 1440 \text{ ways for the boys to be seated}$$

2. The arrangements where the youngest and oldest must not sit together is the total number of arrangements minus the number of arrangements where the oldest and youngest sit together. Therefore, there are:

$$7! - 1440 = 3600 \text{ ways for the boys to be seated}$$

Exercise 10 – 6: Number of choices in a row

1. How many different possible outcomes are there for a swimming event with six competitors?
2. How many different possible outcomes are there for the gold (1st), silver (2nd) and bronze (3rd) medals in a swimming event with six competitors?
3. Susan wants to visit her friends in Pretoria, Johannesburg, Phalaborwa, East London and Port Elizabeth. In how many different ways can the visits be arranged?
4. A head boy, a deputy head boy, a head girl and a deputy head girl must be chosen out of a student council consisting of 18 girls and 18 boys. In how many ways can they be chosen?
5. Twenty different people enter a golf competition. Only the first six of them can win prizes. In how many different ways can the prizes be won?
6. Three letters of the word 'EMPTY' are arranged in a row. How many different arrangements are possible?
7. Pool balls are numbered from 1 to 15. You have only one set of pool balls. In how many different ways can you arrange:
   a) all 15 balls. Write your answer in scientific notation, rounding off to two decimal places.
   b) four of the 15 balls.

8. The captains of all the sports teams in a school have to stand next to each other for a photograph. The school sports programme offers rugby, cricket, hockey, soccer, netball and tennis.

    a) In how many different orders can they stand in the photograph?
    b) In how many different orders can they stand in the photograph if the rugby captain stands on the extreme left and the cricket captain stands on the extreme right?
    c) In how many different orders can they stand if the rugby captain, netball captain and cricket captain must stand next to each other?

9. How many three-digit numbers can be made from the digits 1 to 6 if:

    a) repetition is not allowed?
    b) repetition is allowed?

10. There are two different red books and three different blue books on a shelf.

    a) In how many different ways can these books be arranged?
    b) If you want the red books to be together, in how many different ways can the books be arranged?
    c) If you want all the red books to be together and all the blue books to be together, in how many different ways can the books be arranged?

11. There are two different Mathematics books, three different Natural Sciences books, two different Life Sciences books and four different Accounting books on a shelf. In how many different ways can they be arranged if:

    a) the order does not matter?
    b) all the books of the same subject stand together?
    c) the two Mathematics books stand first?
    d) the Accounting books stand next to each other?

12. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29HS 2. 29HT 3. 29HV 4. 29HW 5. 29HX 6. 29HY
7. 29HZ 8. 29J2 9. 29J3 10. 29J4 11. 29J5

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Worked example 16: Arrangement of letters

QUESTION

If you take the word, 'OMO', how many letter arrangements can we make if:

1. we consider the two O's as different letters?
2. we consider the two O's as identical letters?

SOLUTION

1. Since we consider the two O's as different letters, write the first O as O_1 and the second O_2. The different letter arrangements are as follows:

$$\begin{array}{ccc} O_1MO_2 & MO_1O_2 & O_1O_2M \\ O_2MO_1 & MO_2O_1 & O_2O_1M \end{array}$$

   We can see from writing out all the arrangements that there are 6 different ways for the letters to be arranged. This is not practical if there are a large number of letters. Instead, we can work this out more easily using the fundamental counting principle.

   Using the fundamental counting principle, as there are 3 letters in the word OMO if we treat each O as a separate letter, there are $3! = 6$ different arrangements.

2. If we consider the two O's as identical letters, only 3 arrangements are possible:

   OMO MOO OOM

   We can also work this out using a modified version of the previous identity. We know that if we treat each letter as different, the number of arrangements is $3!$. However, when we have duplicate letters, we have to remove the identical arrangements of these letter from our final answer. So we divide by the factorial of the number of times a letter is repeated.

   In this example, O appears twice so we divide $3!$ by $2!$:

$$\text{number of arrangements} = \frac{3!}{2!} = 3$$

Worked example 17: The number of letter arrangements for a longer word

QUESTION

If you take the word 'BASSOON', how many letter arrangements can you make if:

1. repeated letters are treated as different?
2. repeated letters are treated as identical?
3. the word starts with an O and repeated letters are treated as identical?
4. the word starts and ends with the same letter and repeated letters are treated as identical?

SOLUTION

1. There are 7 letters in the word 'BASSOON'. If we treat each letter as a different letter, there are $7! = 5040$ arrangements.
2. If repeated letters are treated as identical characters, there are two S's and two O's. This is similar to the previous worked example except now we have more than one letter repeated. When more than one letter is repeated, we have to divide the total number of possible arrangements by the product of the factorials of the number of times each letter is repeated.

$$\text{Number of arrangements} = \frac{7!}{2! \times 2!} = 1260 \text{ arrangements}$$

3. If the word starts with an 'O', there are still 6 letters left of which two are S's.

$$\text{Number of arrangements} = \frac{6!}{2!} = 360 \text{ arrangements}$$

4. If the word starts and ends with the same letter, there are two possibilities:
   - S - - - - - S with the letters in between consisting of 'B','A','O','O' and 'N'. Therefore:

$$\text{Number of arrangements} = \frac{5!}{2!} = 60 \text{ arrangements}$$

   - O - - - - - O with the letters in between consisting of 'B','A','S','S' and 'N'. Therefore:

$$\text{Number of arrangements} = \frac{5!}{2!} = 60 \text{ arrangements}$$

   Therefore, the total number of arrangements $= 60 + 60 = 120$.

This allows us to formulate the following:

For a set of n objects, of which n_1 are the same, n_2 are the same $\ldots$, n_k are the same, the number of arrangements $= \dfrac{n!}{n_1! \times n_2! \ldots n_k!}$

Exercise 10 – 7: Number of arrangements of sets containing alike objects

1. You have the word 'EXCELLENT'.
   a) If the repeated letters are regarded as different letters, how many letter arrangements are possible?
   b) If the repeated letters are regarded as identical, how many letter arrangements are possible?
   c) If the first and last letters are identical, how many letter arrangements are there?
   d) How many letter arrangements can be made if the arrangement starts with an L?
   e) How many letter arrangements are possible if the word ends in a T?
2. You have the word 'ASSESSMENT'.
   a) If the repeated letters are regarded as different letters, how many letter arrangements are possible?
   b) If the repeated letters are regarded as identical, how many letter arrangements are possible?
   c) If the first and last letters are identical, how many letter arrangements are there?
   d) How many letter arrangements can be made if the arrangement starts with a vowel?
   e) How many letter arrangements are possible if all the S's are at the beginning of the word?

3. On a piano the white keys represent the following notes: C, D, E, F, G, A, B. How many tunes, seven notes in length, can be composed with these notes if:

   a) a note can be played only once?
   b) the notes can be repeated?
   c) the notes can be repeated and the tune begins and ends with a D?
   d) the tune consists of 3 D's, 2 B's and 2 A's.

4. There are three black beads and four white beads in a row. In how many ways can the beads be arranged if:

   a) same-coloured beads are treated as different beads?
   b) same-coloured beads are treated as identical beads?

5. There are eight balls on a table. Some are white and some are red. The white balls are all identical and the red balls are all identical. The balls are removed one at a time. In how many different orders can the balls be removed if:

   a) seven of the balls are red?
   b) three of the balls are red?
   c) there are four of each colour?

6. How many four-digit numbers can be formed with the digits 3, 4, 6 and 7 if:

   a) there can be repetition?
   b) each digit can only be used once?
   c) if the number is odd and repetition is allowed?

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29J6 2. 29J7 3. 29J8 4. 29J9 5. 29JB 6. 29JC

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10.7 Application to probability problems EMCK5

When needing to determine the probability that an event occurs, and the total number of arrangements of the sample space, S, and the total number arrangements for the event, E, are very large, the techniques used earlier in this chapter may no longer be practical. In this case, the probability may be determined using the fundamental counting principle. The probability of the event, E, is the total number of arrangements of the event divided by the total number of arrangements of the sample space or $\frac{n(E)}{n(S)}$.

Worked example 18: Personal Identification Numbers (PINs)

QUESTION

Every client of a certain bank has a personal identification number (PIN) which consists of four randomly chosen digits from 0 to 9.

1. How many PINs can be made if digits can be repeated?
2. How many PINs can be made if digits cannot be repeated?
3. If a PIN is made by selecting four digits at random, and digits can be repeated, what is the probability that the PIN contains at least one eight?
4. If a PIN is made by selecting four digits at random, and digits cannot be repeated, what is the probability that the PIN contains at least one eight?

SOLUTION

1. If digits can be repeated: you have 10 digits to choose from and you have to choose four times, therefore the number of possible PINs $= 10^4 = 10\,000$.

2. If digits cannot be repeated: you have 10 digits for your first choice, nine for your second, eight for your third and seven for your fourth. Therefore,

$$\text{the number of possible PINs } = 10 \times 9 \times 8 \times 7 = 5040$$

3. Let B be the event that at least one eight is chosen. Therefore the complement of B is the event that no eights are chosen.

   If no eights are chosen, there are only nine digits to choose from. Therefore, $n(\text{not } B) = 9^4 = 6561$.

   The total number of arrangements in the set, as calculated in Question 1, is $10\,000$. Therefore:

$$\begin{aligned} P(B) &= 1 - P(\text{not } B) \\ &= 1 - \frac{n(\text{not } B)}{n(S)} \\ &= 1 - \frac{6561}{10\,000} \\ &= 0{,}3439 \end{aligned}$$

4. Let B be the event that at least one eight is chosen. Therefore the complement of B, is the event that no eights are chosen.

   If no eights are chosen, there are only 9 then 8 then 7 then 6 digits to choose from as we cannot repeat a digit once it is chosen. Therefore,

$$n(\text{not } B) = 9 \times 8 \times 7 \times 6 = 3024$$

   The total number of arrangements in the set, as calculated in Question 1, is $10\,000$. Therefore:

$$\begin{aligned} P(B) &= 1 - P(\text{not } B) \\ & 1 - \frac{n(\text{not } B)}{n(S)} \\ &= 1 - \frac{3024}{10\,000} \\ &= 0{,}6976 \end{aligned}$$

Worked example 19: Number plates

QUESTION

The number plate on a car consists of any 3 letters of the alphabet (excluding the vowels and 'Q'), followed by any 3 digits (0 to 9). For a car chosen at random, what is the probability that the number plate starts with a 'Y' and ends with an odd digit?

SOLUTION

Step 1: Identify what events are counted

The number plate starts with a 'Y', so there is only 1 option for the first letter, and ends with an odd digit, so there are 5 options for the last digit $(1; 3; 5; 7; 9)$.

Step 2: Find the number of events

Use the counting principle. For the second and third letters, there are 20 possibilities (26 letters in the alphabet, minus 5 vowels and 'Q'). There are 10 possibilities for the first and second digits.

$$\text{Number of events} = 1 \times 20 \times 20 \times 10 \times 10 \times 5 = 200\ 000$$

Step 3: Find the total number of possible number plates

Use the counting principle. This time, the first letter and last digit can be anything.

$$\text{Total number of choices} = 20 \times 20 \times 20 \times 10 \times 10 \times 10 = 8\ 000\ 000$$

Step 4: Calculate the probability

The probability is the number of outcomes in the event, divided by the total number of outcomes in the sample space.

$$\text{Probability} = \frac{200\ 000}{8\ 000\ 000} = \frac{1}{40} = 0{,}025$$

Worked example 20: Probability of word arrangements

QUESTION

Refer to worked example 16 for context. If you take the word, 'BASSOON' and you randomly rearrange the letters, what is the probability that the word starts and ends with the same letter if repeated letters are treated as identical?

SOLUTION

If the word starts and ends with the same letter, there are a total number of 120 possible arrangements (from worked example 16). Let this event $= A$.

The total number of possible arrangements if repeated letters are treated as identical $= 1260$ (from worked example 16).

Therefore, the probability of an arrangement beginning and ending with the same letter

$$= \frac{n(A)}{n(S)} = \frac{120}{1260} = 0{,}1$$

Exercise 10 – 8: Solving probability problems using the fundamental counting principle

1. A music group plans a concert tour in South Africa. They will perform in Cape Town, Port Elizabeth, Pretoria, Johannesburg, Bloemfontein, Durban and East London.
   a) In how many different orders can they plan their tour if there are no restrictions?
   b) In how many different orders can they plan their tour if their tour begins in Cape Town and ends in Durban?
   c) If the tour cities are chosen at random, what is the probability that their performances in Cape Town, Port Elizabeth, Durban and East London happen consecutively? Give your answer correct to 3 decimal places.
2. A certain restaurant has the following course options available for a three-course set menu:

STARTERS	MAINS	DESSERTS
Calamari salad	Fried chicken	Ice cream and chocolate sauce
Oysters	Crumbed lamb chops	Strawberries and cream
Fish in garlic sauce	Mutton Bobotie	Malva pudding with custard
	Chicken schnitzel	Pears in brandy sauce
	Vegetable lasagne	
	Chicken nuggets	

   a) How many different set menus are possible?
   b) What is the probability that a set menu includes a chicken course?
3. Eight different pairs of jeans and 5 different shirts hang on a rail.
   a) In how many different ways can the clothes be arranged on the rail?
   b) In how many ways can the clothing be arranged if all the jeans hang together and all the shirts hang together?
   c) What is the probability, correct to three decimal places, of the clothing being arranged on the rail with a shirt at one end and a pair of jeans at the other?
4. A photographer places eight chairs in a row in his studio in order to take a photograph of the debating team. The team consists of three boys and five girls.
   a) In how many ways can the debating team be seated?
   b) What is the probability that a particular boy and a particular girl sit next to each other?
5. If the letters of the word 'COMMITTEE' are randomly arranged, what is the probability that the letter arrangements start and end with the same letter?
6. Four different Mathematics books, three different Economics books and two different Geography books are arranged on a shelf. What is the probability that all the books of the same subject are arranged next to each other?
7. A number plate is made up of three letters of the alphabet (excluding F and S) followed by three digits from 0 to 9. The numbers and letters can be repeated. Calculate the probability that a randomly chosen number plate:
   a) starts with the letter D and ends with the digit 3.
   b) has precisely one D.
   c) contains at least one 5.

8. In the 13-digit identification (ID) numbers of South African citizens:
   - The first six numbers are the birth date of the person in YYMMDD format.
   - The next four digits indicate gender, with 5000 and above being male and 0001 to 4999 being female.
   - The next number is the country ID; 0 is South Africa and 1 is not.
   - The second last number used to be a racial identifier but it is now 8 for everybody.
   - The last number is a control digit, which verifies the rest of the number.

   Assume that the control digit is a randomly generated digit from 0 to 9 and ignore the fact that leap years have an extra day.

   a) Calculate the total number of possible ID numbers.
   b) Calculate the probability that a randomly generated ID number is of a South African male born during the 1980s. Write your answer correct to two decimal places.
9. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29JD 2. 29JF 3. 29JG 4. 29JH 5. 29JJ 6. 29JK
7. 29JM 8. 29JN

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10.8 Summary EMCK6

- The **addition rule** (also called the sum rule) for any 2 events, A and B is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

  This rule relates the probabilities of 2 events with the probabilities of their union and intersection.
- The **addition rule for 2 mutually exclusive events** is

$$P(A \text{ or } B) = P(A) + P(B)$$

  This rule is a special case of the previous rule. Because the events are mutually exclusive, $P(A \text{ and } B) = 0$.
- The **complementary rule** is

$$P(\text{not } A) = 1 - P(A)$$

  This rule is a special case of the previous rule. Since A and $(\text{not } A)$ are complementary, $P(A \text{ or } (\text{not } A)) = 1$.
- The **product rule** for independent events A and B is:

$$P(A \text{ and } B) = P(A) \times P(B)$$

  If two events A and B are **dependent** then:

$$P(A \text{ and } B) \neq P(A) \times P(B)$$

- **Venn diagrams** are used to show how events are related to one another. A Venn diagram can be very helpful when doing calculations with probabilities. In a Venn diagram each event is represented by a shape, often a circle or a rectangle. The region inside the shape represents the outcomes included in the event and the region outside the shape represents the outcomes that are not in the event.

- **Tree diagrams** are useful for organising and visualising the different possible outcomes of a sequence of events. Each branch in the tree shows an outcome of an event, along with the probability of that outcome. For each possible outcome of the first event, we draw a line where we write down the probability of that outcome and the state of the world if that outcome happened. Then, for each possible outcome of the second event we do the same thing. The probability of a sequence of outcomes is calculated as the product of the probabilities along the branches of the sequence.

- **Two-way contingency tables** are a tool for keeping a record of the counts or percentages in a probability problem. Two-way contingency tables are especially helpful for figuring out whether events are dependent or independent.

- The **fundamental counting principle** states that if there are $n(A)$ outcomes for event A and $n(B)$ outcomes for event B, then there are $n(A) \times n(B)$ different possible outcomes for both events.

- When you have n objects to choose from and you choose from them r times, if the number of choices remains the same after each choice, then the total number of possibilities is
$$n \times n \times n \ldots \times n \;\; (r \text{ times}) = n^r$$

- The number of arrangements of n different objects is
$$n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1 = n!$$

- For a set of n objects, of which there are k subsets with repeated objects i.e. n_1 are the same, n_2 are the same, $\ldots, n_k$ are the same, the number of arrangements are
$$\frac{n!}{n_1! \times n_2! \ldots n_k!}$$

Exercise 10 – 9: End of chapter exercises

1. An ATM card has a four-digit PIN. The four digits can be repeated and each of them can be chosen from the digits 0 to 9.
   a) What is the total number of possible PINs?
   b) What is the probability of guessing the first digit correctly?
   c) What is the probability of guessing the second digit correctly?
   d) If your ATM card is stolen, what is the probability, correct to four decimal places, of a thief guessing all four digits correctly on his first guess?
   e) After three incorrect PIN attempts, an ATM card is blocked from being used. If your ATM card is stolen, what is the probability, correct to four decimal places, of a thief blocking the card? Assume the thief enters a different PIN each time.

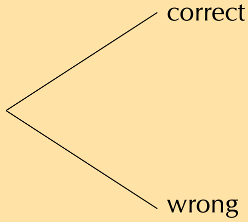
2. The LOTTO rules state the following:

   - Six numbers are drawn from the numbers 1 to 49 - this is called a 'draw'.
   - Numbers are not replaced once drawn, so you cannot have the same number more than once.
   - The order of the drawn numbers does not matter.

   You decide to buy one LOTTO ticket consisting of 6 numbers.

   a) How many different possible LOTTO draws are there? Write your answer in scientific notation, rounding to two digits after the decimal point.

   b) Complete the tree diagram below after the first two LOTTO numbers have been drawn showing the possible outcomes and probabilities of the numbers on your ticket.



   c) What is the probability of getting the first number drawn correctly?

   d) What is the probability of getting the second number drawn correctly if you get the first number correct?

   e) What is the probability of getting the second number drawn correct if you do not get the first number correctly?

   f) What is the probability of getting the second number drawn correct?

   g) What is the probability of getting all 6 LOTTO numbers correct? Write your answer in scientific notation, rounding to two digits after the decimal point.

3. The population statistics of South Africa show that 55% of all babies born are female. Calculate the probability that a couple planning to have children will have a boy followed by a girl and then a boy. Assume that each birth is an independent event. Write your answer as a percentage, correct to two decimal places.

4. Fezile and Vuzi write a Mathematics test. The probability that Fezile will pass the test is $0{,}8$. The probability that Vuzi will pass the test is $0{,}75$. What is the probability that only one of them will pass the test?

5. Landline telephone numbers are 10 digits long. Numbers begin with a zero followed by 9 digits chosen from the digits 0 to 9. Repetitions are allowed.

   a) How many different phone numbers are possible?

   b) The first three digits of a number form an area code. The area code for Cape Town is 021. How many different phone numbers are available in the Cape Town area?

   c) What is the probability of the second digit being an even number?

   d) Ignoring the first digit, what is the probability of a phone number consisting of only odd digits? Write your answer correct to three decimal places.

6. Take the word 'POSSIBILITY'.

   a) In how many way can the letters be arranged if repeated letters are considered identical?

   b) What is the probability that a randomly generated arrangement of the letters will begin with three I's? Write your answer as a fraction.

7. The code to a safe consists of 10 digits chosen from the digits 0 to 9. None of the digits are repeated. Determine the probability of a code where the first digit is odd and none of the first three digits may be a zero. Write your answer as a percentage, correct to two decimal places.

8. Four different red books and three different blue books are to be arranged on a shelf. What is the probability that all the red books and all the blue books stand together on the shelf?

9. The probability that Thandiswa will go dancing on a Saturday night (event D) is $0{,}6$ and the probability that she will go watch a movie is $0{,}3$ (event M). Determine the probability that she will:

    a) go dancing and watch a movie if D and M are independent.
    b) go dancing or watch a movie if D and M are mutually exclusive.
    c) go dancing and watch a movie if $P(D \text{ or } M) = 0{,}7$.
    d) not go dancing or go to a movie if $P(D \text{ and } M) = 0{,}8$.

10. Three boys and four girls sit in a row.

    a) In how many ways can they sit in the row?
    b) What is the probability that they sit in alternating gender positions?

11. The number plate on a car consists of any 3 letters of the alphabet (excluding the vowels, J and Q), followed by any 3 digits from 0 to 9. For a car chosen at random, what is the probability that the number plate starts with a Y and ends with an odd digit? Write your answer as a fraction.

12. There are four black balls and y yellow balls in a bag. Thandi takes out a ball, notes its colour and then puts it back in the bag. She then takes out another ball and also notes its colour. If the probability that both balls have the same colour is $\frac{5}{8}$, determine the value of y.

13. A rare kidney disease affects only 1 in 1000 people and the test for this disease has a 99% accuracy rate.

    a) Draw a two-way contingency table showing the results if $100\ 000$ of the general population are tested.
    b) Calculate the probability that a person who tests positive for this rare kidney disease is sick with the disease, correct to two decimal places.

14. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29JP 2. 29JQ 3. 29JR 4. 29JS 5. 29JT 6. 29JV 7. 29JW
8. 29JX 9. 29JY 10. 29JZ 11. 29K2 12. 29K3 13. 29GT

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Solutions to exercises

1 Sequences and series

Exercise 1 – 1: Arithmetic sequences

1. $26; 30; 34$
2. $-19; -22; -25$
3. $5; 7; 9$
4. $65; 76; 87$
5. $a + 5b; a + 7b; a + 9b$
6. $\frac{3}{2}; 2; \frac{5}{2}$

Exercise 1 – 2: Arithmetic Sequences

1. a) 1
   b) $T_n = 8{,}5 - 1{,}5n$
   c) $T_{21} = -23$
2. a) Yes
   b) 218
   c) $T_{81} = 322$
   d) No
3. a) $T_1 = 5; T_2 = 3; T_3 = 1; T_{10} = -13$
   b)
4. $T_n = \frac{1}{2}n - 1$
7. a) $4; 7; 10; 13$
   b) $d = 3$
   c) $T_n = 3n + 1$
   d) $T_{25} = 76$
   e) $T_{36} = 109$
8. a) $p = 7; q = 7; r = 35; s = 2n + 1$
   b) 801 stitches
9. $p = \frac{1}{2}; T_{15} = 35\frac{1}{2}$
10. $x = 5a - 6$
11. $3s - t; s; -s + t; -3s + 2t; -5s + 3t; -7s + 4t; -9s + 5t; -11s + 6t; -13s + 7t;$

Exercise 1 – 3: Quadratic sequences

1. a) Quadratic sequence
   b) Linear sequence
   c) Quadratic sequence
   d) Linear sequence
   e) Quadratic sequence
   f) Neither
   g) Quadratic sequence
   h) Quadratic sequence
2. $b = 0$ and $c = -2$
3. a) Yes
   b) $-7a^2$
   c) $T_{100} = -197a^2$
4. $b = 2$ and $c = -1$
5. $4; 14; 34; 64; 104; 154$
6. a) -1
   b) 7

Exercise 1 – 4: Constant ratio of a geometric sequence

1. $r = 2$ next terms: $40; 80; 160$
2. $r = 2$ next terms: $\frac{1}{16}; \frac{1}{32}; \frac{1}{64}$
3. $r = 0{,}1$ next terms: $0{,}007; 0{,}0007; 0{,}00007$
4. $r = 3p$ next terms: $27p^4; 81p^5; 243p^6$
5. $r = -10$ next terms: $3000; -30\,000; 300\,000; -3\,000\,000$

Exercise 1 – 5: General term of a geometric sequence

1. $T_n = 5(2)^{n-1}$
2. $T_n = \left(\frac{1}{2}\right)^n$
3. $T_n = 7(0{,}1)^{n-1}$
4. $T_n = p(3p)^{n-1}$
5. $T_n = -3(-10)^{n-1}$

Exercise 1 – 6: Mixed exercises

1. a) $6; 2; \frac{2}{3} \ldots$
   b) Geometric: $r = \frac{1}{2}$
2. $k = 4; m = 10$
3. a) $r = 2$
   b) $a = \frac{1}{4}$
   c) $T_n = \frac{2^n}{8}$
4. a) $r = 2$ and $a = \frac{1}{4}$
   b) $y = \frac{1}{4}$ and $x = \frac{1}{2}$
   c) $T_5 = 4$
5. $T_4 = \sin^3\theta$ and $T_5 = \sin^4\theta$
6. Arithmetic: $5; \frac{1}{4}; -\frac{9}{2}; \ldots$ or $5; 25; 45; \ldots$
   Geometric: $\frac{1}{4}; -\frac{9}{2}; 81; \ldots$ or $25; -45; 81; \ldots$
7. a) $4x^3y$
   b) $r = \frac{2x}{y}$
8. $-1; \frac{1}{3}; -\frac{1}{9}; \frac{1}{27}; -\frac{1}{81}$ or $-1; -\frac{1}{3}; -\frac{1}{9}; -\frac{1}{27}; -\frac{1}{81}$

Exercise 1 – 7: Sigma notation

1. a) $2 + 2 + 2 + 2 = 8$
   b) -1
   c) 34
2. a) 0
   b) $8 + 8 + 8 + 8 = 32$
   c) $15a$
3. a) $a = 4$
   b) $a = \frac{15}{16}$
4. $\sum_{n=1}^{4} (3^{n-3})$
5. $\sum_{i=1}^{25} (18 - 7n)$
6. $\sum_{k=1}^{1000} (2n - 1)$
   $\sum_{k=0}^{999} (2n + 1)$

Exercise 1 – 8: Sum of an arithmetic series

1. $k = 4$
2. a) $n = 10$
   b) $T_6 = 46$
3. a) $n = 13$ or $n = 50$
4. $S_{51} = 4335$
5. $S_{300} = 135750$
6. $\frac{5}{3}$
7. a) $d = 5$ and $a = -9$
   b) $T_{100} = 486$
8. a) $S_9 = 324$
   b) 260
9. $n = 15$
10. $d = 10$
11. $n = 20$

Exercise 1 – 9: Sum of a geometric series

2. a) -2187
   b) -1640
3. $S_4 = 45$
4. $S_{11} = \frac{6141}{512}$
6. $a = 5; r = 2; S_7 = 635$
7. a) $4; 2; 1$
   b) $n = 9$
8. $r = \frac{3}{2}, a = \frac{32}{9}, T_2 = \frac{16}{3}$

Exercise 1 – 10: Convergent and divergent series

1. Arithmetic series: $S_1 = 2; S_2 = 6; S_{10} = 110; S_{100} = 10100$. Divergent.
2. Arithmetic series: $S_1 = -1; S_2 = -3; S_{10} = -55; S_{100} = -5050$. Divergent.
3. Geometric series: $S_1 = \frac{2}{3}; S_2 = \frac{10}{9}; S_{10} = 1{,}965\ldots; S_{100} = 2{,}00\ldots$. Convergent.
4. Geometric series: $S_1 = 2; S_2 = 6; S_{10} = 2046; S_{100} = 2{,}5 \times 10^{30}$. Divergent.

Exercise 1 – 11: Sum to infinity

1. $S_\infty = \frac{2}{3}$
2. $S_\infty = \frac{9}{2}$
3. $-4 < x < 2$
4. $a = \frac{5}{3}, r = \frac{3}{5}$
6.
7. $p = \frac{2}{3}$

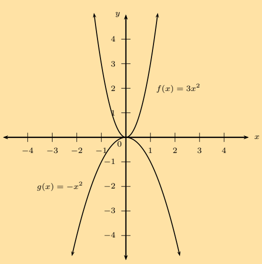
Exercise 1 – 12: End of chapter exercises

1. Infinite arithmetic series
2. a) $S_7 = 28$
   b) $S_{n-1}\frac{n(n-1)}{2}$
3. $S_5 = \frac{121}{2187}$
4. a) $x = 3$
   b) $S_\infty = 8$
5. $\sum_{n=1}^{20} 6\left(\frac{1}{2}\right)^{n-1}$
6. $S_\infty = 15$
7. a) R 250
   b) 30 years
8. $S_\infty = \frac{16}{3}$
9. $n = 10$
10. 150 mm
11. $n = 10$
12. a) R 708,62
    b) R 8553,71
13. a) R 524 288
    b) R 1 048 575
14. $9; 6; 4; \ldots$
15. a) $p = 10$
    c) $T_{10} = 119{,}2$
16. $S_\infty = 162$
17. 34
18. $T_2 = 12$
19. a) Converge: $S_\infty = -\frac{1}{3}$
    b) Diverge: $r = -3$
20. $S_\infty = \frac{5}{3}$
21. $T_{10} = 20$
22. 1 998 953
23. a) 25
    b) Letter: s
24. $\frac{26}{45}$
25. a) $-2 < x < 0$
    b) $f(-\frac{1}{2}) = \frac{2}{3}$
26. $S_n = \frac{a^2(1-a^{2n})}{(1-a^2)}$
27. $T_n = 59$
28. $n = 9$

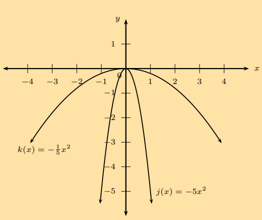
2 Functions

Exercise 2 – 1: Revision

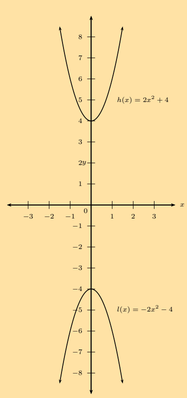
1. a)



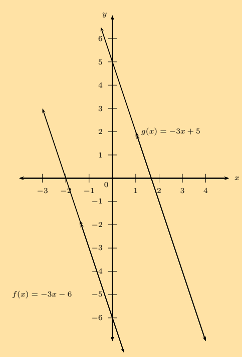
b)



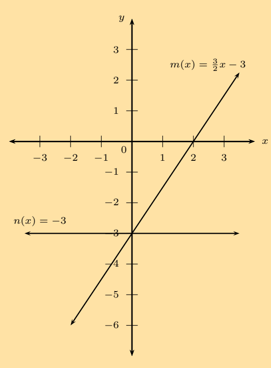
c)



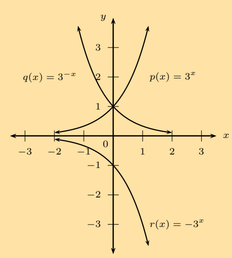
2.



3.



4. a)

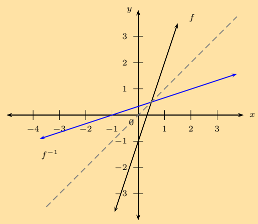


Exercise 2 – 2: Identifying functions

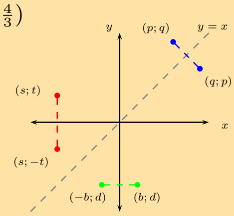
1. a) Yes
   b) Yes
   c) Yes
   d) No
   e) Yes
   f) No
   g) Yes
   h) No
2. a) One-to-one relation: therefore is a function.
   b) One-to-one relation: therefore is a function.
   c) One-to-one relation: therefore is a function.
   d) One-to-many relation: therefore is not a function.
   e) Many-to-one relation: therefore is a function.
3. $d = -500u + 23\,000$

Exercise 2 – 3: Inverse of the function $y = ax + q$

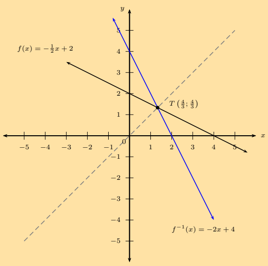
1. $f^{-1}(x) = \frac{1}{5}x - \frac{4}{5}$
2. a) It is a function.
   b) Domain $\{x : x \in \mathbb{R}\}$, Range $\{y : y \in \mathbb{R}\}$
   c) $f^{-1}(x) = -\frac{1}{3}x - \frac{7}{3}$
3. a)



   b) $R\left(3; \frac{4}{3}\right)$
4. a)



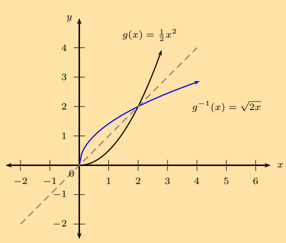
   b) No
5. a) $f(x) = -\frac{1}{2}x + 2$
   b) $f(x) : (4; 0), (0; 2) f^{-1}(x) : (2; 0)$ and $(0; 4)$
   c) $T\left(\frac{4}{3}; \frac{4}{3}\right)$
   d)



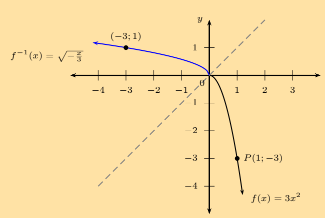
   e) Decreasing function

Exercise 2 – 4: Inverses - domain, range, intercepts, restrictions

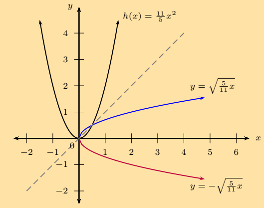
1. a) $y = \pm\sqrt{\frac{4}{3}x} \quad (x \geq 0)$
   b) $y = \pm\sqrt{\frac{1}{2}x} \quad (x \geq 0)$
   c) $y = \pm\sqrt{-5x} \quad (x \leq 0)$
   d) $y = \pm\sqrt{4x} \qquad (x \geq 0)$
2. a) $g^{-1}(x) = \sqrt{2x} \qquad (x \geq 0)$
   b)



   c) Yes
   d) For g :Domain: $x \geq 0$ Range: $y \geq 0$
   For g^{-1} :Domain: $x \geq 0$ Range: $y \geq 0$
   e) $(0; 0)$ or $2; 2$
3. a) $f(x) = -3x^2 \quad (x \geq 0)$
   b) f : domain $x \geq 0$, range $y \leq 0$
   c) $(-3; 1)$
   d) $f^{-1}(x) = \sqrt{-\frac{1}{3}x}, \quad (x \leq 0)$
   e) f^{-1} : domain $x \leq 0$, range $y \geq 0$
   f)



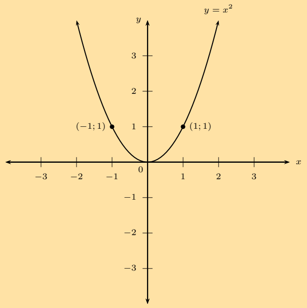
4. a) $y = \pm\sqrt{\frac{5}{11}x} \quad (x \geq 0)$
   b)



   c) $x \geq 0$ or $x \leq 0$
5. a) $a = -\frac{1}{2}, c = 0$ and $m = -\frac{1}{4}$
   b) g : domain: $x \in \mathbb{R}$, range: $y \in \mathbb{R}$ and f^{-1} : domain: $x \geq 0$, range: $y \leq 0$
   c) $x > 4$
   d) $y = 4x^2 \quad (y \geq 0)$
   e) $(0; 0)$ and $\left(-\frac{1}{16}; \frac{1}{64}\right)$

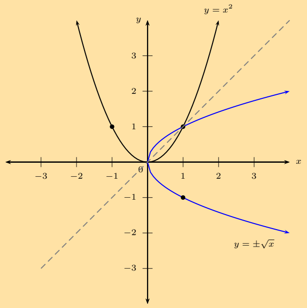
Exercise 2 – 5: Inverses - average gradient, increasing and decreasing functions

1. a)



b) $y = \pm\sqrt{x} \quad (x \geq 0)$

c)



d) No

e) $Q(4;2)$

f) Ave. grad.$_{OP}$ $= 2$ and ave. grad.$_{OQ}$ $= \frac{1}{2}$

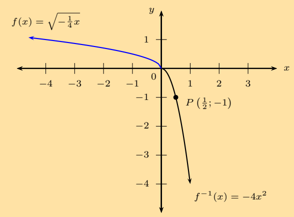
2. a) $f^{-1}(x) = -4x^2$

b) Domain: $x \geq 0$ and Range: $y \leq 0$

c) $f(x) = \sqrt{-\frac{1}{4}x} \quad (x \leq 0)$

d) Domain: $\{x : x \leq 0, x \in \mathbb{R}\}$, Range: $\{y : y \geq 0, y \in \mathbb{R}\}$

e)

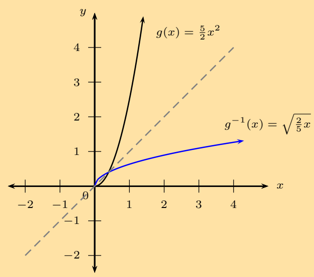


f) Decreasing function

3. a) $g^{-1}(x) = \sqrt{\frac{2}{5}x} \qquad (x \geq 0)$

b) $(0;0)$ and $\left(\frac{2}{5};\frac{2}{5}\right)$

c)



d) Both are increasing functions

e) 1

Exercise 2 – 6: Finding the inverse of $y = b^x$

1. a) $4 = \log_2 16$
   b) $-5 = \log_3\left(\frac{1}{243}\right)$
   c) $3 = \log_{1,7}(4{,}913)$
   d) $x = \log_2 y$
   e) $\log_4 q = 5$
   f) $\log_y 4 = g$
   g) $\log_{(x-4)} 9 = p$
   h) $\log_m 3 = a + 4$
2. a) $2^5 = 32$
   b) $10^{-3} = \frac{1}{1000}$
   c) $10^{-1} = 0{,}1$
   d) $d^b = c$
   e) $5^0 = 1$
   f) $3^{-4} = \frac{1}{81}$
   g) 2
   h) -4

Exercise 2 – 7: Applying the logarithmic law: $\log_a x^b = b\log_a x$

1. $10\log_8 10$
2. $y\log_{16} x$
3. $\frac{\log_3 5}{2}$
4. $z\log_z y$
5. $\frac{1}{x}$
6. q
7. $\frac{3}{4}$
8. -1
9. 15
10. 8
11. 4
12. -3

Exercise 2 – 8: Applying the logarithmic law: $\log_a x = \frac{\log_b x}{\log_b a}$

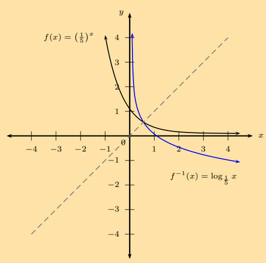
1. a) $\frac{\log_8 4}{\log_8 2}$
   b) $\frac{\log_2 14}{\log_2 10}$
   c) $\frac{\log_2 9 - 1}{\log_2 10}$
   d) $\frac{1}{\log_8 2}$
   e) $\frac{1}{\log_x y}$
   f) $\frac{1 + \log_2 x}{\log_2 10}$
2. a) 1
   b) $\frac{2}{\log 5}$
3. a) $1{,}58$
   b) $6{,}92$

Exercise 2 – 9: Logarithms using a calculator

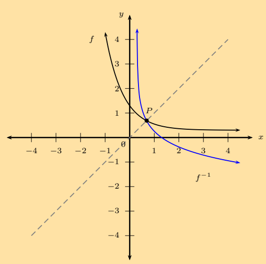
1. a) $0{,}477$
   b) $1{,}477$
   c) $2{,}477$
   d) $-0{,}180$
   e) $-0{,}602$
   f) $2{,}930$
   g) no value
   h) $1{,}262$
   i) -2
   j) $3{,}907$
   k) $1{,}661$
   l) $-2{,}585$
2. a) $3{,}98$
   b) $0{,}01$
   c) $63{,}10$
   d) $100\ 000$
   e) $0{,}32$
   f) $1{,}19$
   g) $2{,}51$
   h) $0{,}06$
   i) $1{,}19$
   j) $0{,}85$
   k) $0{,}06$
   l) $1{,}79$

Exercise 2 – 10: Graphs and inverses of $y = \log_b x$

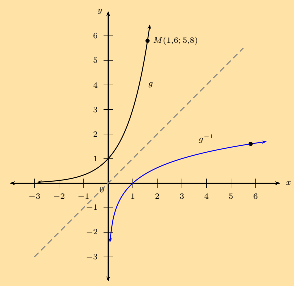
1. a)



   b) $f : (0; 1)$ and $f^{-1} : (1; 0)$
   c)



   d)
2. a) $t = 3$
   b) $g^{-1}(x) = \log_3 x \quad (x > 0)$
   c)



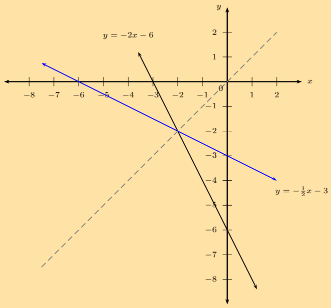
   d) $N(5,8; 1,6)$

Exercise 2 – 11: Applications of logarithms

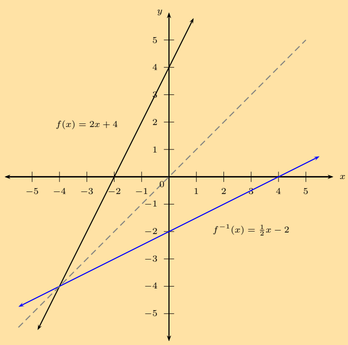
1. 57 years
2. a) Growth $= 36 \times 2^n$
   b) After 14 months

Exercise 2 – 12: End of chapter exercises

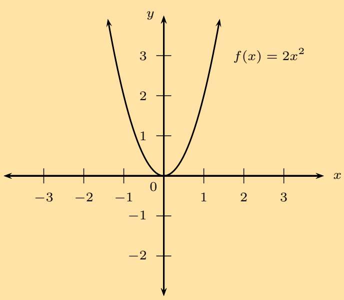
1. a) $h(x) = -2x - 6$
   b) $h^{-1}(x) = -\frac{x}{2} - 3$
   c)



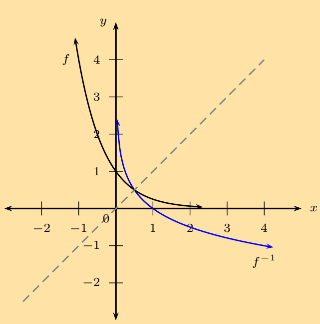
   d) $S(-2; -2)$
   e) x-coordinate is equal to the y-coordinate.
2. a) $f(x) = \frac{1}{2}x - 2$
   b)



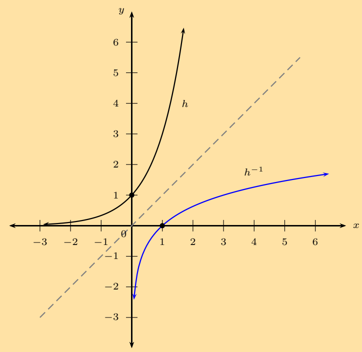
   c) Increasing
3. a)



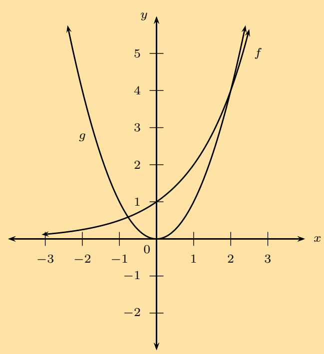
   b) $y = \pm\sqrt{\frac{x}{2}}, \quad x \geq 0$. Domain: $\{x : x \geq 0, x \in \mathbb{R}\}$, Range: $\{y : y \in \mathbb{R}\}$.
4. a)



   b) Yes
   c) $y = \log_{\frac{1}{4}} x$ or $y = -\log_4 x$
   d) $P = \frac{1}{2}$, since the point lies on the line $y = x$.
   e) $G(x) = -\log_4 (x + 2)$ or $G(x) = \log_{\frac{1}{4}} (x + 2)$
   f) Vertical asymptote: $x = -2$
5. a) $h^{-1}(x) = \log_3 x$
   b) Domain: $\{x : x > 0, x \in \mathbb{R}\}$ and range: $\{y : y \in \mathbb{R}\}$.
   c)



   d) $0 < x < 1$
6. a)

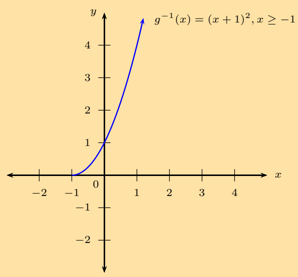


   b) No
   c) Two
7. Graph 1: $y = 3^{-x}$
   Graph 2: $y = -\log_3 x$ or $y = \log_{\frac{1}{3}} x$
   Graph 3: $y = -3^x$
8. b) $a = \frac{1}{3}$
   c) $y = \log_3 x + 2$
   d) $y = \log_3 (x - 1)$
9. a) 10 years
   b) Graph C
10. a) $T_n = 100 \times 2^{n-1}$
    b) 204 800 retweets
    c) 1,75

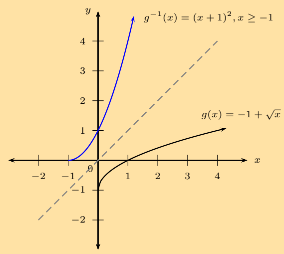
Exercise 2 – 13: Inverses (ENRICHMENT ONLY)

1. a) $g^{-1}(x) = (x+1)^2, \ \ x \geq -1$

   b)



   c)



   d) Yes

   e) Domain: $\{x : x \geq -1, x \in \mathbb{R}\}$, Range: $\{y : y \geq 0, y \in \mathbb{R}\}$.

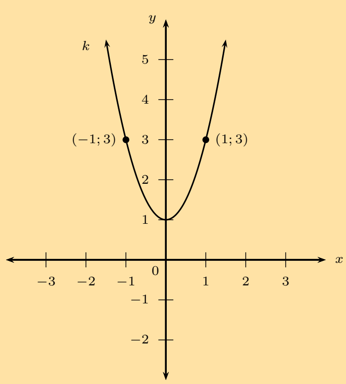
2. a) $y = -2(x-1)^2 + 3$

   b) Maximum value at $(1; 3)$

   c) Domain: $\{x : x \in \mathbb{R}\}$ and range: $\{y : y \leq 3, y \in \mathbb{R}\}$, Axis of symmetry: $x = 1$.

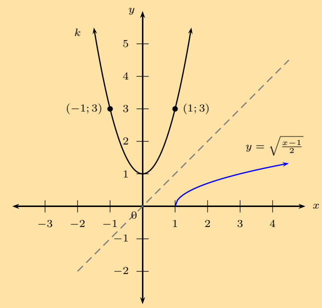
3. a) $q = 1$ or $q = -1$

   b)



   c) $y = \pm\sqrt{\frac{x-1}{2}} \quad (x \geq 1)$

   d)

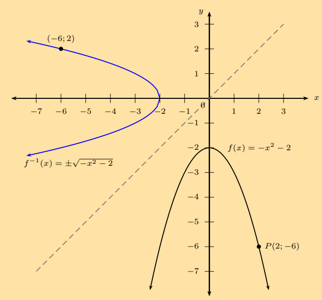


   e) $(3; 1)$

4. a) $f(x) = -x^2 - 2$

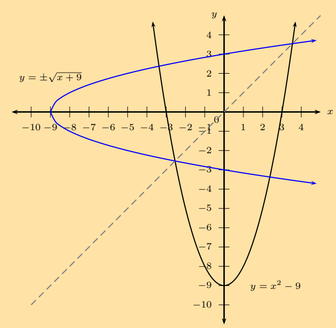
   b) $y = \pm\sqrt{-x-2} \quad (x \leq -2)$

   c)



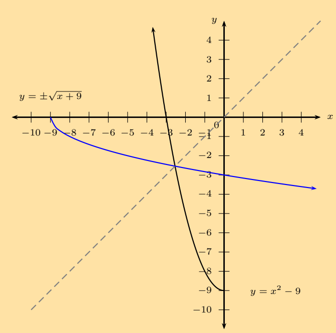
5. a) $y = \pm\sqrt{x+9} \quad (x \geq -9)$

   b)



   c) No

   d)



Exercise 2 – 14: Applying logarithmic law: $\log_a xy = \log_a (x) + \log_a (y)$

1. a) $2\log_8 10$
   b) $1 + \log_2 7$
   c) $3 + \log_2 5$
   e) $1 + \log_2 x + \log_2 y$
   f) $\log_7$
2. a) $\log 30$
   b) 0
   c) $\log_3 12$
   d) $(\log x)(\log 10y)$
   e) Cannot be simplified.
   f) Cannot be simplified.
   g) $\log_a pq$
   h) $\log_a p \times \log_a q$
3. a) $\log xyz$
   b) $\log ab^2c^2d$
   c) $\log 2 + 3$
   d) 1

Exercise 2 – 15: Applying logarithmic law: $\log_a \frac{x}{y} = \log_a x - \log_a y$

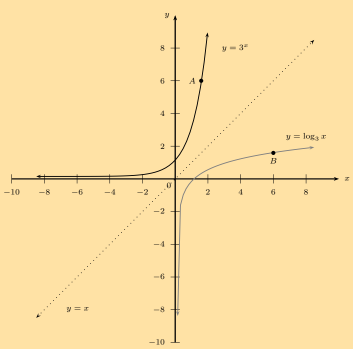
1. a) $2 - \log 3$
   b) $\log_2 15 - 1$
   c) $\log_{16} x - \log_{16} y$
   d) Cannot be simplified.
   e) $1 - \log_5 8$
   f) $\log_x y - \log_x r$
2. a) $-\log 5$
   b) 2
   c) $\log_a \frac{p}{q}$
   d) Cannot be simplified
   e) Cannot be simplified
   f) $\log 3$
3. a) 1
   b) 2
4. Both methods are correct.

Exercise 2 – 16: Simplification of logarithms

1. 9
2. $\log \frac{18}{5}$
3. 2
4. 4

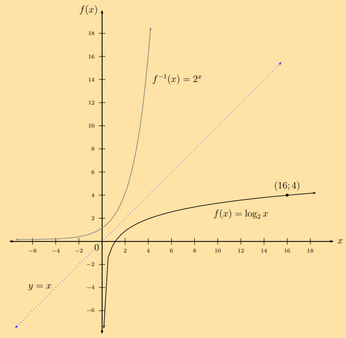
Exercise 2 – 17: Solving logarithmic equations

1. a) $1{,}09$
   b) $3{,}83$
   c) $5{,}66$
   d) $0{,}72$
   e) $-1{,}51$
   f) $65{,}94$
   g) $-0{,}55$
2. a) $y = \log_3 x$
   b) $1{,}6$
   c)



Exercise 2 – 18: Logarithms (ENRICHMENT ONLY)

1. a) False: $\log t + \log d = \log (t \times d)$
   b) False: $q = \log_p r$
   c) True
   d) False: $\log (A) - B$ cannot be simplified further.
   e) True
   f) False: $\log_k m = \frac{\log_p m}{\log_p k}$
   g) True
   h) True
   i) False: bases are different
   j) True
   k) False
   l) False
   m) False
   n) False
2. a) 1
   b) 0
   c) 2
   d) $\log 3$
3. a) 1,7
   b) 1,3
   c) 1,4
   d) $\frac{7}{3}$
   e) 5
4. b) 16
   c)



5. 1,68
6. a) $x < 10, (x \in \mathbb{Z})$
   b) $x < 11$

3 Finance

Exercise 3 – 1: Determining the period of an investment

1. Just over 3 years
2. Just over 5 years and 10 months
3. 3 years and 8 months ago
4. After 7 years
5. 5 years ago
6. 11
7. a) 71 years
   b) less than 5 years
8. 4,5 years

Exercise 3 – 2: Future value annuities

1. a) R 232 539,41
   b) R 210 000
   c) R 22 539,41
2. R 590,27
3. a) R 1 792 400,11
   b) R 382 800
4. R 2 923 321,08
5. a) R 510,85
   b) R 214 557
6. R 1123,28
7. a) 2 years and 211 days
   b) R 377,53

Exercise 3 – 3: Sinking funds

1. a) R 262 094,55
   b) No
2. a) R 104 384,58
   b) R 362 635,97
   c) R 4560,42
3. a) R 230,80
   b) R 1988

Exercise 3 – 4: Present value annuities

1. R 22 383,37
2. R 2420,00
3. R 6712,46
4. a) R 297,93
   b) R 32 176,44
5. a) 8 years
   b) R 2358,88
6. a) R 3917,91
   b) R 332 089,87
7. a) R 1627,05
   b) R 388 743,83
   c) R 18 530,10

Exercise 3 – 5: Analysing investment and loan options

1. a) option A
   b) R 1 937 512,76
2. Bank B

Exercise 3 – 6: End of chapter exercises

1. 6 years
2. R 102 130,80
3. a) R 394,50
   b) R 9796,00
4. 6 years
5. a) R 1 308 370,14
   b) R 468 000
6. a) R 134 243,45
   b) R 67 043,45
7. a) 5,5 years
   b) R 2465,87
8. a) R 1692,54
   b) R 230 651,34
   c) 1,4
9. R 8637,98
10. R 3895,68

4 Trigonometry

Exercise 4 – 1: Revision - reduction formulae, co-functions and identities

1. a) A
   b) A
   c) $\sqrt{1-A^2}$
   d) $-A$
   e) $\frac{A}{\sqrt{1-A^2}}$
2. a) $\sin^2\theta$
   b) $\cos^2\theta$
   c) $P+Q=1$ and $\frac{Q}{P}=\frac{1}{\tan^2\theta}$
3. $-\frac{1}{p}$
4. a) $\sqrt{3}$
   b) $\frac{1}{2}$
5. a) -1
   b) $\tan^2\theta$
   c) $-\cos^2\alpha$
   d) 0
6. a) $\sin\theta$
   b) $-\frac{1}{2}$
9. a) False
   b) False
   c) True
   d) False
   e) True
   f) True

Exercise 4 – 2: Compound angle formulae

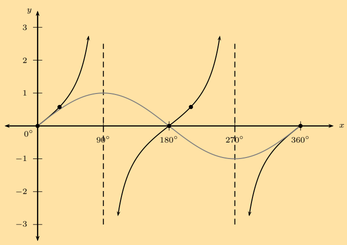
1. a) $\frac{5}{6}$
   b) $\frac{120}{169}$
   c) -1
2. a) $\frac{\sqrt{2}(\sqrt{3}+1)}{4}$
   b) $\frac{\sqrt{2}(1+\sqrt{3})}{4}$
   c) $\frac{\sqrt{2}(\sqrt{3}-1)}{4}$
   d) $2-\sqrt{3}$
   e) $\frac{1}{2}$
   f) 1
   g) $\frac{\sqrt{2}}{2}$
   h) $\frac{\sqrt{3}}{2}$
3. b) $\frac{\sqrt{6}}{2}$
4. $\sqrt{2}$
5. b) $\frac{2-\sqrt{3}}{4}$

Exercise 4 – 3: Double angle identities

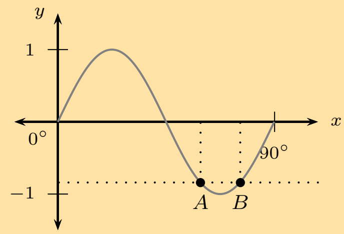
1. a) $-\frac{7}{25}$
   b) $-\frac{24}{25}$
   c) $\frac{24}{7}$
2. a) $-t$
   b) $\sqrt{1-t^2}$
   c) t
   d) $2t^2-1$
   e) $-t$
   f) $2t^2-1$
3. b) $36{,}87°$ or $216{,}87°$
4. a) $\sqrt{\frac{2-\sqrt{2}}{4}}$
   b) $\sqrt{\frac{2-\sqrt{2}}{4}}$

Exercise 4 – 4: Solving trigonometric equations

1. a) $x = 16{,}06° + k\ .\ 180°$ or $x = 73{,}9° + k\ .\ 180°, k \in \mathbb{Z}$
   b) $y = 30° + k\ .\ 120°$ or $y = 90° + k\ .\ 360°, k \in \mathbb{Z}$
   c) $\alpha = 22{,}5° + k\ .\ 90°, k \in \mathbb{Z}$
   d) $p = k\ .\ 360°$ or $p = 36° + k\ .\ 72°, k \in \mathbb{Z}$
   e) $A = 45° + k\ .\ 180°$ or $A = 135° + k\ .\ 180°, k \in \mathbb{Z}$
   f) $x = 51{,}8° + k\ .\ 360°$ or $x = 308{,}2° + k\ .\ 360°, k \in \mathbb{Z}$
   g) $t = 0° + k\ .\ 180°, \quad k \in \mathbb{Z}$
   h) $x = 30° + k\ .\ 360°$
2. a) $x = 0°, 30°, 180°, 210°$ or $360°$
   b)



3. a) $A = 35{,}79° + k\ .\ 90°$ or $A = 31{,}72° + k\ .\ 90°, \quad k \in \mathbb{Z}$
   b) 10 solutions.
4. $A = -300°, -60°, 60°$ or $300°$
5. a) $x = 59{,}26° + k\ .\ 90°$ or $x = 75{,}74° + k\ .\ 90°, k \in \mathbb{Z}$.
   b)



$A(59{,}26°; -0{,}84)$, $B(75{,}74°; -0{,}84)$

7. $x = 60° + k\ .\ 360°$

Exercise 4 – 5: Problems in two dimensions

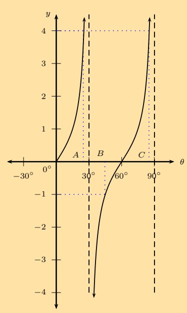
1. a) $A\hat{O}C = 2\theta$
   b) $\cos\theta = \frac{AE}{BE}$
   c) $\sin\theta = \frac{CA}{AE}$
   d) $\sin 2\theta = \frac{CA}{AO}$
3. b) $CA^2 = DC^2 + DA^2 - 2(DC)(DA)\cos\hat{D}$ or $CA^2 = AB^2 + BC^2 - 2(AB)(BC)\cos\hat{B}$
   d) $CA^2 = 188{,}5$ units^2
   e) Area $ABCD = 54{,}0$ units^2
4. a) $AC = 69$ m
   b) $BD = 64$ m

Exercise 4 – 6: Problems in three dimensions

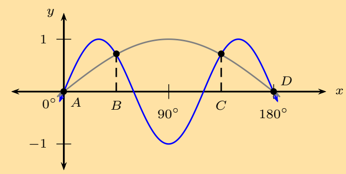
1. a) $BC = 2x\cos\alpha\tan\theta$
   b) $BC = 41{,}40$ m
2. a) $h = \frac{d\sin x\tan z}{\sin(x+y)}$
   b) 15 m
3. b) 16,2 m
   c) Approx. 6 levels
4. c) 1932,3 m^2
5. a) 229 m^2
   c) 148°

Exercise 4 – 7: End of chapter exercises

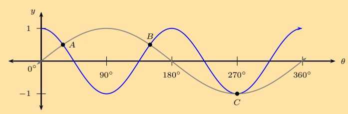
1. a) $\frac{1+\sqrt{3}}{2\sqrt{2}}$
   b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
   c) $2+\sqrt{3}$
   d) $\frac{1}{\sqrt{2}}$
   e) $\frac{1}{\sqrt{2}}$
2. $\cos 2\theta = -0{,}02$ and $\cos 4\theta = -0{,}9992$
3. a) $\frac{31}{49}$
   b) $-\frac{6\sqrt{40}}{31}$
4. $\frac{4}{5}$
5. $\frac{1}{2}$
6. $\theta \in \{-60°; -30°; 30°; 60°\}$
7. a) $\theta = 30° + k \,.\, 360°$ or $\theta = 150° + k \,.\, 360°, k \in \mathbb{Z}$
   b) $x = 30° + k \,.\, 360°,\ x = 150° + k \,.\, 360°, k \in \mathbb{Z},\ x = 135° + k \,.\, 360°$ or $x = 225° + k \,.\, 360°, k \in \mathbb{Z}$
   c) $x = 10° + k \,.\, 360°$ or $x = 116{,}7° + k \,.\, 120°, k \in \mathbb{Z}$
   d) $\alpha = 11{,}3° + k \,.\, 180°$ or $\alpha = 135° + k \,.\, 180°, k \in \mathbb{Z}$
8. b) $\theta = 52{,}53°$ or $\theta = 127{,}47°$
10. b) Restricted values are: $y = 90° + k \,.\, 180°, k \in \mathbb{Z}$
11. a) $\theta = 45° + k \,.\, 60°$ or $\theta = 25{,}32° + k \,.\, 60°, k \in \mathbb{Z}$
    b) $\theta = 25{,}32°, 45°,$ or $85{,}32°$
    c)



    d) $30° < \theta < 45°$
12. b) $x = -180°; -135°; -45°; 0°$ $= 45°; 135°; 180°; 225°; 315°; 360°$
    c)



13. a) $x = 30°; 270°; 150°$
    b)



14. b) 3
    c) $x = 360°$
    d) $x = 29°, 180°, 331°$
    e) $29° \leq x \leq 180°$ or $331° \leq x \leq 360°$
16. $f(\theta) = \frac{3}{2}\sin 2\theta$ and $g(\theta) = -\frac{3}{2}\tan\theta$
18. a) $\hat{U} = 108{,}6°$
    b) $\hat{S} = 71{,}4°$
    c) $R\hat{T}S = 54{,}9°$
19. c) 4 m
20. a) length = 8 cm, breadth = 6 cm, diagonal = 10 cm
    b) 7 cm
    c) $\frac{10}{7}$
21. a) $A\hat{B}C = 90°, A\hat{B}D = 90°$
    d) 8,4 m
22. b) 2 m

5 Polynomials

Exercise 5 – 1: Identifying polynomials

1. a) True
   b) True
   c) False: $f\left(\frac{1}{2}\right) = 0$
   d) True
   e) False: -1
   f) True
2. a) 4
   b) 3
   c) -9
   d) 6
3. a) Cubic polynomial
   b) Quadratic polynomial
   c) Not a polynomial
   d) Not a polynomial
   e) Cubic polynomial
   f) Linear polynomial
   g) Zero polynomial
   h) Polynomial; degree 7
   i) Cubic polynomial
   j) Not a polynomial

Exercise 5 – 2: Quadratic polynomials

1. a) $p = 0$ or $p = -2$
   b) $k = 4$ or $k = -9$
   c) $h = \pm 5$
   d) $x = -7$ or $x = -2$
   e) $y = 4k$ or $y = k$
2. a) $p = -5 - 3\sqrt{3}$ or $p = -5 + 3\sqrt{3}$
   b) $y = -3 \pm \sqrt{7}$
   c) No real solution
   d) $f = -8 \pm 3\sqrt{6}$
   e) $x = -1 \pm \sqrt{\frac{5}{3}}$
3. a) $m = 1$ or $m = -\frac{4}{3}$
   b) No real solution
   c) $y = 2 \pm \sqrt{2}$
   d) $f = \frac{1}{2}$ or $f = -2$
4. a) $(3p - 1)(9p^2 + 3p + 1)$
   b) $2\left(2 + \frac{1}{x}\right)\left(4 - \frac{2}{x} + \frac{1}{x^2}\right)$

Exercise 5 – 3: Cubic polynomials

1. a) $(p - 1)(p^2 + p + 1)$
   b) $(t + 3)(t^2 + 3t + 9)$
   c) $(4 - m)(16 + 4m + m^2)$
   d) $k(1 - 5k)(1 + 5k + 25k^2)$
   e) $\left(2a^2 - b^3\right)\left(4a^4 + 2a^2b^3 + b^6\right)$
   f) $(2 - p - q)(4 + 2p + 2q + p^2 + 2pq + q^2)$
2. a) $a(x) = (x + 1)(x^2 + x + 2) + 5$
   b) $a(x) = (x + 2)(-x^2 + 6x - 17) + 35$
   c) $a(x) = (x - 1)(2x^2 + 5x + 6)$
   d) $a(x) = (x - 1)(x^2 + 3x + 3) + 8$
   e) $a(x) = (x - 1)(x^3 + 3x^2 + 5) + 9$
   f) $a(x) = (x^2 - 2)(5x^2 + 3x + 16) + 7x + 34$
   g) $a(x) = (3x - 1)(x^2 + \frac{2}{3}) + \frac{5}{3}$
   h) $a(x) =$ $(x + 2)(2x^4 - 4x^3 + 9x^2 - 15x + 30) - 64$
3. a) $f(x) = (x + 2)(x + 3) - 5$
   b) $f(x) = (x - 1)(x - 4) - 11$
   c) $f(x) = (x - 1)(2x^2 + 2x + 7) + 3$
   d) $f(x) = (x + 3)(x + 5) + 4$
   e) $f(x) = (x - 1)(x^2 + 3x + 4) - 6$
   f) $f(x) = (x + 3)(2x^2 + x - 1)$
   g) $f(x) = (2x - 1)\left(2x^2 + 3x + 1\right) - 1$
   h) $f(x) = (2x + 3)(x^2 - x + 4) + 10$

Exercise 5 – 4: Remainder theorem

1. a) 11
   b) 7
   c) -25
   d) 11
   e) -4
   f) -26
   g) 2
2. $t = 4$
3. $m = 7$
4. $k = -\frac{3}{2}$
5. $p = -5$
6. $b = -\frac{15}{2}$
7. $h = -6$
8. $m = 2 - n$
9. $p = -5, q = -11$

Exercise 5 – 5: Factorising cubic polynomials

1. -14
2. $a(x) = (x + 1)(x - 2)^2$
3. a) 0
   b) $f(x) = (x - 1)(2x - 1)(x + 2)$
4. $a(x) = (x - 1)(x + 5)(x - 3)$
5. a factor of $f(x)$

Exercise 5 – 6: Solving cubic equations

1. $x = -1$ or $x = 4$ or $x = -4$
2. $n = -2$ or $n = -4$ or $n = 5$
3. $y = -1$ or $y = 4$ or $y = -5$
4. $k = -2$ or $k = -3$ or $k = -4$
5. $x = -2$ or $x = 5$ or $x = -5$
6. $p = 3$ or $p = 2$ or $p = -5$
7. $x = -1$ or $x = 3$ or $x = 4$

Exercise 5 – 7: End of chapter exercises

1. $x = 1$ or $x = -3$
2. $y = -1$ or $y = -2$ or $y = 6$
3. $m = 1$ or $m = -2$ or $m = 2$
4. $x = -2$ or $x = \frac{3+\sqrt{21}}{2}$ or $x = \frac{3-\sqrt{21}}{2}$
5. $x = -1$ or $x = -\frac{1}{2}$ or $x = 3$
6. $x = -1$ or $x = -4$ or $x = 4$
7. b) $x = 2$ or $x = \frac{2}{3}$ or $x = 1$
8. $R = 1$
9. a) $m = 1$ or $m = \frac{1}{8}$ or $m = -2$
   b) $x = 0$ or $x = -3$
10. $R = 40$
11. $p = -\frac{12}{7}$ or $p = 1$
12. $t = 9$ and $Q(x) = x + 5$

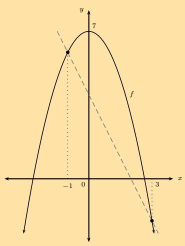
6 Differential calculus

Exercise 6 – 1: Limits

1. a) 0
   b) $\frac{1}{3}$
2. a) 4
   b) 7
   c) $\frac{37}{6}$
   d) Does not exist
   e) 0
   f) Does not exist
   g) 3
   h) 3
   i) $\frac{\sqrt{3}}{6}$

Exercise 6 – 2: Gradient at a point

1. a) -2
   b)



   c) $-2x$
2. $-\frac{3}{a^2}$
3. $y = x - 1$

Exercise 6 – 3: Differentiation from first principles

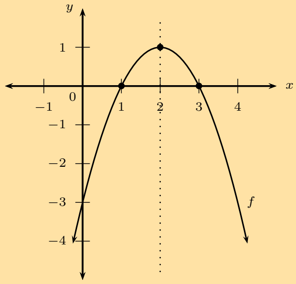
1. a) $\frac{-(x+h)^2-(-x^2)}{h}$
   b) $-2x$
2. $-4x + 3$
3. $\frac{-1}{(x-2)^2}$
4. $g'(3) = -30$
5. $8x - 4$
6. $k'(x) = 30x^2$
7. $f'(x) = nx^{n-1}$

Exercise 6 – 4: Rules for differentiation

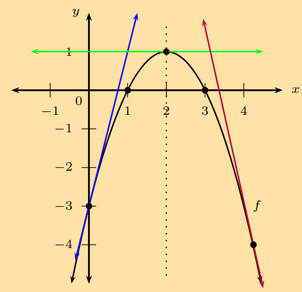
1. a) $6x$
   b) 25
   c) 0
   d) $20x^4$
   e) $-\frac{32}{x^3}$
   f) 0
   g) $4x^3 - 12x$
   h) $2x + 1$
   i) $x^2 - 2x$
   j) $\frac{9}{2}x^{\frac{1}{2}} - 4$
   k) $2x + 7$
   l) $600x^2 - 200x + 40$
   m) $42x^2 - 28x + 7$
2. 1
3. $\frac{1}{2\sqrt{y}}$
4. $2z$
5. $2x - \frac{1}{\sqrt{x^3}} + \frac{3}{x^2}$
6. $\frac{3}{2}\sqrt{x} - \frac{1}{x^4}$
7. $3x^2 - 6 - \frac{9}{x^2}$
8. $\frac{1}{2}$
9. $6\theta^2 - 12 - \frac{18}{\theta^2}$
10. $\frac{5}{2}t^{\frac{3}{2}} + \frac{9}{2}3t^{\frac{1}{2}} + \frac{3}{2t^{\frac{1}{2}}} - \frac{1}{2t^{\frac{3}{2}}}$

Exercise 6 – 5: Equation of a tangent to a curve

1. $y = 13x - 23$
2. a) $\left(-\frac{5}{6}; -\frac{13}{12}\right)$
   b) $(-3; -2)$
3. a) $(1; 1)$
   b) $\left(\frac{3}{4}; \frac{1}{4}\right)$
4. a)



   b) $i) y = 4x - 3;\ \ ii) y = 1;\ \ iii) y = -4{,}5x + 15{,}0625$
   c)

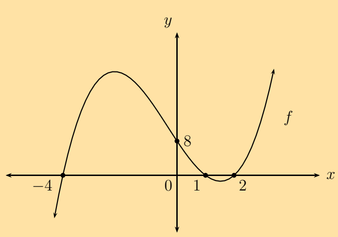


Exercise 6 – 6: Second derivative

1. a) 10
   b) $48x$
   c) 2
   d) $20x^3 - 6x$
   e) $6x - 2$
   f) $\frac{-60}{x^4}$
   g) $-\frac{1}{4\sqrt{x^3}} + 10$
2. $f'(x) = 20x + 15;\ \ f''(x) = 20$
3. $-\frac{4}{3\sqrt[3]{x^4}}$
4. a) $g'(x) = -6 + 24x - 24x^2;\ \ g''(x) = 24 - 48x$
   b) $g'(x)$: parabolic/quadratic; $g''(x)$: linear
   c) 0

Exercise 6 – 7: Intercepts

1. a) $(-4; 0)$, $(1; 0)$, $(2; 0)$ and $(0; 8)$
   b)

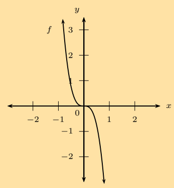


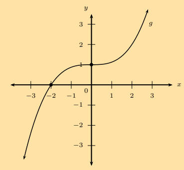
   c) Increasing
2. a) $(-5; 0)$, $(-3; 0)$, $(3; 0)$ and $(0; 45)$
   b) $(-1; 0)$, $\left(\frac{1}{4}; 0\right)$, $(2; 0)$ and $\left(0; \frac{1}{2}\right)$
   c) $(1; 0)$, $(\sqrt{12}; 0)$, $(-\sqrt{12}; 0)$ and $(0; 12)$
   d) $(0; 0)$, $(-4; 0)$ and $(4; 0)$
   e) $(0; 6)$, $(-1; 0)$, $(3 - \sqrt{3}; 0)$ and $(3 + \sqrt{3}; 0)$
3. $(-5; 0)$, $(0; 0)$ and $(2; 0)$

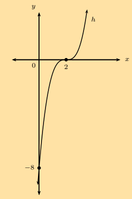
Exercise 6 – 8: Stationary points

1. $\left(\frac{5}{2}; \frac{1}{4}\right)$
2. $x = -2$ or $x = 3$
3. a) $(1; 0)$
   b) (0;1) and $\left(\frac{10}{3}; -\frac{473}{27}\right)$
   c) None

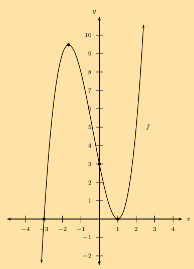
Exercise 6 – 9: Concavity and points of inflection

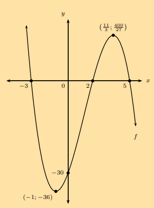
1. 

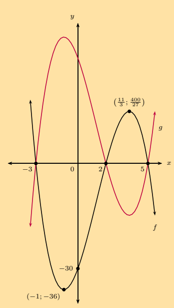
2. 

3. 

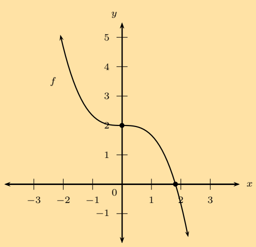
Exercise 6 – 10: Mixed exercises on cubic graphs

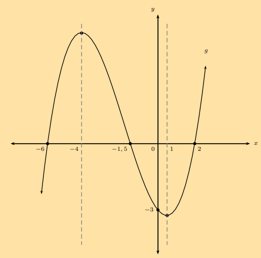
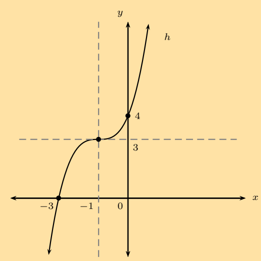
1. a) $f(x) = (x-1)(x+3)(x-1)$
   b) $(0;3)$; $(1;0)$; $(-3;0)(1;0)$ and $\left(-\frac{5}{3}; \frac{256}{27}\right)$
   c) 

2. a) 

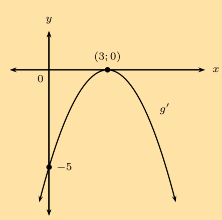
   b) 

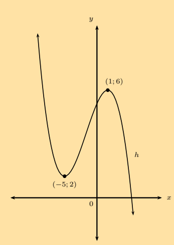
3. a) $f(x) = x^3 - 9x^2 + 24x - 20$
   b) $A(4;-4)$

4. a) 
   b) i.$x > \sqrt[3]{6}$; ii.$x \in R$, $x \neq 0$ and iii.$x > 0$

5. a) 
   b) 

6. a) 2 units
   b) 1 unit
   c) 5 units
   d) 12 units
   e) $C = \left(-\frac{2}{3}; -\frac{400}{27}\right)$ and $F = (4;36)$
   f) 7 units
   g) 12
   h) $y = 15x - 15$

7. 

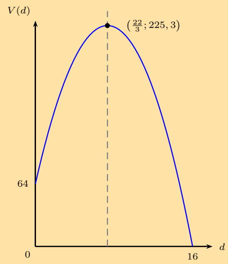
8. 

Exercise 6 – 11: Solving optimisation problems

1. $\frac{20}{3}$ and $\frac{40}{3}$
2. b) $x = 10$ cm
3. 1 unit
4. $x = 5$ m and $y = 10$ m
5. b) $x \approx 7{,}9$ cm and $h =\approx 12{,}0$ cm

Exercise 6 – 12: Rates of change

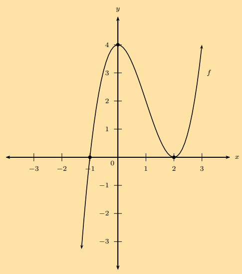
1. a) -4 kℓ per day
   b) Decreasing
   c) 16 days
   d) $7\frac{1}{3}$ days
   e) 225,3 kℓ
   f)



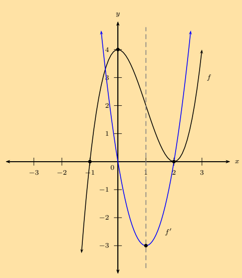
2. a) 1 m
   b) 18 m.s^{-1}
   c) 9 m.s^{-1}
   d) 28 metres
   e) -6 m.s^{-2}
   f) 3 m.s^{-1}
   g) 0 m.s^{-1}
   h) 6,05 s
   i) $-18{,}3$ m.s^{-1}
3. $a = 6$ m.s^{-2}
4. a) $T'(t) = 4 - t$
   b) $(4; 10]$

Exercise 6 – 13: End of chapter exercises

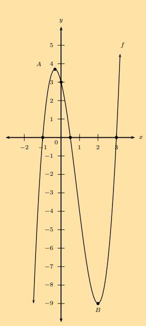
1. $f'(x) = -2x + 2$
2. $f'(x) = -\frac{1}{x^2}$
3. 3
4. a) $-3 - 10x$
   b) $8x - 2$
   c) $8x + \frac{1}{3x^2}$
   d) $\frac{5}{2\sqrt{x^3}}$
5. a) $f'(x) = 4x - 1$
   b) $(2; 6)$
6. $g'(2) = \frac{255}{8}$
7. a) $y = \frac{x}{2} + \frac{3}{2}$
   b) 9
8. a) $p'(t) = \frac{1}{5\sqrt[5]{t^2}}$
   b) $k'(n) = 6 + \frac{15}{n^2} + \frac{20}{n^3}$
9. $y = \frac{1}{2\sqrt{x}} - \frac{5}{x^2}$
10. a) $\frac{dy}{dx} = 3x^2$
    b) $\frac{dx}{dy} = \frac{1}{3\sqrt[3]{y^2}}$
11. a) 0
    c) $y = 3x(x - 2)$
    d)



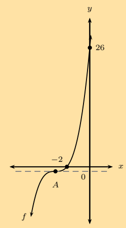
    e) $(-1; 0)$ and $(3; 4)$
    f)



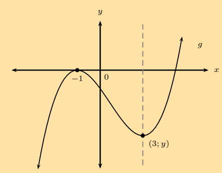
    g) $f''(x) = 6x - 6$
12. a) $(-1; 0)$, $\left(\frac{1}{2}; 0\right)$ and $(3; 0)$
    b) $\left(-\frac{1}{3}; \frac{100}{27}\right)$ and $(2; -9)$
    c)



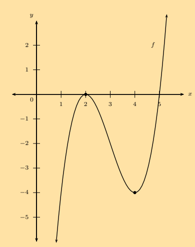
    d) $k = 3{,}7$ or $k = -9$
    e) $y = -8x + 4$
13. a) $f(x) = x^3 + 9x^2 + 27x + 26$
    b) -1
    c)



14. a) $x = -1$ and $x = 3$
    b) $g'(0) = -2$
    c)



15. $h(x) = -x + 13$
16. b) $y = 9x - 23$
    c) $C(-1; -32)$
    d) $(-1; 0)$ and $(3; 0)$
17. a)



    b) $y = -4$
    c) $(3; -2)$
18. 2
19. a) 86 mm
    b) $-3{,}125$ mm per minute
    c) After 4 minutes
20. 24 m.s^{-2}
21. $10 + 5\sqrt{2}$ units
22. a) $H = \frac{1}{\pi r^2}$
    c) $d = 0{,}94$ m
23. a) $V = \frac{1}{12}(10\pi d^2 - \pi d^3)$
    b) $r = 3{,}34$ cm and $h = 3{,}34$ cm
    c) Maximum volume $= 38{,}79$ cm^3
24. a) $-11{,}5$ m per hour
    b) Decreasing
    c) 06h49

Exercise 7 – 1: Revision

1. a) $5\sqrt{10}$; $M = \left(\frac{1}{2}; \frac{7}{2}\right)$; $m = 3$; $y = 3x + 2$
   b) $\sqrt{306}$; $M = \left(\frac{5}{2}; \frac{3}{2}\right)$; $m = \frac{3}{5}$; $y = \frac{3}{5}x$
   c) $\sqrt{5h^2 - 20hk + 20k^2}$; $M = \left(\frac{h+2k}{2}; -3k\right)$; $m = -2$; $y = -2x + h - k$
   d) $\sqrt{104}$; $M = (1; 4)$; $m = 5$; $y = 5x - 1$
2. $A(7; 0)$
3. $r = -1$ or $r = 23$
4. a) $y = 2x + 3$
   b) $y = -x - 1$
   c) $y = \frac{1}{3}x - 2$
   d) $y = -4$
   e) $y = -4x + 1$
   f) $x = 5$
   g) $x = \frac{3}{4}$
   h) $y = 2px + 3$
   i) $y = 4x - \frac{3}{5}$
   j) $y = -2$
   k) $y = \frac{7}{2}x + 5$
   l) $y = -5x$

Exercise 7 – 2: Inclination of a straight line

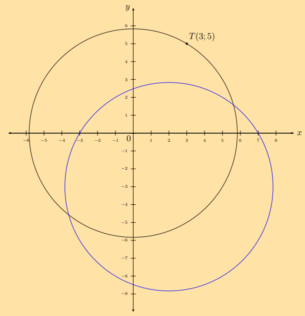
1. a) 36,9°
   b) 26,6°
   c) 45°
   d) Horizontal line
   e) 60°
   f) 68,2°
   g) 153,4°
   h) 56,3°
2. 86,6°
3. $y = -3x + 5$
4. $y = \frac{3}{2}x + 2$
5. $y = -5x - 1$
6. $y = x - 4$
7. a) $y = -2x + 7$
   b) $\left(\frac{7}{2}; 0\right)$
   c) $\theta = 116{,}6°$
   d) $m = \frac{1}{2}$
   e) $B\hat{A}C = 90°$
   f) $y = -2x$
   g) $\left(\frac{1}{2}; -\frac{3}{2}\right)$
   h) $y = -2x - \frac{1}{2}$
8. a)
   b) $I\,(6; 9)$
9. a) $y = \frac{3}{2}x + 2$
   b) $T\hat{S}V = 45°$

Exercise 7 – 3: Equation of a circle with centre at the origin

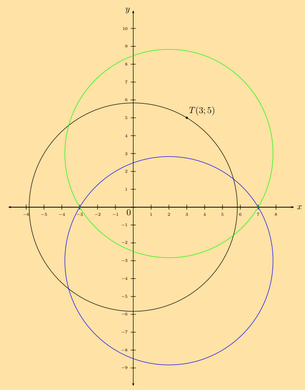
1. a) $r = 4$, $\left(1; \sqrt{15}\right)$ and $\left(1; -\sqrt{15}\right)$
   b) $r = 10$, $\left(2; \sqrt{96}\right)$ and $\left(2; -\sqrt{96}\right)$
   c) $r = 3$, $\left(1; \sqrt{8}\right)$ and $\left(1; -\sqrt{8}\right)$
   d) $r = \sqrt{20}$, $(2; 4)$ and $(2; -4)$
   e) $r = \frac{3}{2}$, $\left(1; \frac{\sqrt{5}}{2}\right)$ and $\left(1; -\frac{\sqrt{5}}{2}\right)$
   f) $r = \frac{\sqrt{10}}{3}$, $\left(1; \frac{1}{3}\right)$ and $\left(1; -\frac{1}{3}\right)$
2. a) $x^2 + y^2 = 25$
   b) $x^2 + y^2 = 11$
   c) $x^2 + y^2 = 34$
   d) $x^2 + y^2 = \frac{25}{4}$
   e) $x^2 + y^2 = 225$
   f) $x^2 + y^2 = p^2 + 9q^2$
   g) $x^2 + y^2 = 1$
   h) $x^2 + y^2 = 29t^2$
   i) $x^2 + y^2 = \frac{49}{4}$
   j) $x^2 + y^2 = \frac{16}{9}$
3. a) Yes: $x^2 + y^2 = 8$
   b) No, cannot be written in the form $x^2 + y^2 = r^2$
   c) No, cannot be written in the form $x^2 + y^2 = r^2$.
   d) Yes: $x^2 + y^2 = \sqrt{6}$
   e) Yes: $x^2 + y^2 = 11$
   f) No, cannot be written in the form $x^2 + y^2 = r^2$.
   g) Yes: $x^2 + y^2 = 9$
4. $\left(\sqrt{3}; 4\right)$ and $\left(\sqrt{3}; -4\right)$
5. a) $A\left(6\sqrt{10}; 2\sqrt{10}\right)$ or $A\left(-6\sqrt{10}; -2\sqrt{10}\right)$
   b) $A\left(4\sqrt{5}; 8\sqrt{5}\right)$ or $A\left(-4\sqrt{5}; -8\sqrt{5}\right)$
6. a) $x^2 + y^2 = 13$
   b)
   c)
   d) $PQ = 2\sqrt{13}$ units.
   e) $y = -\frac{3}{2}x$
   f) $y = \frac{2}{3}x + \frac{13}{3}$

Exercise 7 – 4: Equation of a circle with centre at $(a; b)$

1. a) Yes
   b) Yes
   c) No, coefficients of x^2 term and y^2 term are different.
   d) No, cannot be written in general form $(x-a)^2 + (y-b)^2 = r^2$
   e) Yes
   f) No, r^2 must be greater than zero.
2. a) $x^2 + (y-4)^2 = 9$
   b) $x^2 + y^2 = 25$
   c) $(x+2)^2 + (y-3)^2 = 40$
   d) $(x-p)^2 + (y+q)^2 = 6$
   e) $\left(x+\frac{1}{2}\right)^2 + \left(y-\frac{3}{2}\right)^2 = 10$
   f) $(x-1)^2 + (y+5)^2 = 26$
3. a) Centre: $(0; 2)$, $r = 5$ units
   b) Centre: $\left(-\frac{1}{2}; 0\right)$, $r = 2$ units
   c) Centre: $(2; -1)$, $r = \sqrt{10}$ units
   d) Centre: $(-1; 3)$, $r = 5$ units
   e) Centre: $(-3; -4)$, $r = \sqrt{30}$ units
   f) Centre: $\left(\frac{1}{3}; 2\right)$, $r = \sqrt{8}$ units
   g) Centre: $(-4; 5)$, $r = \sqrt{20}$ units
   h) Centre: $(3; -3)$, $r = \sqrt{12}$ units
4. $x^2 + (y+4)^2 = 20$ or $x^2 + (y-4)^2 = 20$
5. $(x+2)^2 + (y+2)^2 = 25$
6. a) $(x-4)^2 + (y-4)^2 = 13$
   b) $M\left(\frac{7}{2}; \frac{13}{2}\right)$
   c) $m_{MN} \times m_{KL} = -5 \times \frac{1}{5} = -1$
   d) $P(2; 1)$
   e) $y = \frac{3}{2}x - 2$
7. $(x-1)^2 + (y+3)^2 = 37$
8. a) $x^2 + y^2 = 34$
   b) $(x-2)^2 + (y+3)^2 = 34$
   c)

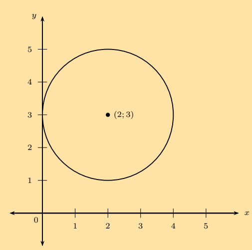


   d)



Centre of the shifted circle: $(2; 3)$

9.



Exercise 7 – 5: Equation of a tangent to a circle

1. a) $m = -\frac{2}{3}$
   b) $m_{\perp} = \frac{3}{2}$
2. a) $m = \frac{5}{3}$
   b) $m_{\text{tangent}} = -\frac{3}{5}$
3. $y = 7x + 19$
4. $y = -\frac{3}{4}x - \frac{1}{2}$
5. $y = 2x - 8$
6. a) $(x+4)^2 + (y-8)^2 = 136$
   b) 18
   c) $y = \frac{3}{5}x + 24$
7. a) $P(-4; -2)$ and $Q(2; 4)$
   b) $PQ = 6\sqrt{2}$
   c) $M(-1; 1)$
   d) $m_{PQ} \times m_{OM} = -1$
   e) $y = -2x - 10$ and $y = -\frac{1}{2}x + 5$
   f) $S(-10; 10)$
   h) $y = -2x + 10$ and $y = -2x - 10$

Exercise 7 – 6: End of chapter exercises

1. a) $(x)^2 + (y - 5)^2 = 25$
   b) $(x - 2)^2 + y^2 = 16$
   c) $(x + 5)^2 + (y - 7)^2 = 324$
   d) $(x + 2)^2 + y^2 = 9$
   e) $(x + 5)^2 + (y + 3)^2 = 3$
2. a) $(x - 2)^2 + (y - 1)^2 = 4$
   b) $(0; 1)$ and $(2; 3)$
3. a) $(x + 3)^2 + (y + 2)^2 = 20$
   b) $(x - 3)^2 + (y - 1)^2 = 17$
4. a) $(-9; 6)$, $r = 6$ units
   b) $(2; 9)$, $r = \sqrt{2}$ unit
   c) $(-5; -7)$, $r = \sqrt{12}$ units
   d) $(0; -4)$, $r = \sqrt{23}$ units
   e) $(2; -3)$, $r = 2$ units
5. a) $(0; 16), (0; -4), (-8; 0), (8; 0)$
   b) $(0; 0), (-8; 0)$
6. a) $(-3; 6)$ and $r = 5$ units
   b) $(-2; 4)$ and $r = \sqrt{20}$ units
   c) $(0; -4)$ and $r = \sqrt{23}$ units
   d) $(3; 0)$ and $r = 5$ units
   e) $(\frac{5}{2}; -\frac{3}{2})$ and $r = \frac{\sqrt{31}}{2}$ units
   f) $(3n; -5n)$ and $r = \sqrt{43}n$ units
7. a) $m = \frac{1}{12}$
   b) $m_{\perp} = -12$
8. a) $(0; 4), (2; 4)$
   b) $m = \frac{1}{3}$
9. $y = -\frac{1}{5}x + \frac{14}{5}$
10. $y = -\frac{7}{9}x - \frac{22}{3}$
11. $y = \frac{1}{2}x - 4$
12. a) $(x + 4)^2 + (y - 2)^2 = 61$
    b) $p = 3$
    c) $y = -\frac{6}{5}x - 15$
13. a) $y = -\frac{1}{4}x + \frac{17}{4}$
    b) $y = -\frac{3}{4}x + \frac{25}{4}$
    c) $y = -\frac{4}{3}x + 9$
    d) $y = -\frac{3}{2}x + \frac{21}{2}$
14. $y = x - 10$ and $y = x + 10$
15. a) $y = -x + 8$
    b) $B(5; 3)$
    c) $(x - 4)^2 + (y - 4)^2 = 2$
    d) The circle must be shifted 4 units down and 4 units to the left.
    e) $(x - 6)^2 + (y - 5)^2 = 18$

8 Euclidean geometry

Exercise 8 – 1: Revision

1. $z = 30°$
2. a) $OD = 3$ cm
   b) $AD = 8$ cm
   c) $AB = 4\sqrt{5}$cm
3. a) $R\hat{Q}S, Q\hat{S}O$
   b) $P\hat{O}S = 2a$
4. a) $O\hat{D}C = 35°$
   b) $C\hat{O}D = 110°$
   c) $C\hat{B}D = 55°$
   d) $B\hat{A}D = 90°$
   e) $A\hat{D}B = 45°$
6. $m = \frac{1}{4}n$
7. $y = 17$ mm
9. a) $A\hat{B}T = 90° - \frac{x}{2}$
   b) $O\hat{B}A = \frac{x}{2}$
   c) $\hat{C} = 90° - \frac{x}{2}$

Exercise 8 – 2: Ratio and proportion

1. a) $p = 5$
   b) $p = 7$
   c) $p = \frac{1}{2}$
   d) $p = 2$
2. Red $= 60$ sweets, blue $= 40$ sweets and yellow $= 60$ sweets,
3. Substance $B = 35{,}71$ kg
5. a) $(a + b) : b$
   b) $\frac{a}{2a} = \frac{1}{2}$
   c) $\frac{2a+b}{3a+b}$
   d) $2a : b$
6. $\frac{\text{area } ABF}{\text{area } ABCD} = \frac{9}{25}$
7. $AC = 20$ m, $CB = 20$ m
8. $TQ = 30$ m, $PT = 15$ m
9. a) $1{,}2$ m^2
   b) Perimeter: 414,8 cm; Area 11 700 cm^2
   c) 108×108cm^2

Exercise 8 – 3: Proportionality of polygons

1. a) $MN = 17{,}15$ cm
   b) Area $\triangle OPQ = 44{,}12$ cm^2
2. $t = 12$ cm, $p = 18$ cm, $q = 30$ cm
3. $\sqrt{5}x^2$
4. a) $FH = 19$ mm
   b) 76 mm^2

Exercise 8 – 4: Proportionality of triangles

1. a) $\frac{BC}{AB} = \frac{3}{2}$
   b) $\frac{AB}{AC} = \frac{2}{5}$
   c) 30 mm
2. $QP = 16$ cm and $PR = 10$ cm
5. a) "The line joining **the mid-points of two sides** of a triangle is **parallel** to the third side and equal to **half the length of the third side.**"
   c) Converse: a line through the mid-point of one side of a triangle, parallel to a second side, bisects the third side.

Exercise 8 – 5: Proportion theorem

1. a) $\frac{1}{4}$
   b) $\frac{9}{16}$
3. 4,87

Exercise 8 – 6: Similar polygons

1. a) Similar, all angles $= 90^\circ$ and sides are in the same proportion.
   b) Not enough information is given. Corresponding angles are equal.
   c) Not similar, sides are not in the same proportion.
2. a) True
   b) False
   c) False
   d) True
   e) False
   f) True
   g) False
   h) True

Exercise 8 – 7: Similarity of triangles

1. b) 1,6 cm
2. c) 21,6 mm
3. $AE = 2$ cm, $EC = 12{,}3$ cm and $BE = 2{,}7$ cm

Exercise 8 – 8: Similarity of triangles

1. Yes
2. $ED = 4$ cm
4. c) $FE = 20$ cm

Exercise 8 – 9: Theorem of Pythagoras

1. $6\sqrt{3}$cm
2. $a = 1$ unit and $b = 2\sqrt{3}$ units
3. b) 3 units

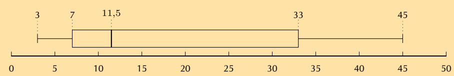
Exercise 8 – 10: End of chapter exercises

1. $SV = 17{,}5$ units
2. $\frac{DS}{SZ} = \frac{5}{3}$
3. $IJ = 25{,}7$ m and $KJ = 22{,}4$ m
4. $FH = 23{,}5$ cm
5. $\frac{HJ}{KI} = \frac{40}{9}$
6. $BC = 8{,}3$ m, $CF = 16{,}7$ m, $CD = 8{,}7$ m, $CE = 5{,}6$ m, $EF = 11{,}1$ m
7. $XZ = 49$ cm and $YZ = 46{,}96$ cm
9. c) $2\sqrt{3}$ cm
12. a) $> W\hat{R}S = 90°$
    b) $\hat{W} = 40°$
    c) $\hat{P}_1 = 40°$
14. a) $\hat{A} = \hat{D}_4 = \hat{E}_2 = \hat{D}_2 = x$

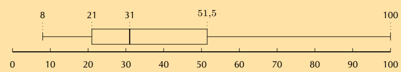
9 Statistics

Exercise 9 – 1: Revision

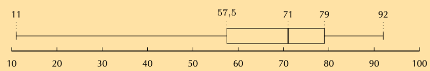
1. a) skewed right
   b) symmetric
   c) skewed left
   d) skewed right
   e) skewed left
   f) symmetric
2. a) mean = $18{,}5$; five number summary (3; 7; $11{,}5$; 33; 45); skewed right.



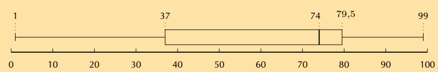
   b) mean = $40{,}83$; five number summary (8; 21; 31; $51{,}5$; 100); skewed right



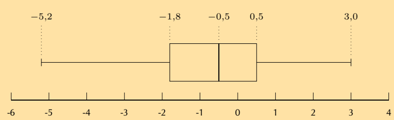
   c) mean = $65{,}4$; five number summary (11; $57{,}5$; 71; 79; 92); skewed left



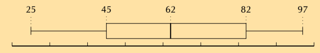
   d) mean = $40{,}83$; five number summary (1; 37; 74; $79{,}5$; 99); skewed left



   e) mean = $-0{,}6$; five number summary ($-5{,}2$; $-1{,}8$; $-0{,}5$; $0{,}5$; 3); close to symmetric/slightly skewed left



   f) mean = $61{,}83$; five number summary (25; 45; 62; 82; 97) ; symmetric



3. a) $\bar{x} = 5{,}9$; $\sigma^2 = 12{,}54$; $\sigma = \pm 3{,}54$; 63%
   b) $\bar{x} = 5{,}5$; $\sigma^2 = 7{,}65$; $\sigma = \pm 2{,}77$; 70%
   c) $\bar{x} = 43$; $\sigma^2 = 735{,}17$; $\sigma = \pm 27{,}11$; 67%

4. a)
   - Mean = 5
   - $\sigma^2 = 9$
   - $\sigma = \pm 3$

   b)
   - Mean = 8,58
   - $\sigma^2 = 30{,}91$
   - $\sigma = \pm 5{,}56$

   c)
   - Mean = 3,88
   - $\sigma^2 = 1{,}47$
   - $\sigma = \pm 1{,}21$

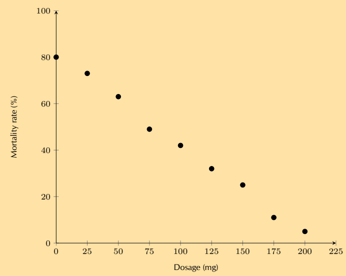
   d)
   - Mean = $-1{,}66$
   - $\sigma^2 = 11{,}47$
   - $\sigma = \pm 3{,}39$

   e)
   - Mean = 54,07
   - $\sigma^2 = 1406{,}07$
   - $\sigma = \pm 37{,}50$

5. a) A: 3,84. B: 3,82.
   b) A: $\pm 0{,}121$. B: $\pm 0{,}184$.
   c) Supermarket A
6. a) 47,87
   b) $\pm 0{,}27$
   c) 3
7. 13 and 20
8. 1,72 m
9. a) 47,5%
   b) 29

Exercise 9 – 2: Intuitive curve fitting

1. a) quadratic
   b) exponential
   c) linear
   d) linear
   e) exponential
   f) quadratic
2. a) strong, positive linear function
   b) $y = 0{,}6x + 1$ - learner answer may vary
   c) 16 - learner answer may vary
   d) 40 - learner answer may vary
3. a) Exponential
   b) 33 554 432
4. a) Quadratic
   b) temperature $= 18{,}33$, yield $= 6{,}17$
5. a)



   b) strong, negative linear
   c) $y = 0{,}38x + 80$ - learner answer may vary
   d) 210,53 mg
   e) Extrapolation - trend different outside of data range

Exercise 9 – 3: Least squares regression analysis

1. a) $\hat{y} = 0{,}62 + 0{,}57x$
   b) $\hat{y} = -6{,}57 + 0{,}45x$
   c) $\hat{y} = -22{,}61 + 1{,}70x$
2. a) $\hat{y} = 9{,}07 + 1{,}26x$
   b) $\hat{y} = -29{,}09 + -5{,}84x$
   c) $\hat{y} = 9{,}45 + 1{,}33x$
   d) $\hat{y} = -12{,}44 + -3{,}71x$
   e) $\hat{y} = -1{,}94 + 3{,}25x$
   f) $\hat{y} = -5{,}64 + 0{,}72x$
   g) $\hat{y} = 3{,}52 + -0{,}13x$
   h) $\hat{y} = 14{,}55 + -0{,}03x$
   i) $\hat{y} = 16{,}94 + 0{,}20x$
   j) $\hat{y} = 5{,}14 + -7{,}03x$
3. a) $\hat{y} = 0{,}19 + 5{,}70x$
   b) $\hat{y} = 18{,}62 - 0{,}09x$
   c) $\hat{y} = 3{,}61 + 2{,}33x$
4. c) $\hat{y} = 858{,}48 + 164{,}06x$
   d) R 3319,42
   e) 13
5. a)

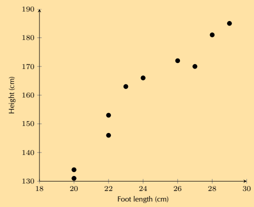
English % (x)	Mathematics % (y)	xy	x^2
28	35	980	784
33	36	1188	1089
30	34	1020	900
45	45	2025	2025
45	50	2250	2025
55	40	2200	3025
65	50	3250	4225
70	65	4550	4900
76	85	6460	5776
65	70	4550	4225
85	80	6800	7225
90	90	8100	8100
$\sum = 742$	$\sum = 740$	$\sum = 46\,673$	$\sum = 47\,324$

b) $\hat{y} = 6{,}01 + 0{,}89x$

c) $50{,}51\%$

d) $77{,}52\%$

6. a)



b) Strong (or fairly strong), positive, linear trend

c) $\hat{y} = 27{,}56 + 5{,}50x$

e) $146{,}36$ cm

f) $29{,}53$ cm

Exercise 9 – 4: Correlation coefficient

1. a) $r = -0{,}29$, negative, weak

b) $r = 0{,}68$, positive, moderate

c) $r = 0{,}91$, positive, very strong

c) $r = 0{,}00$, no correlation

d) $r = 0{,}14$, positive, very weak.

e) $r = 0{,}83$, positive, strong.

2. a) $r = -0{,}95$, negative, very strong.

b) $r = -0{,}40$, negative, weak.

3. a) $-0{,}48$, negative, weak.

b) $0{,}50$, positive, moderate.

4. a)

City	Degrees N (x)	Average temp. (y)	xy	x^2	$(x-\bar{x})^2$	$(y-\bar{y})^2$
Cairo	43	22	946	1849	15,21	0,64
Berlin	53	19	1007	2809	193,21	14,44
London	40	18	720	1600	0,81	23,04
Lagos	6	32	192	36	1095,61	84,64
Jerusalem	31	23	713	961	65,61	0,04
Madrid	40	28	1120	1600	0,81	27,04
Brussels	51	18	918	2601	141,61	23,04
Istanbul	39	23	897	1521	0,01	0,04
Boston	43	23	989	1849	15,21	0,04
Montreal	45	22	990	2025	34,81	0,64
Total:	391	228	8492	16 851	1562,9	173,6

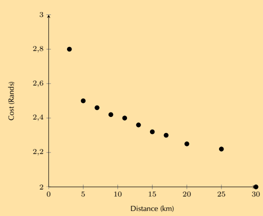
b) $\hat{y} = 33{,}38 - 0{,}27x$

d) $-0{,}81$

e) strong, negative, linear

f) $31{,}04$

5. a)



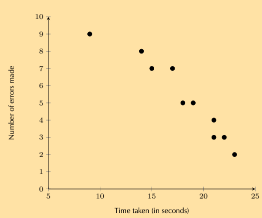
b) $\hat{y} = 2{,}67 - 0{,}02x$

c) $r = -0{,}92$

d) very strong, negative, linear

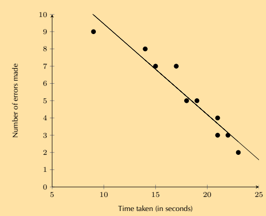
e) 46 kilometres

6. a)



b) When more time is taken to complete the task, the learners make fewer errors.

c)

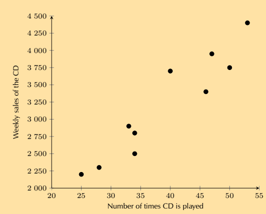


d) $r = -0{,}96$

e) 8

f) strong, negative

7. a)



b) $\hat{y} = 293{,}06 + 74{,}28x$

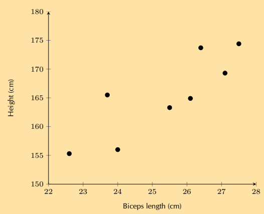
c) $r = 0{,}95$

d) 3650 (to the nearest 50)

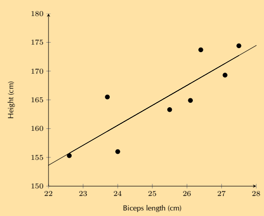
e) very strong, positive

Exercise 9 – 5: End of chapter exercises

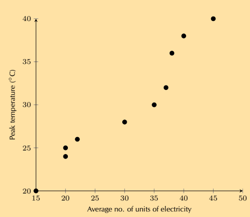
1. a) 16%
   b) $81{,}5\%$
2. 4,69 machine hours
3. a) $\hat{y} = 86\ 893{,}33 + 3497{,}14x$
   b) R 111 373,31, R 114 870,45.
   c) 13 months
4. a) $\hat{y} = 601{,}28 + 3{,}59x$
   b) r=0,86. Strong, positive, linear.
   c) 528 burgers
   d) R 2360,38
5. a)



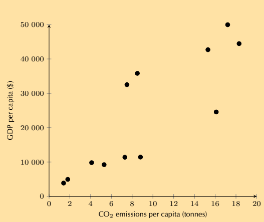
   b) $\hat{y} = 77{,}32 + 3{,}47x$
   c)



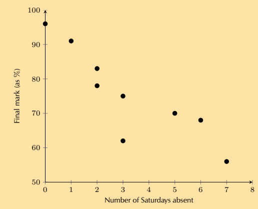
   d) $r = 0{,}85$
   e) strong, positive, linear
6. a) Yes, $r = 0{,}98$
   b) 11
7. a)



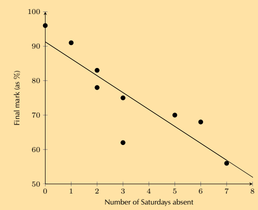
   b) $\hat{y} = 11{,}36 + 0{,}61x$
   c) 0,96
   d) Very strong, positive, linear
   e) 55 units, extrapolation
   f) quadratic
   g) temperature $= 16{,}54$, units $= 9{,}44$
8. a)



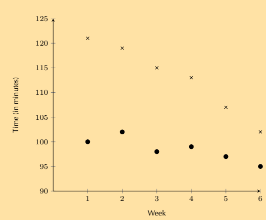
   b) $y = 2500x + 1000$ - learner answer may vary
   c) $\hat{y} = 1134{,}00 + 2393{,}74x$
   d) $r = 0{,}85$
   e) strong, positive, linear
   f) 0,24 tonnes
9. a)



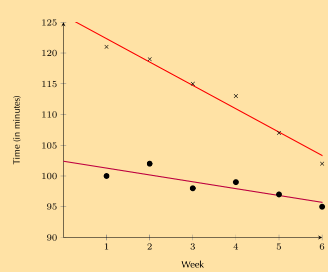
   b)



   c) $r = -0{,}87$
   d) The greater the number of Saturdays absent, the lower the mark.
   e) 72%
10. a)



   b) Both data sets show negative, linear trends. The trend in Grant's data appears to be more rapidly decreasing than the trend in Christie's data.
   c)



   d)
   e) No
   f) ninth week

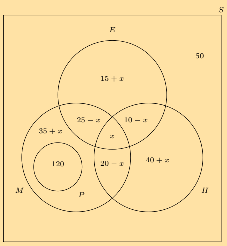
10 Probability

Exercise 10 – 1: The product and addition rules

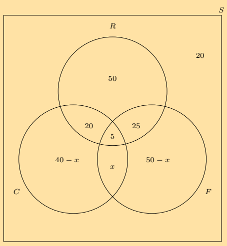
1. a) Dependent
   b) Independent
   c) Independent
   d) Independent
   e) Dependent
2. a) No
   b) Yes
3. a) Dependent
   b) Independent
4. a) No
   b) Yes
5. a) $\frac{1}{36}$
   b) $\frac{1}{4}$
   c) $\frac{25}{36}$
   d) $\frac{5}{18}$
   e) $\frac{11}{36}$
6. a) $\frac{6}{35}$
   b) $\frac{12}{35}$
   c) $\frac{17}{35}$
   d) Injuries, suspensions etc.
7. a) $\frac{1}{3}$
   b) $\frac{2}{3}$
   c) $\frac{2}{9}$
8. a) $\frac{2}{3}$
   b) $\frac{4}{15}$
   c) $\frac{1}{3}$
   d) Yes
9. a) No
   b) Yes
10. a) 0,5
    b) $\frac{5}{7}$
11. a) $\frac{1}{3}$
    b) $\frac{1}{36}$
    c) Yes
12. a) Learner dependent.
    b) 0,44
    c) 0,56
13. a) $\frac{4}{15}$
    b) $\frac{2}{5}$
    c) $\frac{7}{9}$
    d) $\frac{2}{9}$
    e) $\frac{8}{45}$
14. 55%
15. a) $\frac{2}{3}$
    b) $\frac{1}{3}$
    c) $\frac{1}{6}$
    d) $\frac{2}{3}$
    e) Independent

Exercise 10 – 2: Venn and tree diagrams

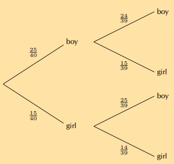
1. a) 115
   b) 16
   c) $\frac{73}{115}$
   d) $\frac{68}{115}$
   e) $\frac{1}{23}$
   f) $\frac{20}{23}$
   g) $\frac{81}{115}$
   h) $\frac{39}{115}$
   i) $\frac{24}{115}$
2. a)



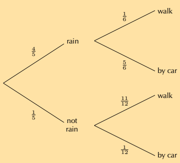
   b) 5
   c) $\frac{5}{32}$
   d) $\frac{33}{64}$
3. a)



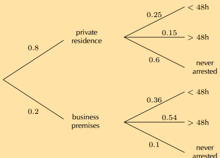
   b) 10
   c) $\frac{1}{10}$
   d) $\frac{17}{20}$
   e) No
4. a)



   b) $\frac{25}{52}$
   c) $\frac{45}{52}$
   d) dependent
5. a)



   b) $\frac{2}{15}$
   c) $\frac{41}{60}$
6. a)



   b) 0,48
   c) 0,5
   d) 0,875
   e) i) 4; ii) 7

Exercise 10 – 3: Contingency tables

1. a) Variables: gender and no. of accidents. Purpose: to determine if gender is related to the number of accidents a driver is involved in.

   b)

	$\leq$ 2 accidents	$>$ 2 accidents	Total
Female	210	90	300
Male	140	60	200
Total	350	150	500

   c) Yes

2. a) $\frac{3}{4}$

   b) $\frac{1}{8}$

   c) 81

   d)

	Malaria	No malaria	Total
Male	27	189	216
Female	81	567	648
Total	108	756	864

3. a) 120

   b) 280

   c) 84

   d)

	Reaction time $< 1{,}5$ s	Reaction time $> 1{,}5$ s	Total
≤ 40 years	84	36	120
> 40 years	196	84	280
Total	280	120	400

4. a)

	Flu	No flu	Total
Placebo	228	60	288
Treatment	12	252	264
Total	240	312	552

   b) $\frac{13}{23}$

   c) $\frac{11}{23}$

   d) $\frac{21}{46}$

   e) Dependent

   f) $\frac{21}{22}$

   g) $\frac{5}{24}$

   h) Yes

   i) No

5. a)

	Sick	Healthy	Total
Positive	999	9	1008
Negative	1	8991	8992
Total	1000	9000	10 000

   b) 0,01

   c) 0,0001

Exercise 10 – 4: Number of possible outcomes if repetition is allowed

1. 60
2. $1{,}0995 \times 10^{12}$
3. 100 000
4. 8 000 000
5. 2880
6. a) 175 760 000

   b) 60 761 421

   c) 3 114 752
7. 314

Exercise 10 – 5: Factorial notation

1. a) 6
   b) 720
   c) 12
   d) 40 320
   e) 120
   f) 738
   g) 359
   h) $\frac{1}{15}$
   i) -28
   j) 216
   k) 72
2. a) 239 500 800
   b) $1{,}49 \times 10^{-12}$
   c) 574 560
   d) 960
   e) 20 736

Exercise 10 – 6: Number of choices in a row

1. 720
2. 120
3. 120
4. 612
5. 27 907 200
6. 60
7. a) $1{,}31 \times 10^{12}$
   b) 32 760
8. a) 720
   b) 24
   c) 144
9. a) 120
   b) 216
10. a) 120
    b) 48
    c) 24
11. a) 39 916 800
    b) 13 824
    c) 725 760
    d) 967 680

Exercise 10 – 7: Number of arrangements of sets containing alike objects

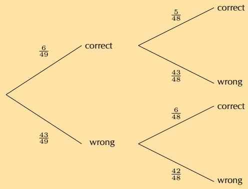
1. a) 362 880
   b) 30 240
   c) 3360
   d) 6720
   e) 3360
2. a) 3 628 800
   b) 75 600
   c) 11 760
   d) 22 680
   e) 360
3. a) 5040
   b) 823 543
   c) 16 807
   d) 210
4. a) 5040
   b) 35
5. a) 8
   b) 56
   c) 70
6. a) 256
   b) 24
   c) 128

Exercise 10 – 8: Solving probability problems using the fundamental counting principle

1. a) 5040
   b) 120
   c) 0,114
2. a) 72
   b) 0,5
3. a) 6 227 020 800
   b) 9 676 800
   c) 0,513
4. a) 40 320
   b) 0,25
5. $\frac{1}{12}$
6. $\frac{1}{210}$
7. a) $\frac{1}{240}$
   b) $\frac{1}{8}$
   c) 0,271
8. a) 9 239 076 000
   b) 0,025

Exercise 10 – 9: End of chapter exercises

1. a) 10 000
   b) $\frac{1}{10}$
   c) $\frac{1}{10}$
   d) $\left(\frac{1}{10}\right)^4 = 0{,}0001$
   e) 0,9997
2. a) $1{,}01 \times 10^{10}$
   b)



   c) $\frac{6}{49}$
   d) $\frac{5}{48}$
   e) $\frac{6}{48}$
   f) $\frac{6}{49}$
   g) $7{,}15 \times 10^{-8}$
3. 11,14%
4. 0,35
5. a) 1 000 000 000
   b) 10 000 000
   c) $\frac{1}{2}$
   d) 0,002
6. a) 3 326 400
   b) $\frac{1}{165}$
7. 38,89%
8. $\frac{2}{35}$
9. a) 0,18
   b) 0,9
   c) 0,2
   d) 0,2
10. a) 5040
    b) $\frac{1}{35}$
11. $\frac{1}{38}$
12. 12
13. a)

	Sick	Healthy	Total
Positive	99	999	1098
Negative	1	98 901	98 902
Total	100	99 900	100 000

    b) 0,09